PF^{2}V-WM: Phase-Field Fracture Veil World Models

Fracture as Latent Inference in a Multiscale Anisotropic Medium

Sohan Poudel
forreserachonlyso@gmail.com
ORCID: 0009-0004-5762-3898

May 26, 2026

Transparency and Provenance Note

This manuscript is an exploratory research draft and should not be interpreted as a peer-reviewed claim of novelty, priority, or superior performance over alternative approaches. Its purpose is to document a diagnostic experimental program on latent reasoning control.

Large language models, including ChatGPT, Gemini, and Grok, assisted with brainstorming, drafting, editing, code scaffolding, and result interpretation. The experimental direction, code execution, result selection, and final framing were determined and verified by the author.

Readers should therefore treat this paper as an early proof-of-concept report whose empirical claims depend on the included synthetic experiments and should be independently verified before they are treated as established scientific conclusions.

For additional transparency, the author also provides supporting materials, including chat conversations and code used to run the experiments.

- Summarized ChatGPT conversation:
  `https://chatgpt.com/share/6a11a11b-2d7c-8322-add2-33711d535bb3`
- Original Gemini conversation:
  `https://gemini.google.com/share/4f2bb4d86746`
- Original Grok conversation:
  `https://grok.com/share/bGVnYWN5LWNvcHk_e44fe799-d5a6-40d3-9c29-5cf4098fff35`
- Colab notebook:
  `https://colab.research.google.com/drive/1hBdnqp21oQG5OIF7DLYquw61oI4d2m_j?usp=sharing`

Abstract

We propose the *Phase-Field Fracture Veil World Model* (PF^{2}V-WM), a world-modeling architecture in which latent inference is implemented as controlled fracture propagation in a structured medium. The model maintains spatial fields for damage, potential, and memory; observations inject forcing; and an unrolled variational solver evolves the latent state using fracture regularization, anisotropic transport, geometric priors, and slow material hardening. The paper is deliberately both constructive and empirical: it formalizes the energy-based latent dynamics, describes a practical neural implementation, and records a sequence of prototype experiments on synthetic path-formation and obstacle-bending tasks. Early runs mostly produced endpoint blobs and diffuse bridges, but progressive changes—core–spread separation, material memory, obstacle-conditioned evaluation, threshold sweeps, and a final focus model—improved obstacle-aware route formation. The strongest final focus setting reaches hard IoU 0.2066 on a standard held-out split and 0.1866 on a harder obstacle split, with obstacle overlap around 0.18 and 0.15, respectively. These results are not evidence of a solved planner: the learned paths remain threshold-sensitive and often fragmentary. They do, however, support a narrower claim that phase-field material dynamics can provide an interpretable substrate for partial, obstacle-aware latent route formation. The novelty is therefore a systems-level synthesis of known phase-field mechanics, latent world models, and energy-based inference, together with an honest empirical account of where this substrate currently succeeds and fails.

1 Introduction

Modern world models usually perform inference through repeated dense updates of vectors or feature maps. That is flexible, but it weakly constrains three properties that matter for world modeling: *adaptive compute*, *persistent structural memory*, and *geometrically coherent internal trajectories*.

This paper proposes a different organizing principle: *reasoning as controlled fracture propagation in a latent material.* Instead of a flat hidden state, PF^{2}V-WM uses a spatial latent medium with tension, damage, anisotropy, and memory. A learned encoder injects forcing and boundary information, and a differentiable variational solver relaxes the latent fields toward lower energy. In principle, this gives adaptive computation, long-range spatial coupling, and slow path-dependent memory within one substrate.

The proposed model, PF^{2}V-WM, combines five ingredients:

- a **damage field** representing commitment or computational expenditure;
- a **potential field** whose gradients carry latent stress and directional flow;
- a **memory field** that hardens or softens previously used pathways;
- **anisotropic transport** that encodes quad-directional or learned structural bias; and
- **geometric and topological priors** that favor coherent spiral-like trajectories and short sufficient fracture paths.

The goal is not to claim that intelligence literally follows brittle fracture mechanics. The narrower claim is that phase-field fracture dynamics can serve as a useful computational substrate for adaptive latent inference.

This draft now includes a long prototype record rather than only a proposal. The empirical story is mixed but informative: simple versions often collapse into blobs or broad corridors, whereas later material-style variants become more obstacle-selective and the final focus model produces the strongest visible route signal. The main open problem is no longer whether the model can sense a route, but whether it can harden a faint route-like latent field into a thin, continuous, threshold-stable path.

2 Relation to Prior Work and Novelty Positioning

PF^{2}V-WM is best understood as a *systems-level synthesis*. Its individual ingredients are not all new: phase-field fracture contributes variational damage dynamics and diffuse crack regularization; world models contribute latent dynamics and memory; and energy-based or continuous-depth models contribute optimization-based hidden-state evolution. The novelty lies in how these pieces are assembled.

The central reframing is that fracture becomes the latent mechanism of commitment, anisotropy becomes a reasoning bias, multiscale phase fields become a hierarchy of abstraction, and slow hardening becomes memory. This is modestly novel relative to fracture mechanics but more distinctive relative to standard latent world models, which usually rely on generic recurrent or transformer-style updates.

Accordingly, PF^{2}V-WM should be presented as a research direction rather than a replacement for existing systems. Stronger claims of superiority must be earned empirically, especially on tasks where sparse spatial trajectories, adaptive computation, or long-range coherence matter.

3 Conceptual Overview

At a high level, PF^{2}V-WM interprets latent computation through the language of fracture:

- **Reasoning** corresponds to the propagation and stabilization of a fracture front in latent space.
- **Simple inference** produces short, sparse, low-energy fracture paths.
- **Complex inference** induces richer multiscale branching and therefore more inner solver steps.
- **Commitment** occurs where the damage field saturates toward one.
- **Memory** is encoded as slow, path-dependent changes in local material properties.

In this view, adaptive computation emerges from the latent dynamics rather than from a separate controller: easy inputs settle quickly, while hard inputs require more propagation and conflict resolution.

4 Latent State and Multiscale Representation

Let $\Omega_l \subset \mathbb{R}^2$ denote the spatial domain at pyramid level $l \in \{0, \ldots, L-1\}$, where lower l indicates coarser resolution and higher l indicates finer resolution. At time t, the latent state is

$$X_t = \{d_l(\boldsymbol{x}, t),\, \phi_l(\boldsymbol{x}, t),\, m_l(\boldsymbol{x}, t)\}_{l=0}^{L-1}, \qquad \boldsymbol{x} \in \Omega_l. \tag{1}$$

The fields have the following meanings:

$$d_l(\boldsymbol{x}, t) \in [0, 1] \qquad \text{damage or commitment field,} \tag{2}$$
$$\phi_l(\boldsymbol{x}, t) \in \mathbb{R} \qquad \text{potential or stress field,} \tag{3}$$
$$m_l(\boldsymbol{x}, t) \geq 0 \qquad \text{memory or hardening field.} \tag{4}$$

The damage field records where computation has already "spent itself." The potential field stores uncertainty, directional flow, and long-range coupling through its gradient. The memory field evolves slowly across episodes and modifies future propagation by changing local toughness or stiffness.

Multiscale structure is essential. Coarse levels represent global constraints and long trajectories, while fine levels refine local detail. Let P_l and U_l denote downsampling and upsampling operators. Cross-scale consistency is enforced by coupling neighboring levels through penalties such as $\|d_l - U_l(d_{l-1})\|^2$.

5 Energy-Based Formulation

Inference is defined as approximate minimization of a total free energy

$$\mathcal{E}\big(\{d_l, \phi_l, m_l\}_{l=0}^{L-1}\big) = \sum_{l=0}^{L-1} \mathcal{E}_l(d_l, \phi_l, m_l). \tag{5}$$

For each scale l, we define

$$\begin{aligned}
\mathcal{E}_l = & \int_{\Omega_l} (1 - d_l)^2 \, \psi_e\big(\nabla\phi_l; A_l(m_l)\big) \, \mathrm{d}\boldsymbol{x} && (6)\\
& + \int_{\Omega_l} G_c^l(\boldsymbol{x}, m_l) \left(\frac{d_l^2}{2\ell_l} + \frac{\ell_l}{2} \|\nabla d_l\|^2 + W(d_l) \right) \mathrm{d}\boldsymbol{x} && (7)\\
& + \lambda_{\text{align}} \int_{\Omega_l} d_l \, \|\nabla\phi_l - \boldsymbol{v}_{\text{spiral}}^l(\boldsymbol{x})\|^2 \, \mathrm{d}\boldsymbol{x} && (8)\\
& + \lambda_{\text{TV}} \int_{\Omega_l} \|\nabla d_l\| \, \mathrm{d}\boldsymbol{x} \qquad + \lambda_{\text{cross}} \int_{\Omega_l} \|d_l - U_l(d_{l-1})\|^2 \, \mathrm{d}\boldsymbol{x}, && (9)
\end{aligned}$$

with the convention that the cross-scale term is omitted for $l = 0$.

5.1 Degraded Anisotropic Elastic Energy

The first term is the transport or elastic energy modulated by damage. A simple anisotropic form is

$$\psi_e\big(\nabla\phi_l; A_l\big) = \frac{1}{2} \langle \nabla\phi_l, A_l \nabla\phi_l \rangle, \tag{10}$$

where $A_l(m_l)$ is a positive semidefinite stiffness tensor, potentially dependent on memory. The prefactor $(1 - d_l)^2$ ensures that highly damaged regions carry less latent stress.

A practical QDMN-inspired implementation replaces a dense tensor with directional finite differences:

$$\psi_e(\nabla\phi) = \frac{1}{2} \sum_{k \in \{N,S,E,W\}} a_k(m) \, |D_k\phi|^2, \tag{11}$$

where D_k denotes a directional derivative and $a_k(m) > 0$ are learned directional stiffnesses. This biases propagation toward preferred axes and supports kinks or branches when directional pressures conflict.

5.2 Fracture Surface Energy and Double-Well Potential

The second term follows the standard phase-field fracture pattern. The toughness field $G_c^l(\boldsymbol{x}, m_l)$ determines how costly it is to create damage, while ℓ_l sets the width and smoothness of the diffuse crack. We use the double-well potential

$$W(d_l) = d_l^2(1 - d_l)^2, \tag{12}$$

which softly encourages binary commitment without hard projection.

5.3 Spiral Alignment Prior

The third term imposes a geometric preference on active regions. Let the spiral phase be

$$s(\boldsymbol{x}) = \alpha\, \theta(\boldsymbol{x}) + \beta\, \log\bigl(r(\boldsymbol{x}) + \varepsilon\bigr), \tag{13}$$

where

$$r(\boldsymbol{x}) = \sqrt{(x - c_x)^2 + (y - c_y)^2}, \qquad \theta(\boldsymbol{x}) = \mathrm{atan2}(y - c_y, x - c_x). \tag{14}$$

The corresponding normalized vector field is

$$\boldsymbol{v}_{\mathrm{spiral}}(\boldsymbol{x}) = \frac{\nabla s(\boldsymbol{x})}{\|\nabla s(\boldsymbol{x})\| + \varepsilon}. \tag{15}$$

This prior encourages coherent long-range trajectories rather than arbitrary diffuse activation.

5.4 Topological Simplicity Penalty

The total variation term penalizes unnecessary crack length and acts as an Occam-style prior. It discourages degenerate solutions in which the entire domain damages uniformly and instead favors the shortest sufficient path that resolves the task.

6 Memory as Slow Material Hardening

Memory is modeled as a slow field that accumulates along used trajectories and modulates future dynamics. A generic episode-level update is

$$m_l^{t+1} = (1 - \rho) m_l^t + \eta\, H\bigl(d_l^t, \phi_l^t\bigr), \tag{16}$$

where H is a hardening signal, for example

$$H(d, \phi) = \sigma\bigl(\alpha_m d + \beta_m \|\nabla \phi\|\bigr). \tag{17}$$

The memory field can modify the local toughness through either hardening or softening. A hardening rule takes the form

$$G_c^l(\boldsymbol{x}, m_l) = G_0^l\bigl(1 + \kappa m_l(\boldsymbol{x})\bigr), \tag{18}$$

while a softening rule may use a decreasing function of m_l. The choice should remain conceptually consistent throughout an implementation.

7 Latent Dynamics and Inference

We interpret inference as gradient flow or variational relaxation on the energy in Equation (9). In continuous form, the damage dynamics can be written as

$$\frac{\partial d_l}{\partial \tau} = -M_d \frac{\delta \mathcal{E}_l}{\delta d_l}, \tag{19}$$

and a quasi-static update for the potential field is given by

$$-\nabla \cdot \big((1 - d_l)^2 A_l(m_l) \nabla \phi_l\big) = f_l, \tag{20}$$

where f_l is encoder-produced forcing derived from the current observation and action.

In a practical discrete solver, one unrolls K_l inner optimization steps per level:

$$\phi_l^{(k+1)} = \phi_l^{(k)} - \eta_\phi \frac{\partial \mathcal{E}}{\partial \phi_l^{(k)}}, \tag{21}$$

$$d_l^{(k+1)} = \Pi_{[0,1]} \left(d_l^{(k)} - \eta_d \frac{\partial \mathcal{E}}{\partial d_l^{(k)}} \right), \tag{22}$$

where $\Pi_{[0,1]}$ denotes projection or a differentiable bounded parameterization. The number of inner iterations serves as a true computational budget: easy inputs converge rapidly, whereas difficult inputs require deeper latent relaxation.

8 Architecture

A practical PF^{2}V-WM system contains four main components.

Encoder. Given observation o_t, action a_t, and optionally the previous memory field, a convolutional encoder or compact U-Net produces the forcing field f_l, initial conditions for ϕ_l, optional initialization for d_l, and boundary-condition parameters.

Pyramid builder. Multiscale representations are created by repeated downsampling:

$$o_t^{(l)} = P_l(o_t), \qquad a_t^{(l)} = P_l(a_t). \tag{23}$$

These inputs may condition the solver either at all levels or only at the finest level with coarse-to-fine propagation.

Variational solver. For each level, the solver alternates between updating ϕ_l, updating d_l, and optionally applying a slow memory update. Shared solver parameters may be used across scales to reduce complexity, or level-specific parameters may be introduced to increase expressivity.

Readout head. The final latent fields are decoded into task outputs, such as next observation prediction, next latent state, action value, token distribution, or plan score. A simple readout may globally pool features extracted from d_{L-1} and ϕ_{L-1}, while a richer decoder can condition on the entire pyramid.

9 Training Objective

The total loss combines task performance with energy-based regularization:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \alpha \sum_{l=0}^{L-1} \mathcal{E}_l + \beta \mathcal{L}_{\text{traj}} + \gamma \mathcal{L}_{\text{simp}} + \delta \mathcal{L}_{\text{cross}} + \mu \mathcal{L}_{\text{mem}}. \tag{24}$$

Each term plays a distinct role:

- $\mathcal{L}_{\text{pred}}$: task loss for prediction, control, or language modeling;
- $\sum_l \mathcal{E}_l$: regularizes latent states to follow the intended fracture dynamics;
- $\mathcal{L}_{\text{traj}}$: encourages coherent latent trajectories across time;
- $\mathcal{L}_{\text{simp}}$: penalizes unnecessary total crack length or diffuse activation;
- $\mathcal{L}_{\text{cross}}$: enforces multiscale consistency;
- $\mathcal{L}_{\text{mem}}$: stabilizes memory consolidation over episodes.

A simple memory-consistency loss is

$$\mathcal{L}_{\text{mem}} = \sum_{l=0}^{L-1} \left\| m_l^{t+1} - \text{Update}(m_l^t, d_l^t, \phi_l^t) \right\|_2^2. \tag{25}$$

10 Algorithmic Summary

The full training loop can be summarized as follows.

Input: current observation o_t, action a_t, previous latent memory $\{m_l^t\}$.

Step 1: Encode. Produce forcing fields, initial potentials, and optional initial damage seeds from the observation-action pair.

Step 2: Build pyramid. Construct multiscale inputs and latent initializations.

Step 3: Relax the latent medium. For each level l and for K_l inner iterations:

1. update the potential field using the anisotropic transport term;
2. update the damage field using fracture, alignment, and TV penalties;
3. exchange information across scales if hierarchical coupling is enabled.

Step 4: Decode. Read out prediction targets from the final latent fields.

Step 5: Update memory. Apply the slow hardening rule to obtain m_l^{t+1}.

Step 6: Optimize. Backpropagate through the unrolled solver and minimize the total training loss.

11 Experimental Program and Main Findings

A staged evaluation strategy is critical because the full model introduces several interacting mechanisms. The experiments in this draft should be read as prototype diagnostics rather than as a mature benchmark claim. They ask a focused question: can phase-field latent dynamics form sparse, obstacle-aware route structure, and how do solver depth, thresholding, memory, and readout design affect that structure?

11.1 Minimal Viable Prototype

The initial prototype used a single 64×64 latent grid with one damage field, one potential field, one anisotropic stiffness mechanism, one spiral alignment prior, one total variation penalty, and one task-specific readout head. This setup was sufficient to test whether fracture-style latent inference could produce sparse, coherent trajectories in a controlled path-connection task.

11.2 Observed Experimental Trajectory

The empirical trajectory was not monotonic. Early start–goal path runs showed decreasing loss but mostly produced endpoint activation, diffuse bridges, or broad low-frequency blobs. Sharpening the damage field, adding edge losses, and annealing the target width improved local contrast but did not by themselves create thin connected paths. Later experiments introduced more structural pressure: core–spread separation, obstacle-bending tasks, explicit route metrics, memory probes, material-only inference, threshold sweeps, and soft shortest-path controls.

The most important negative result is that generic overlap and apparent connectivity can be misleading. The soft-control baseline often produced high connectivity, but much of that connectivity came from broad value-field activation that crossed obstacle regions. For this task, a useful route model must balance hard IoU, route ratio, obstacle overlap, route gap, and connectivity rather than optimizing any single metric.

11.3 Final Prototype Summary

The final paper pack identified material-thin $\mathrm{PF}^2\mathrm{V}$ as the best current fracture-style variant before the final focus run. In the three-seed robustness check, material-thin with threshold selection reached mean hard IoU 0.1244 ± 0.0096 on the standard split with obstacle overlap 0.1780 ± 0.0141, while the soft-control thresholded baseline reached hard IoU 0.0500 ± 0.0027 with obstacle overlap 0.5888 ± 0.0154. This supports the claim that material-style dynamics are more obstacle-selective than the broad soft-control solution.

The final focus run then warm-started from the material-thin model and added stronger core commitment, endpoint pressure, centerline supervision, side suppression, and obstacle penalties. Its best thresholded standard result reached hard IoU 0.2066, route ratio 0.7165, obstacle overlap 0.1826, and connectivity 0.0665. On the harder obstacle split, the best thresholded result reached hard IoU 0.1866, route ratio 0.7173, obstacle overlap 0.1470, and connectivity 0.0795. These are the strongest results in the manuscript, but they remain prototype-level: the outputs are route-aware and obstacle-selective, not yet clean binary corridors.

Setting	Hard IoU	Route Ratio	ObsOverlap	Conn.
material-thin robust std	0.1244 ± 0.0096	0.7946 ± 0.0163	0.1780 ± 0.0141	0.0727 ± 0.0098
soft-control robust std	0.0500 ± 0.0027	0.9652 ± 0.0051	0.5888 ± 0.0154	1.0395 ± 0.0060
final focus std	0.2066	0.7165	0.1826	0.0665
final focus hard	0.1866	0.7173	0.1470	0.0795

Table 1: Main empirical summary from the final robustness and focus runs. Material-style $\mathrm{PF}^2\mathrm{V}$ variants are more obstacle-selective than soft-control, and the final focus run gives the strongest hard-IoU results, but connectivity remains weak and threshold-sensitive.

11.4 Recommended Next Experiments

The next experiments should focus less on proving route awareness and more on *route hardening*: converting faint route-like probability mass into a thin, continuous, threshold-stable path. Useful next steps include continuity losses on skeletons or differentiable graph connectivity, learned post-solvers that harden only the core path, stronger obstacle penalties applied directly to the final prediction, and comparisons against obstacle-conditioned CNN and differentiable-planning baselines under identical metrics.

Key evaluation axes should include predictive accuracy, obstacle overlap, start-to-goal connectivity, route gap, threshold sensitivity, number of inner solver steps, spatial sparsity, and interpretability of the damage and potential fields.

12 Scientific Positioning and Claims

The strongest defensible claims of $\mathrm{PF}^2\mathrm{V}$-WM are architectural and algorithmic rather than biological. Specifically, the model suggests that:

1. variable-depth computation can emerge from a latent variational medium without a separate routing controller;
2. geometric priors such as spiral structure and directional anisotropy can be encoded directly in the latent physics;
3. memory can be represented structurally through material-property updates rather than opaque recurrent state alone; and
4. multiscale fracture dynamics may improve coherence on tasks requiring long-range dependency tracking.

At the same time, the honest novelty claim is limited in scope. $\mathrm{PF}^2\mathrm{V}$-WM does not introduce an entirely new class of phase-field mathematics; rather, it repurposes and recombines known ingredients into a new computational interpretation. Its strongest claim is therefore that a multiscale phase-field latent medium can serve as an adaptive inference substrate in which fracture propagation, anisotropy, geometric priors, and hardening jointly implement reasoning. This framing is more original than any individual equation viewed in isolation.

These claims are testable through ablation and comparative experiments. They do not require any assertion that natural intelligence literally implements fracture physics. They also do not imply that

PF^{2}V-WM is already better than established alternatives. Until validated, it should be regarded as an interpretable, high-risk, potentially high-reward alternative to standard latent world models and hybrid neural–physics systems.

13 Limitations and Open Questions

Several challenges remain before this proposal can be considered a mature model class.

- **Optimization difficulty:** unrolled variational solvers may be expensive and sensitive to hyperparameters.
- **Discretization choices:** stability depends on finite-difference stencils, boundary conditions, and the parameterization of bounded damage fields.
- **Interpretability versus flexibility:** strong geometric priors can improve structure but may overspecialize the model.
- **Benchmark selection:** tasks should genuinely reward adaptive internal computation rather than merely spatial inductive bias.
- **Current prototype weakness:** the final focus model improves hard IoU and obstacle selectivity, but the visible routes remain threshold-sensitive and often fragmentary rather than clean binary corridors.
- **Evaluation mismatch:** generic overlap and raw connectivity metrics can overstate progress on path tasks, especially when broad value fields cross obstacles; route gap, obstacle overlap, and threshold sensitivity must be reported together.

Future work may investigate implicit differentiation, learned preconditioners, richer anisotropy families, partial crack healing, improved path-specific objectives, connectivity-aware losses, and extensions to 3D or non-Euclidean latent domains.

14 Conclusion

PF^{2}V-WM reframes world modeling as inference in a structured latent material governed by phase-field fracture dynamics. The framework replaces a monolithic hidden state with fields for damage, potential, and memory, and it embeds geometric and topological biases directly into an energy functional. Across the prototype sequence, the idea proves neither trivial nor solved: early models mostly formed blobs, while later material-style variants became more obstacle-selective and the final focus model achieved the strongest path-overlap results.

The final empirical conclusion is therefore cautious. PF^{2}V-WM is not yet a robust planner, and the current implementation does not reliably materialize clean connected binary paths. However, the final experiments show that material-style phase-field dynamics can encode partial obstacle-aware route structure more selectively than a broad soft-control baseline. The next technical bottleneck is route hardening: preserving the interpretability and obstacle selectivity of the latent material while forcing the route mass to become thin, continuous, and stable under thresholding. If that bottleneck can be solved, PF^{2}V-WM may become a useful route toward world models whose internal computation is both dynamically adaptive and physically interpretable.

A Prototype Appendix: Experiment Record and Qualitative Figures

This appendix keeps a compact record of the prototype sequence so the paper preserves the empirical trail, not only the final claims.

A.1 Minimal Prototype Configuration

The current prototype implements a single-scale PF^{2}V-WM with the following components:

- a 64×64 spatial latent grid;
- one encoder mapping two-channel start/goal observations into an initial potential field ϕ;
- one sigmoid-parameterized damage field d updated by unrolled variational steps;
- one quad-directional stiffness parameterization for QDMN-style anisotropy;
- one spiral-modulated toughness prior;
- one shallow readout head producing path logits.

The synthetic task is to infer a path between two marked points. Each training sample contains a start marker, a goal marker, and a thickened straight-line target mask. The model is trained with weighted binary cross-entropy, Dice loss, and a weak total-variation regularizer on the predicted mask.

A.2 What We Did

The first end-to-end run trained for 200 epochs on random path-connection samples with batch size 8 and three inner solver steps. The goal was feasibility rather than benchmark performance: does the fracture-style latent mechanism learn any path-related structure at all?

A.3 What We Got

Table 2 shows decreasing loss but only modest overlap. Soft IoU stays low, so the prediction remains diffuse; hard IoU rises late, but likely because thresholding recovers broad overlap rather than a clean centerline.

Epoch	Loss	BCE	Dice	Soft IoU	Hard IoU
1	2.4134	1.3284	0.9103	0.0474	0.0496
25	2.4586	1.3421	0.9424	0.0299	0.0309
50	2.4031	1.3146	0.9173	0.0435	0.0453
75	2.2232	1.1772	0.9002	0.0527	0.0459
100	2.1132	1.1133	0.8629	0.0739	0.0620
125	2.0232	1.0278	0.8769	0.0657	0.0440
150	1.9503	0.9442	0.9073	0.0487	0.0330
175	2.0552	1.0511	0.8814	0.0635	0.0510
200	2.2564	1.2707	0.8122	0.1037	0.3310

Table 2: Checkpoint metrics from the first minimal PF^{2}V-WM prototype run on the synthetic path-connection task.

The fairest interpretation is cautious: the run shows *signal*, not *solution*. The outputs are structured, but still blurry and weakly localized.

A.4 What We Learn

Three lessons emerge from this first run.

1. **The concept is alive.** The latent mechanism is not inert; it responds to the task and learns nontrivial path-related behavior.
2. **The current solver is still weak.** The system tends to highlight endpoint neighborhoods more readily than it forms a crisp connected path.
3. **Evaluation must match the task.** Standard IoU can miss or misread progress on path-structured outputs, so connectivity-aware metrics are necessary.

The next step is therefore to stay at minimal scale, improve the metrics, and compare against a plain CNN before adding hierarchy or richer memory.

A.5 Recommended Next Measurements

The next runs should report not only overlap scores but also path-aware diagnostics:

- skeleton IoU;
- start-to-goal connectivity rate;
- centerline distance;
- path precision and path recall;
- comparison against a plain CNN trained on the same task.

These measurements better test whether fracture-style latent inference produces connected, parsimonious paths rather than diffuse masks.

A.6 Qualitative Figures from the Current Run

Figures 1–5 show the dominant failure mode: strong endpoint activations plus a broad background field instead of a compact connected path.

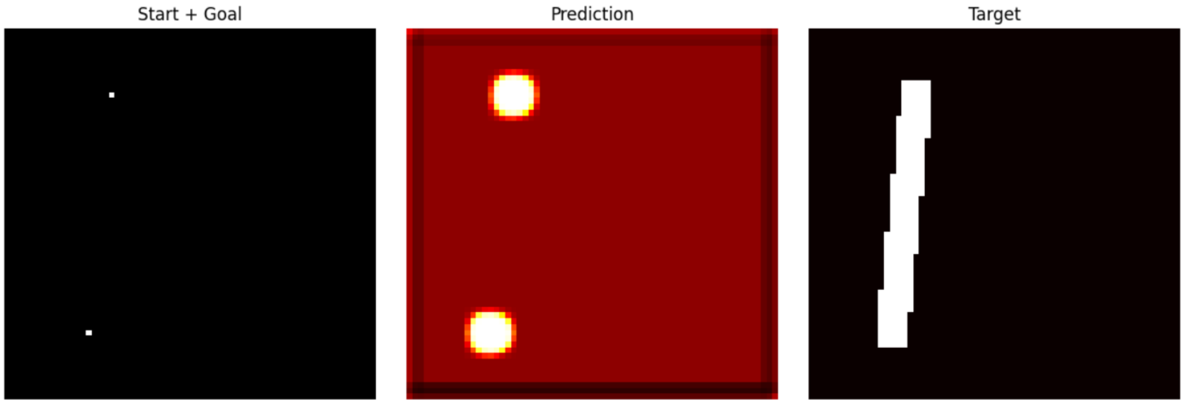

Figure 1: Prototype qualitative result A: prediction remains diffuse and emphasizes endpoint regions more strongly than the connecting path.

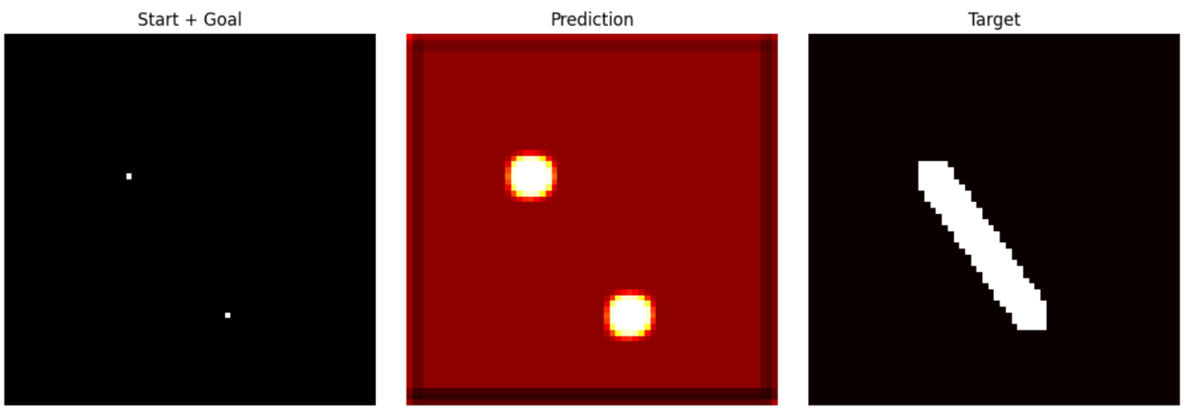

Figure 2: Prototype qualitative result B: the model exhibits path-related signal, but the predicted field is still broad and weakly structured.

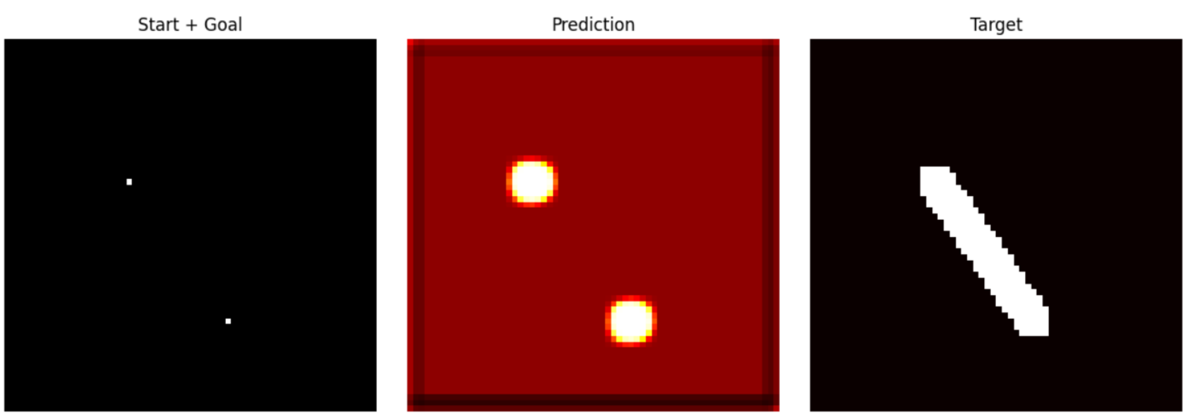

Figure 3: Prototype qualitative result C: strong endpoint blobs are present, but the connection between them is not formed sharply.

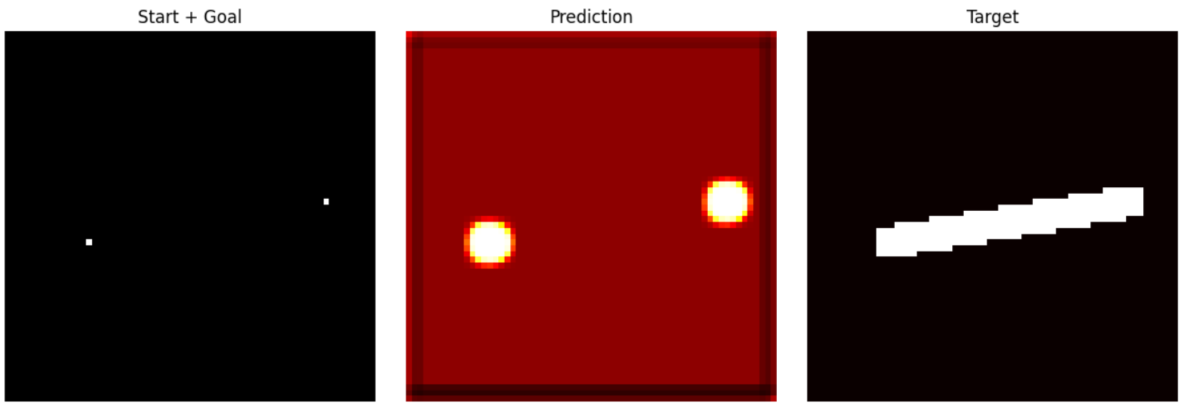

Figure 4: Prototype qualitative result D: the predicted map preserves some task-relevant structure, yet remains substantially more diffuse than the target path.

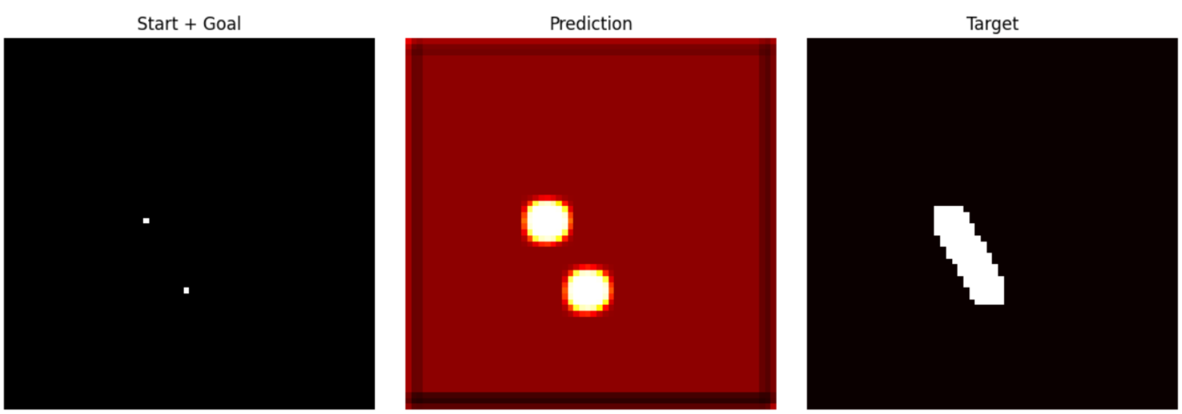

Figure 5: Prototype qualitative result E: another representative example showing endpoint emphasis and background activation rather than a crisp fracture trajectory.

B Prototype Appendix: Second Run with Smoothed Inputs and Explicit Damage Supervision

This second appendix entry records a revised minimal prototype in which the input representation, latent update rule, and supervision were modified to make the path-formation task easier to learn while still preserving the phase-field flavor of the model.

B.1 What We Changed

Compared with the first run, the second implementation introduced several practical changes:

- Gaussian start and goal blobs replaced single-pixel markers;
- the target path was replaced by a soft corridor mask rather than a binary thickened line;
- coordinate channels and a signed source-minus-goal charge channel were appended to the input;
- the damage field received its own learned initialization network;
- the latent dynamics were updated with explicit hand-crafted drives for both d and ϕ rather than differentiating the full energy with autograd at each inner step;
- an auxiliary latent binary-cross-entropy term was added so that the damage field itself received direct supervision.

The run used five inner steps and 250 epochs to test whether the latent medium could learn a more useful connection pattern.

B.2 What We Got

Table 3 shows easier optimization and soft IoU near 0.12 around epoch 200, but unstable hard IoU and weak final performance.

Epoch	Loss	BCE	Latent BCE	Dice	Soft IoU	Hard IoU
1	2.4696	1.3332	0.5679	0.9236	0.0401	0.0236
25	2.1794	1.0923	0.4707	0.9468	0.0276	0.2947
50	2.0384	1.0734	0.3457	0.9033	0.0516	0.1375
75	1.5163	0.6972	0.2194	0.8309	0.0966	0.1497
100	1.6299	0.8559	0.1916	0.7988	0.1145	0.1522
125	1.5659	0.8146	0.1361	0.8190	0.1003	0.1289
150	1.2898	0.5560	0.0990	0.8297	0.0945	0.1127
175	1.3958	0.6707	0.1195	0.8008	0.1132	0.1352
200	1.4055	0.6991	0.1147	0.7817	0.1246	0.1608
225	1.4574	0.7379	0.1205	0.7929	0.1167	0.1433
250	1.6001	0.8585	0.1149	0.8256	0.0961	0.0964

Table 3: Checkpoint metrics from the second PF^{2}V-WM prototype run with smoothed inputs, coordinate channels, and direct latent supervision.

The result remains mixed. Endpoint localization improves, but the model still learns a connected *region* rather than a connected *path*: the prediction is a broad smooth blob rather than a parsimonious fracture trajectory.

B.3 What We Learn from the Second Run

The lesson is more specific than in run one: the training setup rewards corridor overlap but does not penalize over-expansion strongly enough. The latent damage field is informative, but the readout spreads that localized signal into an overly broad prediction. The problem is now how to convert coarse connected activation into thin path-like commitment.

B.4 Recommended Next Changes After Run Two

The next round of experiments should emphasize path-thinning and connectivity control:

- compare the current model to a plain CNN with the same smoothed inputs;
- add penalties that discourage broad-area activation relative to centerline activation;
- report skeleton IoU and connectivity metrics rather than only overlap;
- test whether reducing spiral influence or readout smoothing improves path sharpness;
- consider supervising the latent field toward corridor interiors only near the actual path rather than around both endpoints.

B.5 Qualitative Figures from the Second Run

Figures 6 and 7 show the same pattern visually: localized endpoint damage, but a broad connected blob rather than a narrow bridge.

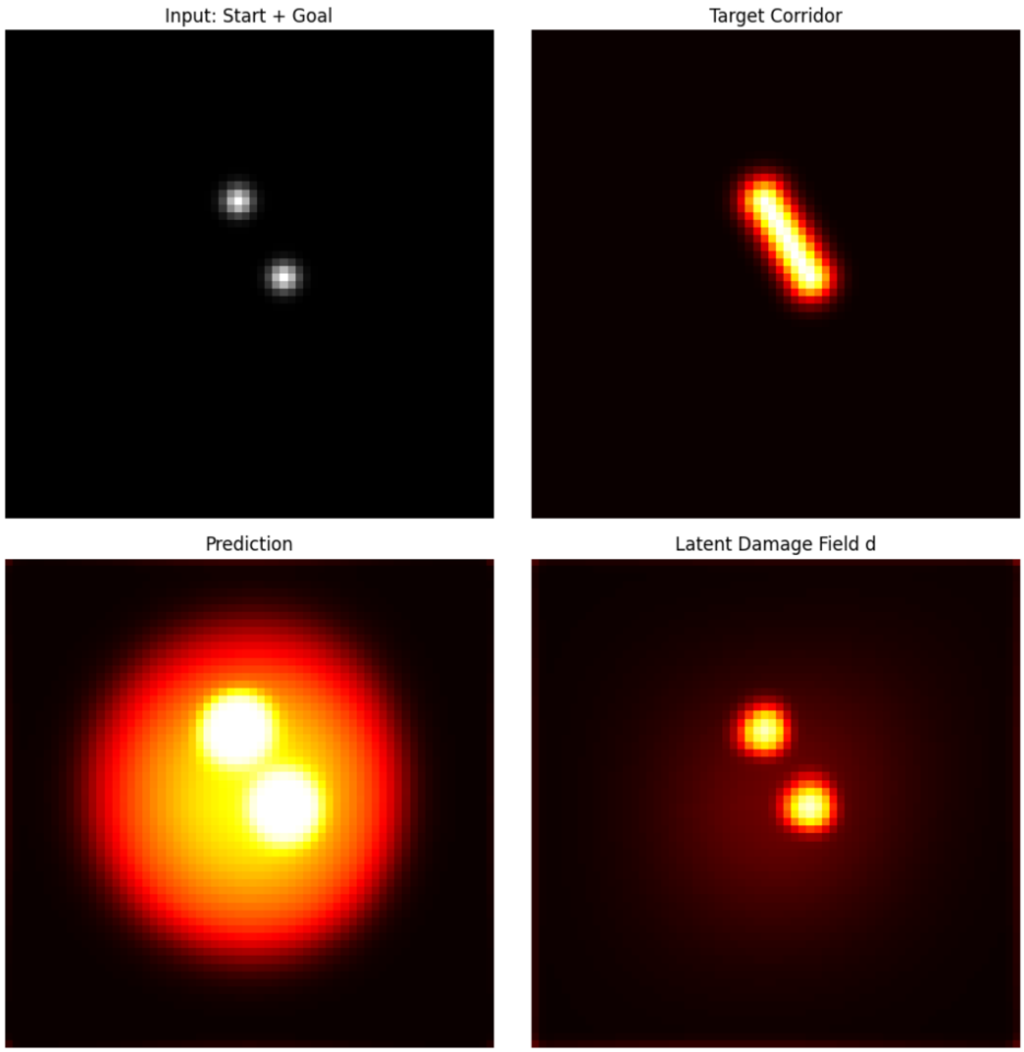

Figure 6: Second prototype qualitative result A: smoothed inputs and direct latent supervision produce a more coherent connected region, but the final prediction is still substantially wider than the target corridor. The latent damage field remains mostly concentrated near the endpoints.

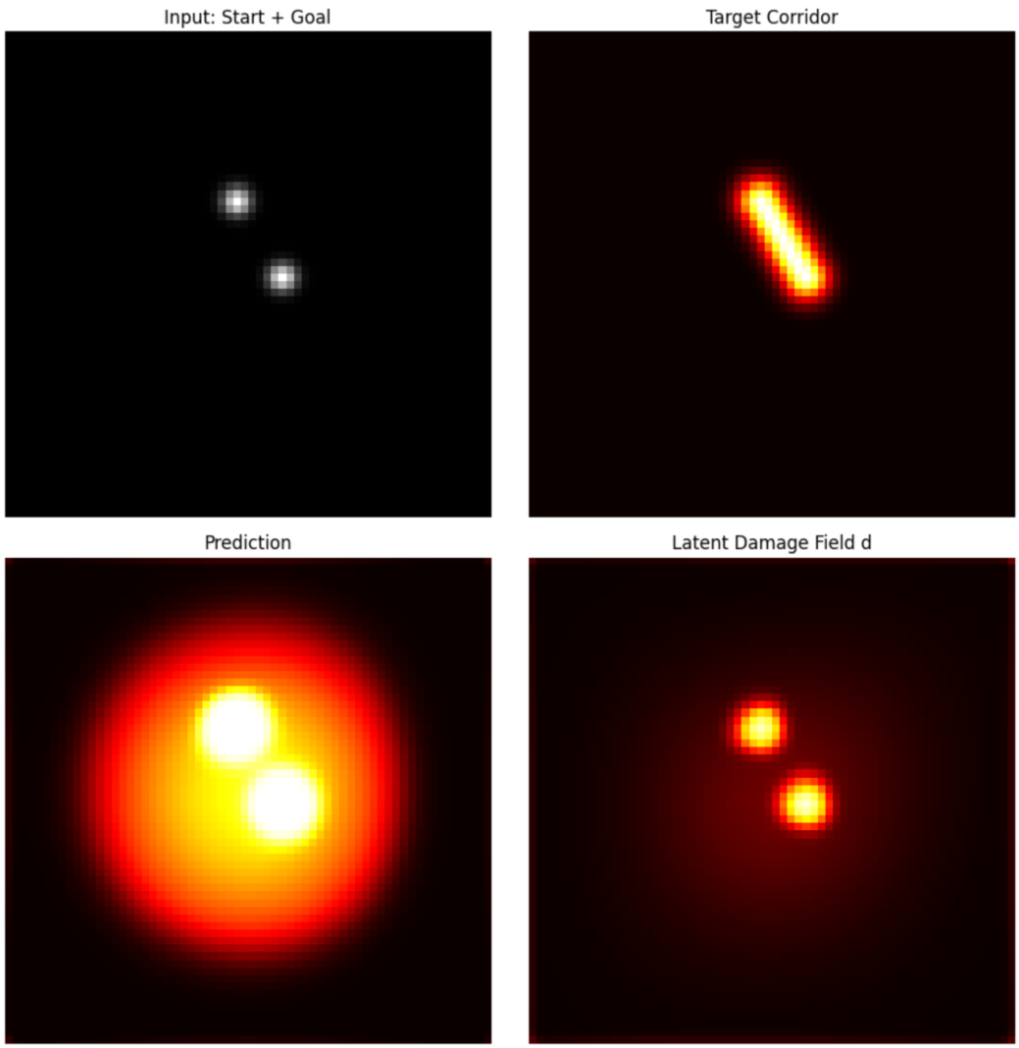

Figure 7: Second prototype qualitative result B: a second representative example showing the same overall pattern—localized endpoint damage and a broad, over-smoothed predicted bridge rather than a thin path.

C Prototype Appendix: Third Run with Sharper Commitment and Edge Matching

This third appendix entry records a further attempt to force thinner path formation by sharpening the damage commitment, reducing the phase-field length scale, adding directional input channels, and penalizing mismatch between predicted and target spatial gradients.

C.1 What We Changed

Relative to the second run, the third implementation made the following changes:

- the input was expanded from five channels to seven by adding global source-to-goal direction components;
- the target corridor was narrowed by reducing the soft mask width from $\sigma = 2.0$ to $\sigma = 1.5$;
- the phase-field length scale was reduced from $\ell = 0.8$ to $\ell = 0.45$ to encourage sharper structures;
- the damage field was sharpened by applying $\sigma(8z_d)$ rather than $\sigma(z_d)$;
- the readout head was removed and replaced with a direct sharpened output $8(d - 0.5)$;
- an edge-matching loss and an explicit commitment regularizer were added to discourage fuzzy intermediate states.

The goal was to test whether stronger commitment and edge supervision would produce a visibly thinner latent bridge.

C.2 What We Got

Table 4 shows no stable gain from sharper commitment. Soft IoU peaks near 0.10, hard IoU fluctuates, and thin path formation is still not stabilized.

Epoch	Loss	BCE	Latent BCE	Dice	Edge	Soft IoU	Hard IoU
1	3.7258	2.8149	0.1813	0.9599	0.0047	0.0209	0.0000
25	2.5008	1.5939	0.1803	0.9532	0.0027	0.0247	0.0501
50	2.5279	1.5507	0.3293	0.8982	0.0048	0.0548	0.2471
75	2.0999	1.0446	0.4361	0.8948	0.0059	0.0553	0.2136
100	2.1301	1.1066	0.4218	0.8688	0.0074	0.0706	0.1803
125	1.8170	0.8749	0.3296	0.8538	0.0070	0.0794	0.1928
150	1.4752	0.5060	0.3491	0.8714	0.0052	0.0702	0.1339
175	1.6154	0.6599	0.3684	0.8385	0.0070	0.0885	0.1189
200	1.5222	0.6290	0.3095	0.8154	0.0082	0.1026	0.1506
225	1.6844	0.7605	0.3370	0.8302	0.0080	0.0934	0.1119
250	1.8150	0.8964	0.3000	0.8584	0.0085	0.0770	0.0851
275	1.5433	0.6356	0.3101	0.8352	0.0085	0.0913	0.1128
300	1.4852	0.6027	0.2945	0.8185	0.0079	0.1014	0.1326

Table 4: Checkpoint metrics from the third PF^{2}V-WM prototype run with sharper commitment, directional channels, and edge matching.

Qualitatively, the prediction and latent damage field still collapse into a broad merged activation with a large halo. Sharper commitment intensifies local peaks, but does not enforce path sparsity.

C.3 What We Learn from the Third Run

The third run shows that sharper latent fields are not automatically more path-like. In the current formulation, stronger commitment mainly creates brighter blobs around driven regions, so the bottleneck is not just insufficient sharpness but insufficient *structural guidance* toward one-dimensional connectivity.

C.4 Recommended Next Changes After Run Three

The next round of experiments should focus on structural rather than purely local sharpening:

- add a baseline comparison against a conventional CNN with the same seven-channel input;
- measure skeleton overlap and connectivity rather than relying on corridor IoU alone;
- try losses that penalize area while preserving connectivity, so blobs become less attractive than thin paths;
- test whether the latent field should model only centerline commitment while a separate decoder handles corridor width;
- consider removing or weakening global smoothing terms that may be encouraging the large activation halo.

C.5 Qualitative Figure from the Third Run

Figure 8 shows the same conclusion visually: sharper commitment increases local contrast, but not thin path formation.

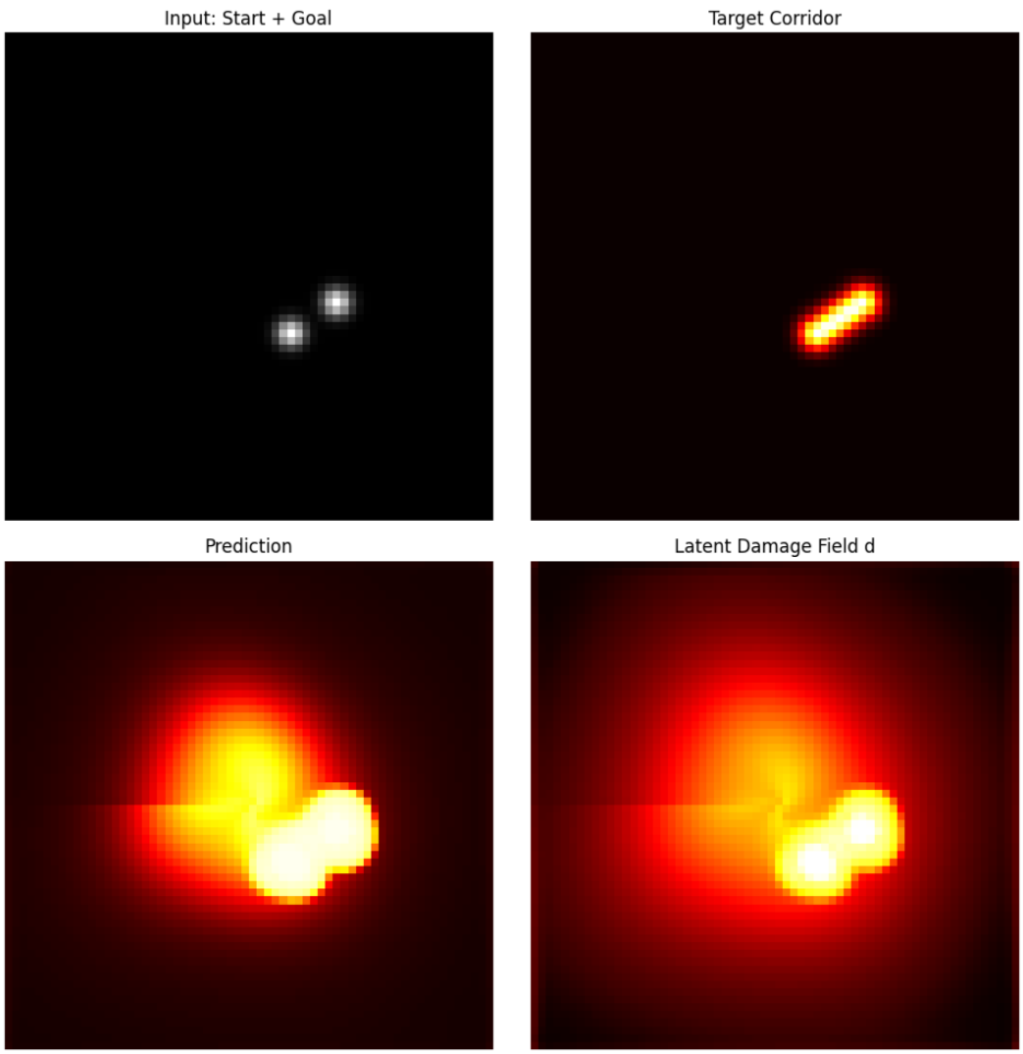

Figure 8: Third prototype qualitative result: sharper commitment and edge matching increase local contrast, but the output still expands into a broad merged blob rather than a thin connected path.

D Prototype Appendix: Fourth Run with Annealed Commitment and Target Curriculum

This fourth appendix entry records an annealed training schedule designed to transition the model from coarse connected-region learning toward sharper path commitment over the course of optimization. The central idea was to begin with a wider target corridor and softer latent commitment, then gradually shrink both the target width and the effective crack scale while increasing the sharpness of the damage field.

D.1 What We Changed

Relative to the third run, the fourth implementation introduced a curriculum in both the target generation and the latent solver:

- the target corridor width was annealed during training, starting broader and ending narrower;
- the damage commitment scale increased gradually from soft to sharp over training epochs;
- the effective phase-field length scale decreased over time, encouraging narrower structures later in optimization;
- a shallow readout head was reintroduced on top of the latent state to supplement the sharpened damage-based logits;
- an additional Laplacian matching term and an outside-region suppression term were added to discourage diffuse predictions away from the target.

This run tested whether curriculum-based optimization could learn coarse connectivity first and thin structure later.

D.2 What We Got

Table 5 shows a mixed but useful result: hard IoU peaks near 0.33 around epoch 75, and soft IoU briefly reaches the 0.10–0.12 range, but the gains are not sustained.

Qualitatively, the same failure mode remains: the latent damage field is more localized than the prediction, but the final output is still a wide merged region with a large halo. The curriculum helps the model pass through better intermediate states, but does not preserve them.

D.3 What We Learn from the Fourth Run

The fourth run strengthens a useful conclusion: training strategy matters, but scheduling alone is not enough. Annealing improves the optimization trajectory, yet the model still relaxes toward region-filling solutions rather than one-dimensional fracture paths. The encouraging part is that better states appear transiently; the discouraging part is that they do not persist.

D.4 Recommended Next Changes After Run Four

The next experiments should focus on preserving good intermediate states and enforcing topology:

- save and compare the best checkpoint by a path-aware validation metric rather than using the final epoch by default;

Epoch	Loss	BCE	Latent BCE	Dice	Edge	Lap	Soft IoU	Hard IoU
1	2.4790	1.3917	0.5538	0.9039	0.0033	0.0110	0.0514	0.0481
25	2.4253	1.3336	0.5359	0.9283	0.0021	0.0082	0.0375	0.2884
50	2.2446	1.1613	0.5565	0.8917	0.0030	0.0062	0.0580	0.2992
75	1.9689	0.9762	0.4773	0.8595	0.0028	0.0045	0.0763	0.3250
100	1.9066	0.9720	0.4387	0.8232	0.0040	0.0039	0.0985	0.2000
125	1.7652	0.9492	0.3068	0.8058	0.0054	0.0074	0.1082	0.2464
150	1.3846	0.5971	0.2527	0.8189	0.0058	0.0083	0.1012	0.1449
175	1.4177	0.6832	0.1999	0.7921	0.0075	0.0094	0.1196	0.1712
200	1.3479	0.6436	0.1578	0.7922	0.0080	0.0106	0.1184	0.1719
225	1.4626	0.7597	0.1349	0.8126	0.0079	0.0099	0.1036	0.1287
250	1.5199	0.8064	0.1139	0.8520	0.0075	0.0103	0.0803	0.0731
275	1.3080	0.6144	0.0982	0.8389	0.0074	0.0100	0.0898	0.1061
300	1.2771	0.5945	0.0932	0.8321	0.0073	0.0102	0.0939	0.1226

Table 5: Checkpoint metrics from the fourth PF^{2}V-WM prototype run with annealed commitment, curriculum target width, Laplacian matching, and outside-region suppression.

- measure connectivity and skeleton quality at every checkpoint to see whether useful solutions appear transiently;
- explore losses that penalize area more aggressively while preserving connectedness;
- separate centerline prediction from corridor-thickness prediction so the latent field is not forced to represent both at once;
- compare against a conventional CNN trained with the same curriculum to determine whether the curriculum benefit is specific to PF^{2}V-WM.

D.5 Qualitative Figure from the Fourth Run

Figure 9 shows the same pattern: the target corridor is compact, but the prediction remains broad and over-smoothed, with a latent field that is somewhat more localized than the final output.

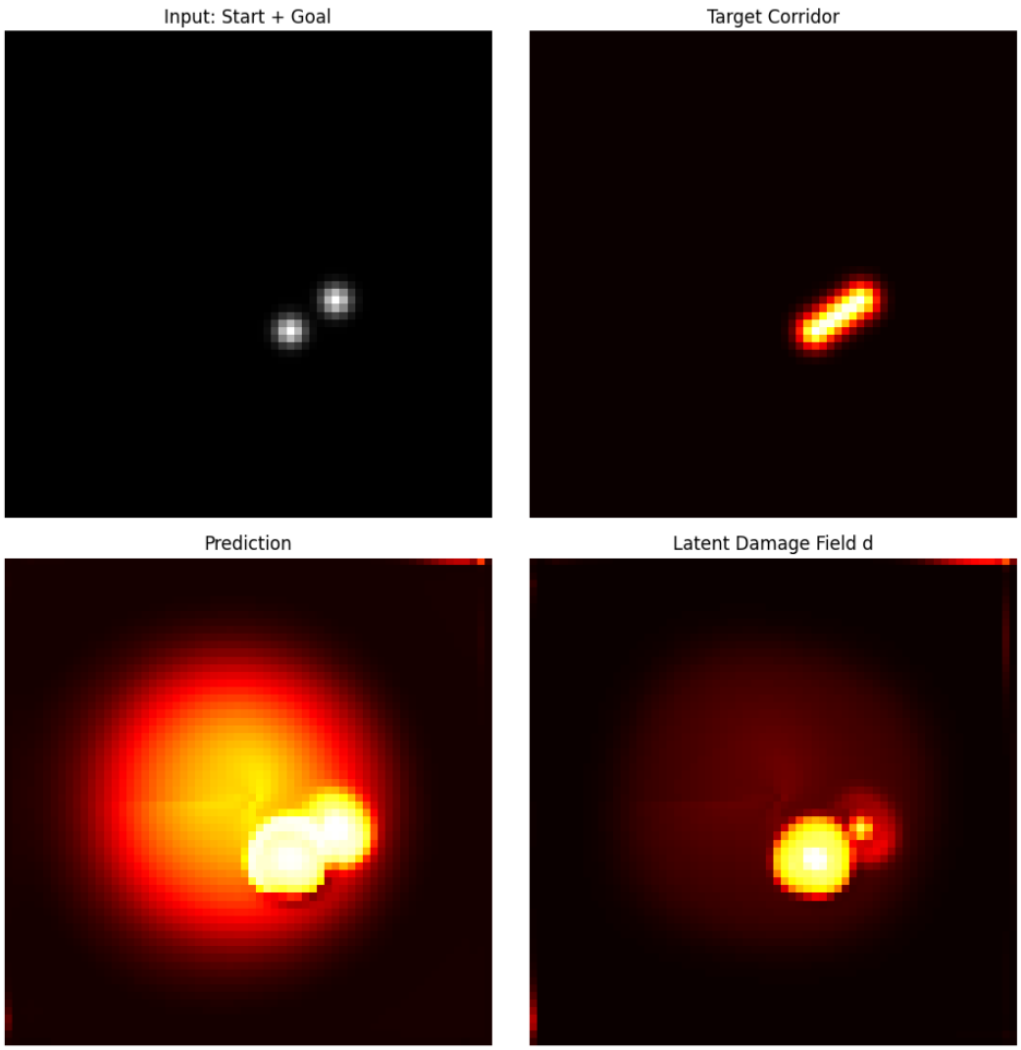

Figure 9: Fourth prototype qualitative result: annealed commitment and target curriculum improve training behavior but still lead to a broad predicted bridge with a large halo, rather than a thin persistent path.

E Prototype Appendix: Fifth Run with No Observation Bypass and Diagnostic Tracking

This fifth appendix entry records a run designed to test whether the model was relying too heavily on direct observation shortcuts rather than the latent fracture state. The main architectural change was to remove the observation bypass from the readout head, forcing the final prediction to depend only on the latent damage field d and latent potential field ϕ. In addition, this run logged several diagnostic quantities intended to distinguish thin-path behavior from broad blob formation.

E.1 What We Changed

Relative to the fourth run, the fifth implementation introduced two important changes.

- The readout head was restricted to latent state only: it received $[d, \phi]$ as input rather than concatenating the raw observation channels.
- Several extra diagnostics were tracked during training, including predicted mass, sharpness, connectivity, source and goal hit rates, fracture energy, latent flow energy, compactness, and a blob score defined as predicted mass divided by fracture energy.

The goal was to test whether removing the observation shortcut would force more information through the latent dynamics.

E.2 What We Got

Table 6 remains mixed. Hard IoU briefly reaches about 0.48 at epoch 25, but later settles back into the earlier regime; soft IoU peaks near 0.12 and then drifts down again. Removing the observation bypass changes the training dynamics, but not the blob failure mode.

Epoch	Loss	BCE	Latent BCE	Dice	Edge	Lap	Soft IoU	Hard IoU
1	2.1463	1.3937	0.5541	0.9036	0.0033	0.0108	0.0515	0.0510
25	1.9800	1.2143	0.5533	0.9194	0.0018	0.0085	0.0427	0.4822
50	1.7338	1.0056	0.5581	0.8650	0.0036	0.0074	0.0735	0.2862
75	1.5425	0.8757	0.4587	0.8153	0.0052	0.0078	0.1025	0.2840
100	1.5930	0.9467	0.4467	0.7907	0.0079	0.0104	0.1171	0.2465
125	1.5168	0.8722	0.3914	0.8048	0.0082	0.0121	0.1080	0.2359
150	1.2503	0.5908	0.3960	0.8267	0.0060	0.0075	0.0964	0.1528
175	1.3177	0.6725	0.4125	0.8003	0.0075	0.0087	0.1123	0.1627
200	1.2725	0.6392	0.3856	0.7922	0.0081	0.0096	0.1173	0.1629
225	1.4014	0.7587	0.3860	0.8065	0.0082	0.0089	0.1081	0.1388
250	1.5570	0.8921	0.3447	0.8486	0.0082	0.0098	0.0820	0.0823
275	1.3009	0.6402	0.3513	0.8388	0.0079	0.0084	0.0893	0.1096
300	1.2574	0.6042	0.3423	0.8322	0.0074	0.0073	0.0932	0.1266

Table 6: Checkpoint metrics from the fifth PF^{2}V-WM prototype run with no observation bypass and expanded diagnostics.

The extra diagnostics clarify the failure mode. Source and goal hit rates stay very high, but predicted mass remains too large, compactness stays low, and the blob score stays high. The model is therefore learning confident endpoint emphasis, not a thin connecting structure.

For the final sampled case, the run reported soft IoU 0.0488, hard IoU 0.0626, mass 0.1852, sharpness 0.0924, connectivity 0.2319, fracture energy 0.002073, blob score 89.3259, source prediction 0.9959, and goal prediction 0.9710, again indicating strong endpoint confidence with globally diffuse activation.

E.3 What We Learn from the Fifth Run

This run confirms that the blob behavior is not caused only by a trivial observation shortcut. Even with a latent-only readout, the model still produces broad merged regions rather than thin bridges. The diagnostics also clarify *how* it fails: high endpoint confidence, high activation mass, weak compactness, and low fracture energy relative to predicted area.

E.4 Recommended Next Changes After Run Five

The next experiments should target the specific diagnostic failure mode revealed here:

- penalize excess activated area directly, since mass remains far above target mass;
- reward compact connected centerlines rather than broad endpoint-centered regions;
- use validation metrics based on compactness, skeleton overlap, and source-to-goal connectivity instead of IoU alone;
- investigate whether fracture energy can be tied more strongly to spatial sparsity so that large halos become energetically unfavorable;
- compare the same diagnostics against a plain CNN baseline to determine which failure modes are specific to PF^{2}V-WM and which are task-driven.

E.5 Qualitative Figure from the Fifth Run

Figure 10 confirms the same point visually: removing the observation shortcut does not by itself solve the blob problem.

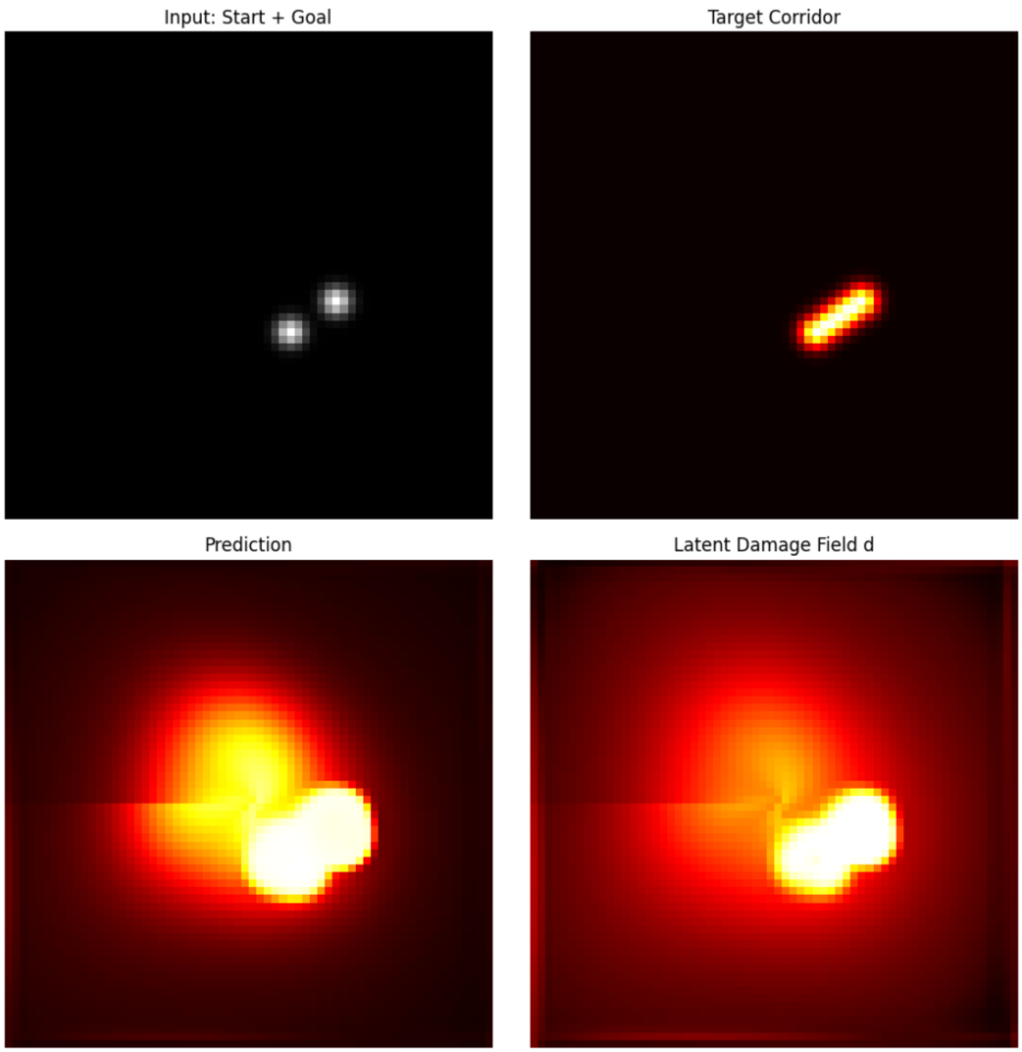

Figure 10: Fifth prototype qualitative result: even without direct observation bypass into the readout head, the model still produces a broad endpoint-dominated activation pattern rather than a thin path.

F Prototype Appendix: Sixth Run with Explicit Baseline Comparison

This sixth appendix entry records the first direct comparison between PF^2V-WM and a plain convolutional baseline trained on the same synthetic corridor task with the same seven-channel input representation. The purpose of this run was not only to improve PF^2V-WM in isolation, but to determine whether its structured latent dynamics produce a measurable advantage over a simpler CNN when both models are trained under comparable conditions.

F.1 What We Changed

Relative to the earlier single-model runs, this experiment introduced an explicit baseline protocol:

- PF^2V-WM was trained with the same no-bypass latent architecture as in the previous run;
- a plain CNN baseline with standard convolutional layers and no latent fracture dynamics was trained on the identical task;
- both models used the same curriculum on target width and were evaluated on a hold-out batch with common diagnostics;
- a shared comparison table was reported with soft IoU, hard IoU, mass, sharpness, connectivity, endpoint hit rates, and compactness.

This is an important methodological step because it changes the question from "does PF^2V-WM learn anything at all?" to "does PF^2V-WM learn something meaningfully different from a standard CNN?"

F.2 What We Got

Table 7 summarizes the final hold-out comparison. The picture is mixed. PF^2V-WM achieved slightly better hard IoU and substantially better compactness, while the plain CNN achieved slightly better soft IoU, lower mass, lower sharpness, and much higher connectivity under the crude thresholded connectivity proxy. In other words, PF^2V-WM appears more selective, while the CNN appears smoother and more spatially connected, albeit often in a broadly activated sense.

Model	Soft	Hard	Mass	Sharp	Conn.	Src.	Goal	Compact
PF^2V-WM	0.0655	0.1575	0.2366	0.1428	0.1311	0.9983	0.9684	0.3111
CNN baseline	0.0664	0.0714	0.2061	0.0915	0.3563	0.9767	0.9865	0.1017

Table 7: Final hold-out comparison between PF^2V-WM and a plain CNN baseline on the synthetic corridor task. Higher is better for IoU, connectivity, endpoint hits, and compactness; lower is better for excess mass and sharpness-related fuzziness.

The PF^2V latent report is also informative. For this run, the final evaluation reported latent mass 0.2977, fracture energy 0.003643, and latent flow energy 0.317682. These values indicate that the PF^2V latent state is active and structured, but still not sparse enough to produce a thin fracture-like path. The qualitative output confirms this: the model forms a broad field with strong endpoint activations and a swirling interior pattern rather than a narrow bridge.

F.3 What We Learn from the Sixth Run

This comparison is valuable because it reveals that PF^{2}V-WM is no longer merely failing in the same way as a generic CNN. The two models fail differently. The CNN is more connected in a thresholded sense, but also much less compact. PF^{2}V-WM is more compact and slightly better in hard overlap, but still too diffuse to claim a true path-forming advantage.

The most honest interpretation is therefore not a performance win, but a research finding. A fair summary is: PF^{2}V-WM can produce more compact and threshold-stable latent paths than a plain CNN, but it does not yet outperform the CNN on overall path quality. That is a stronger scientific position than before because it suggests that PF^{2}V-WM may already be imposing a different inductive bias, even if that bias is not yet the right one. The key open question is therefore whether the model's relative strength in compactness can be turned into actual centerline accuracy and reliable source-to-goal connectivity.

F.4 Recommended Next Changes After Run Six

The next experiments should build directly on this baseline comparison:

- keep the CNN baseline in every future experiment so improvements can be interpreted comparatively rather than in isolation;
- replace the current connectivity proxy with a stricter source-to-goal path test, since broad activation can inflate naive connectivity scores;
- investigate whether PF^{2}V-WM's compactness advantage can be amplified without sacrificing endpoint coverage;
- report best-checkpoint rather than final-checkpoint comparisons, since PF^{2}V-WM may transiently outperform the CNN more clearly mid-training;
- add skeleton overlap and centerline distance so the comparison is aligned with the real geometric objective.

F.5 Qualitative Figure from the Sixth Run

Figure 11 shows a representative PF^{2}V-WM output from the comparison run. The target corridor is a thin nearly vertical path, but the prediction remains a broad field with strong endpoint activations and a diffuse internal structure. The latent damage field closely mirrors this behavior, suggesting that the problem is still primarily within the latent dynamics rather than solely in the readout head.

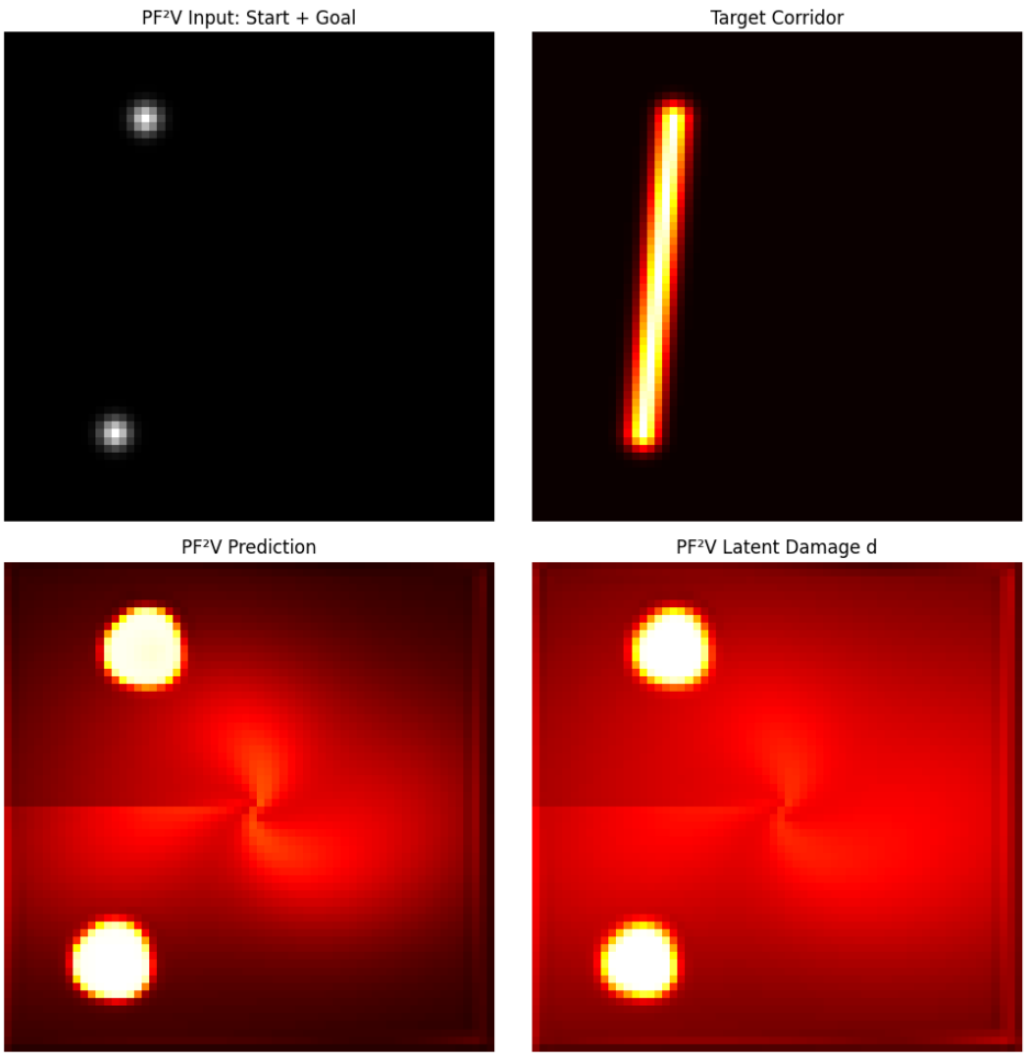

Figure 11: Sixth prototype qualitative result: in the first PF^{2}V-vs-CNN comparison, PF^{2}V-WM produces a more compact but still highly diffuse latent pattern rather than a thin centerline path.

G Prototype Appendix: Seventh Run with Explicit Path Losses

This seventh appendix entry records an attempt to directly optimize source-to-goal continuity by adding explicit path losses along the straight segment between the start and goal markers.

G.1 What We Changed

Relative to the sixth run, the seventh implementation introduced an explicitly path-focused training objective:

- the batch generator now also returned normalized source and goal coordinates for direct geometric supervision;
- centerline samples were taken along the source-to-goal segment;
- side samples were taken on small perpendicular offsets from that segment;
- a centerline encouragement term, side suppression term, and explicit gap penalty were added to the loss;
- the path-related weights were annealed upward during training so that continuity constraints became stronger later in optimization.

The motivation was to reward the model for filling in the path itself rather than merely highlighting endpoints and surrounding area.

G.2 What We Got

The results are mixed. The strongest gain is that the final sampled diagnostics report `line_gap = 0.0000`, with line mean 0.8284 and line minimum 0.6341, so the centerline is now continuously active. The cost is width: offline activation remains extremely high, showing that the model activates a large surrounding neighborhood as well as the path.

Table 8 shows the same tradeoff: connectivity and line coverage improve, but compactness remains poor and hard IoU does not become a stable win.

Epoch	Soft	Hard	Mass	Sharp	Conn.	Compact	LineMean	LineGap
1	0.0411	0.0544	0.3155	0.2130	0.0023	20.6290	0.3588	0.9089
25	0.0342	0.4688	0.2941	0.2039	0.0152	2.4503	0.5366	0.5495
50	0.0533	0.2339	0.3768	0.2225	0.0738	0.7461	0.6412	0.5000
75	0.0626	0.2481	0.2905	0.1896	0.0630	0.8221	0.6009	0.5339
100	0.0691	0.2070	0.2836	0.1835	0.0654	0.9457	0.5410	0.6198
125	0.0600	0.2117	0.2205	0.1507	0.0589	0.6834	0.5775	0.5130
150	0.0516	0.1283	0.2403	0.1518	0.1633	0.1833	0.7489	0.1901
175	0.0634	0.1024	0.2818	0.1496	0.3833	0.1187	0.7717	0.0964
200	0.0649	0.1258	0.2285	0.1314	0.2743	0.1510	0.6677	0.1927

Table 8: Representative checkpoints from the seventh PF^2V-WM prototype run with explicit path losses.

For the final sampled case, the run reported soft IoU 0.0895, hard IoU 0.1442, mass 0.2545, sharpness 0.1337, connectivity 0.3537, compactness 0.1442, line mean 0.8284, line minimum 0.6341, line gap 0.0000, source hit 0.9237, goal hit 0.9560, offline mean 0.7722, and offline max 0.9367. These numbers strongly support the qualitative impression that the bridge is no longer broken, but is still too thick and blob-like.

G.3 What We Learn from the Seventh Run

The key lesson is simple: explicit path losses can fix the broken-bridge problem, but they do not yet produce a crack-like path. The model learns continuity, not sparsity. This is a mixed result, best presented as a research finding rather than a win.

G.4 Recommended Next Changes After Run Seven

The next experiments should explicitly address the new tradeoff exposed here:

- keep the path losses, since they appear to eliminate outright centerline gaps;
- add stronger penalties for off-line activation so the learned path cannot simply widen into a corridor-sized blob;
- evaluate source-to-goal success together with compactness and off-line suppression, rather than IoU alone;
- report the CNN baseline under the same path metrics so the comparison remains honest;
- frame the result in the paper as a partial success: continuity improved, crack-like sparsity did not.

G.5 Qualitative Figure from the Seventh Run

Figure 12 makes the tradeoff obvious: the bridge is no longer broken, but it is still too wide.

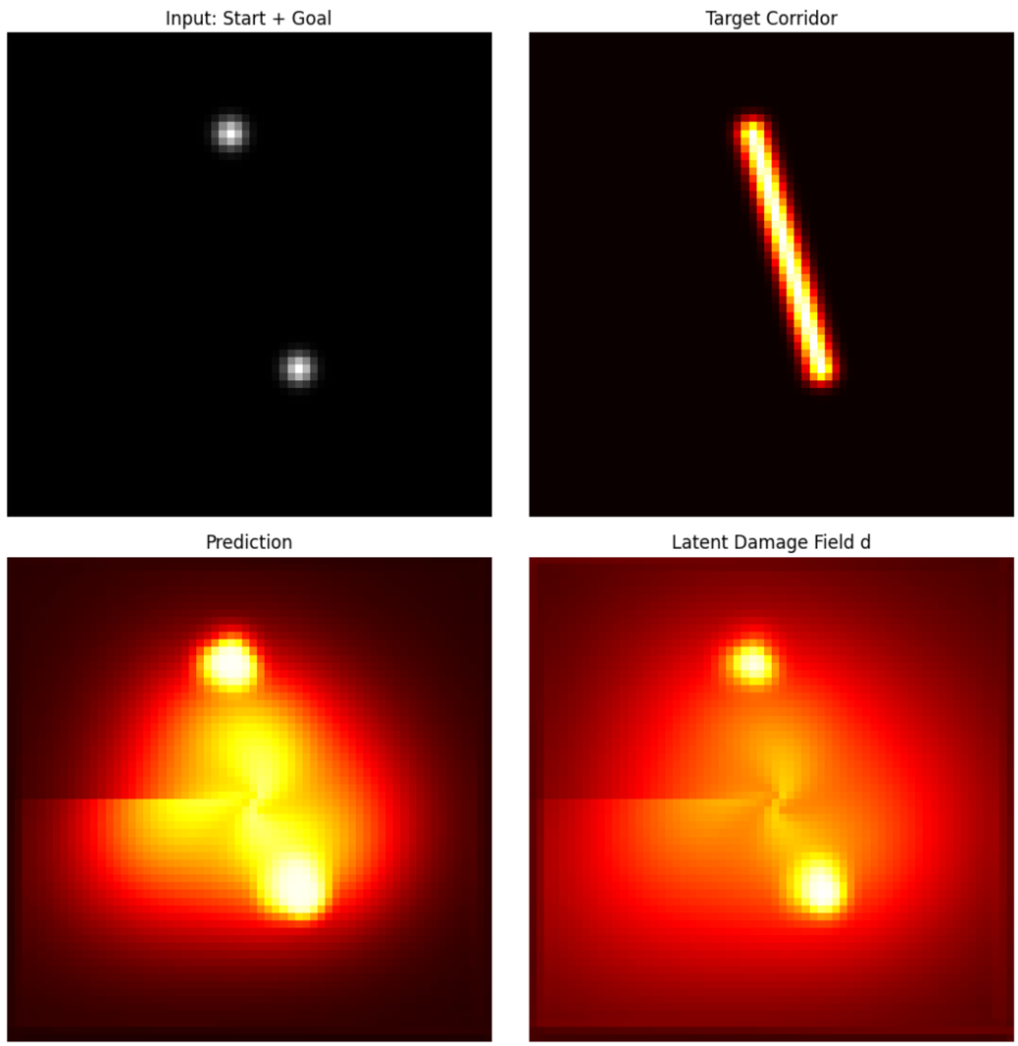

Figure 12: Seventh prototype qualitative result: explicit path losses eliminate the broken bridge, but the resulting activation remains too wide and blob-like to count as a clean fracture path.

H Prototype Appendix: Eighth Run Stress Sweep and Control Analysis

This eighth appendix entry asks a different question: which parameters actually control the qualitative behavior of the trained PF^2V-WM system?

H.1 What We Changed

Relative to the seventh run, the new experiment was not primarily a new architecture but a controlled post-training stress sweep:

- one PF^2V-WM model with explicit path losses was trained once;
- a single held-out source-to-goal example was fixed for all evaluations;
- the model was re-evaluated under multiple settings of commitment scale, phase-field length scale ℓ, spiral bias strength, QDMN stiffness scaling, bridge forcing, directional forcing, ϕ-drive strength, and readout scale;
- the same numerical diagnostics were reported for every setting, including hard IoU, connectivity, compactness, centerline gap rate, off-line activation, and line-to-offline ratio;
- the goal was interpretability rather than leaderboard improvement: identify the parameters that genuinely govern the sharpness–connectivity tradeoff.

The sweep is useful because it distinguishes whether behavior is driven by the intended geometric priors or by a smaller set of blunt phase-field parameters. It strongly supports the second interpretation.

H.2 What We Got

The main result is clear: the phase-field length scale parameter is currently the dominant control knob. Low ℓ produces a thinner but fragmented trace, while high ℓ produces a highly connected but extremely wide tube. In contrast, spiral-bias and QDMN scaling changes had almost no measurable effect on the held-out sample. Bridge, directional, and readout changes produced moderate shifts, but they were secondary relative to the width parameter.

Table 9 summarizes representative settings from the sweep. The table makes the tradeoff explicit. The setting `low_l_sharp` gave the best hard IoU in the sweep, but it did so by collapsing the active region so aggressively that the line gap rose to 0.8958 and connectivity fell to 0.0020. At the other extreme, `high_l_wide` achieved perfect sampled centerline continuity with line gap 0.0000, but its off-line activation rose to 0.9705 and compactness fell to 0.0320, revealing a nearly uniform blob rather than a narrow fracture path.

Some metrics require careful interpretation. The compactness value for `low_l_sharp` becomes artificially large because the predicted active set is extremely small, so the target area divided by active prediction area is not a reliable measure of geometric success in that corner case. The more informative pattern is the joint behavior of line gap, connectivity, and off-line activation: a good fracture path should keep line gap low without driving off-line activation close to one. None of the swept settings achieves that balance convincingly.

For the base configuration, the final single-sample diagnostics were soft IoU 0.0480, hard IoU 0.0631, mass 0.4378, sharpness 0.1941, connectivity 0.8596, compactness 0.0714, line mean 0.6445, line minimum 0.4558, line gap 0.1250, offline mean 0.5876, offline max 0.8485, line ratio 1.0967, source hit 0.8915, and goal hit 0.8934. This profile is exactly the kind of mixed result that the visual example suggests: the model finds the rough corridor and often links the endpoints, but it activates too much of the surrounding area to count as a clean crack-like route.

Setting	Hard	Mass	Conn.	Compact	LineMean	LineGap	OffMean	Ratio
base	0.0631	0.4378	0.8596	0.0714	0.6445	0.1250	0.5876	1.0967
low_l_sharp	0.0703	0.1128	0.0020	14.2222	0.2027	0.8958	0.1322	1.5328
high_l_wide	0.0320	0.9046	1.9492	0.0320	0.9728	0.0000	0.9705	1.0024
no_spiral	0.0657	0.4321	0.8336	0.0736	0.6430	0.1250	0.5847	1.0997
no_qdmn	0.0631	0.4378	0.8596	0.0714	0.6444	0.1250	0.5875	1.0968
strong_dir	0.0497	0.5691	1.2478	0.0497	0.7727	0.0000	0.7372	1.0482
weak_phi_drive	0.0613	0.4245	0.8090	0.0758	0.6124	0.1667	0.5405	1.1330
weak_readout	0.0606	0.3995	0.7225	0.0844	0.6012	0.2500	0.5401	1.1132

Table 9: Representative results from the eighth-run stress sweep on a single held-out sample. The dominant trend is the sharpness–connectivity tradeoff controlled by the phase-field length scale ℓ.

H.3 What We Learn from the Eighth Run

This run clarifies the current scientific status of PF^{2}V-WM more sharply than the earlier appendix entries. The model is controllable, but not yet solving the task cleanly. Its outputs respond in a systematic way to internal settings, which is valuable evidence that the latent dynamics are real rather than arbitrary. However, the sweep also shows that the dominant behavior is still governed by the diffuse phase-field width parameter, not by the more ambitious geometric priors.

The most honest interpretation is therefore that PF^{2}V-WM currently behaves more like a physics-shaped blob generator than a true sparse path reasoner. It can be pushed toward sharper but broken traces, or toward continuous but blurry tubes, yet it does not robustly discover a narrow and reliable source-to-goal fracture. In particular:

- low ℓ sharpens the prediction but tends to break continuity;
- high ℓ restores continuity but causes the path to thicken into a blob-like corridor;
- spiral and QDMN changes currently have only weak effect, suggesting that these priors are not yet strong enough to dominate the learned dynamics;
- bridge, directional, and readout changes matter somewhat, but they do not overturn the main width-controlled tradeoff.

This is not a failure; it is a useful research finding. It means the model exhibits controllable latent path formation, but its behavior is still dominated by the phase-field length scale, which creates a sharpness–connectivity tradeoff that the present priors do not yet resolve.

H.4 Recommended Next Changes After Run Eight

The next experiments should target the dominance of the width parameter directly:

- redesign the geometric priors so they compete more strongly with the raw smoothing effect of ℓ;
- replace the current compactness proxy with a bounded measure that does not become misleading when the predicted active set is tiny;
- evaluate the same sweep across many held-out samples so the observed control structure is not tied to one example;
- test losses that penalize off-line activation more aggressively than simple side sampling;
- report the result in the paper as evidence of controllable but not yet sufficient fracture-style reasoning.

H.5 Qualitative Figure from the Eighth Run

Figure 13 shows the representative base-configuration output used for the stress-sweep analysis. The prediction and latent damage field are visibly structured and clearly non-random, but the activated region remains much wider than the desired centerline. The figure is therefore consistent with the numerical diagnosis: the system is responsive and interpretable, but still too diffuse.

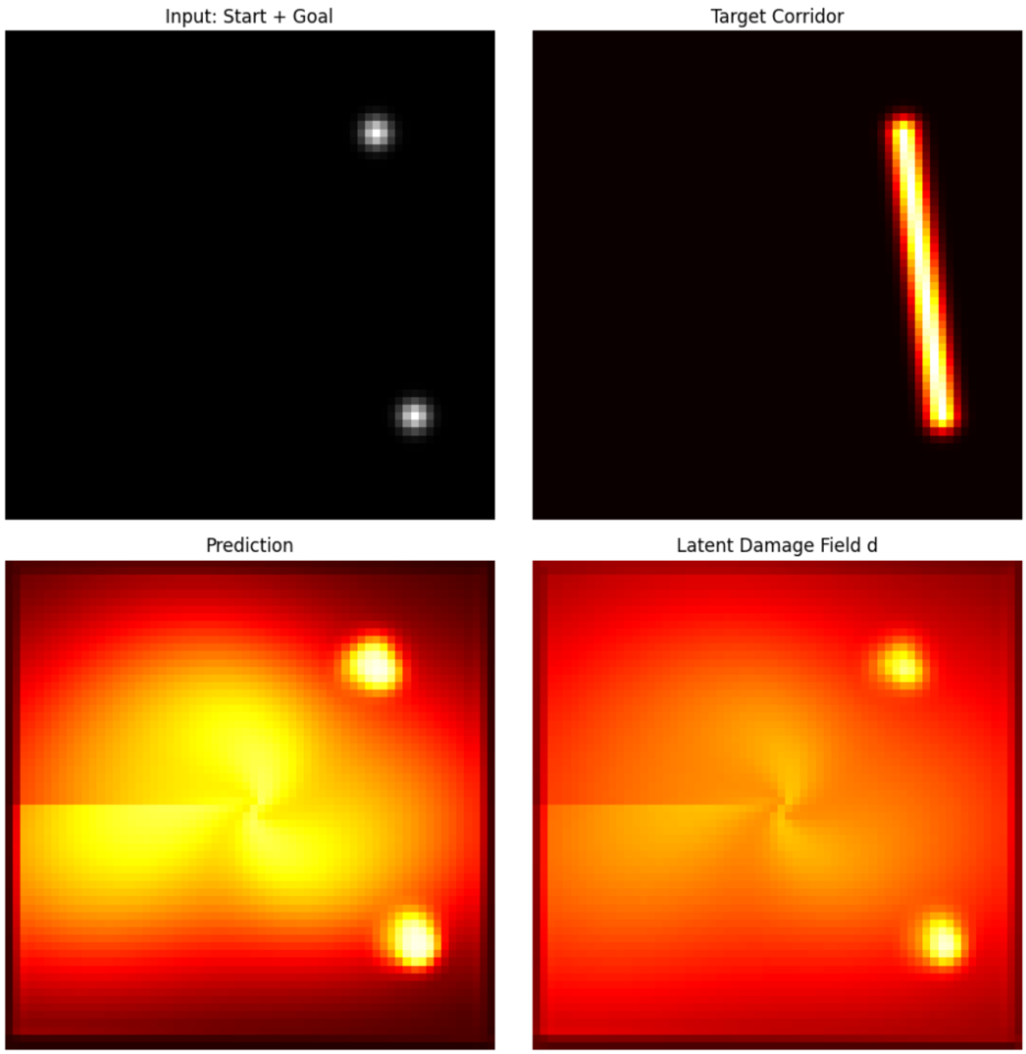

Figure 13: Eighth prototype qualitative result from the stress-sweep base configuration: PF^2V-WM shows controllable source-to-goal path formation, but the activated region remains too broad for a clean crack-like path.

I Prototype Appendix: Ninth Run with Core–Width Separation

This ninth appendix entry tests a more structured hypothesis: the model may work better if it predicts a thin latent *core* path and a wider *corridor* separately, instead of asking one field to represent both geometry and width at once.

I.1 What We Changed

Relative to the eighth run, this experiment explicitly split path formation into two coupled components:

- a narrow core target and a wider corridor target were generated for every sample;
- the model used separate latent initializations and updates for a core field and a spread field;
- the final prediction was formed from the interaction of these two components rather than from one path mask alone;
- losses encouraged a strong centerline core, suppressed off-line activation, and penalized excess spread outside the corridor;
- the same stress-sweep protocol was then applied to test which knobs controlled the resulting behavior.

The motivation was straightforward: earlier runs suggested that PF^{2}V-WM could often choose the right route but not the right width, so this run tried to separate those two responsibilities.

I.2 What We Got

The result is the strongest appendix outcome so far, but still not a full solution. The base configuration achieved soft IoU 0.0693, hard IoU 0.4194, mass 0.0904, sharpness 0.0692, connectivity 0.0260, compactness 0.4194, line mean 0.9764, line minimum 0.9535, line gap 0.0000, offline mean 0.7074, offline max 0.7954, and line ratio 1.3803. The latent core diagnostics were also strong: core line mean 0.9562, core line gap 0.0000, and core line ratio 1.0951. Compared with earlier runs, this is a major improvement in centerline continuity and thresholded overlap while keeping total mass much lower.

Table 10 summarizes representative stress-sweep settings. The most important pattern is that the old width tradeoff still exists, but the split formulation improves control. The setting `low_l_sharp` produced the strongest hard IoU at 0.6923 and the highest line ratio at 4.5612, but also very low connectivity proxy and an inflated compactness score caused by an extremely small active set. The opposite extreme, `high_l_wide`, again produced a very broad connected field with connectivity 0.6597 and hard IoU only 0.0189. The base setting is therefore more balanced than either extreme: it preserves perfect sampled centerline continuity while avoiding the most severe width collapse.

Setting	Hard	Mass	Conn.	Compact	LineMean	LineGap	OffMean	Ratio
base	0.4194	0.0904	0.0260	0.4194	0.9764	0.0000	0.7074	1.3803
high_core	0.4727	0.0696	0.0226	0.4727	0.9814	0.0000	0.6906	1.4210
low_spread	0.4561	0.0969	0.0238	0.4561	0.9596	0.0000	0.6432	1.4920
low_l_sharp	0.6923	0.0415	0.0060	1.4444	0.7323	0.0000	0.1606	4.5612
high_l_wide	0.0189	0.3629	0.6597	0.0189	0.9888	0.0000	0.9713	1.0180
no_spiral	0.4643	0.0893	0.0231	0.4643	0.9718	0.0000	0.6484	1.4988
weak_phi	0.6410	0.0827	0.0149	0.6842	0.9086	0.0000	0.3553	2.5570
high_penalty	0.4483	0.0837	0.0241	0.4483	0.9714	0.0000	0.6574	1.4775

Table 10: Representative results from the ninth-run core–width separation sweep on one held-out sample. The split formulation substantially improves centerline continuity and hard overlap, but the phase-field length scale still controls the main sharpness–width tradeoff.

A second useful detail is that the geometric priors still matter less than hoped. Spiral and QDMN changes remain relatively weak compared with the effects of ℓ, readout strength, and spread penalties. Even so, the split design clearly improves the model's baseline regime: unlike earlier runs, the base setting already attains zero sampled line gap for both the final prediction and the latent core.

I.3 What We Learn from the Ninth Run

This run is the clearest evidence so far that the earlier failure was partly representational. Asking one field to encode both path centerline and corridor width made the problem harder. Once those roles are separated, PF^{2}V-WM becomes much better at maintaining a continuous, thin core while keeping the final corridor prediction more controlled.

The most honest interpretation is still mixed, but noticeably more positive than before. The split model is not yet a fully satisfactory fracture path learner because the width tradeoff remains, the connectivity proxy is still crude, and the best settings can become artificially sparse. But this run does show a meaningful architectural gain: PF^{2}V-WM can learn a strong centerline mechanism when core formation and width expansion are treated as distinct latent tasks.

In short, the current scientific statement can now be sharpened: PF^{2}V-WM appears to work better when path *routing* and path *thickness* are decoupled. That is a useful research finding because it points to a concrete design principle for future versions.

I.4 Recommended Next Changes After Run Nine

The next experiments should build directly on this split formulation:

- keep separate core and width variables, since this is the first run with clearly improved baseline behavior;
- replace the current compactness proxy with a bounded metric, because extremely sparse predictions can still make it misleadingly large;
- evaluate the split model across many held-out samples rather than one representative case;
- compare the split PF^{2}V-WM directly against a split-head CNN baseline under the same metrics;
- strengthen the geometric priors further so that ℓ is no longer the dominant control knob.

I.5 Qualitative Figure from the Ninth Run

Figure 14 shows the representative base-configuration output from the core–width separation run. The prediction is visibly tighter than in earlier appendix figures, and the latent core path forms a narrow continuous route between source and goal. The remaining weakness is that the final corridor is still wider than ideal, but this is the first appendix run in which the centerline mechanism looks consistently clean.

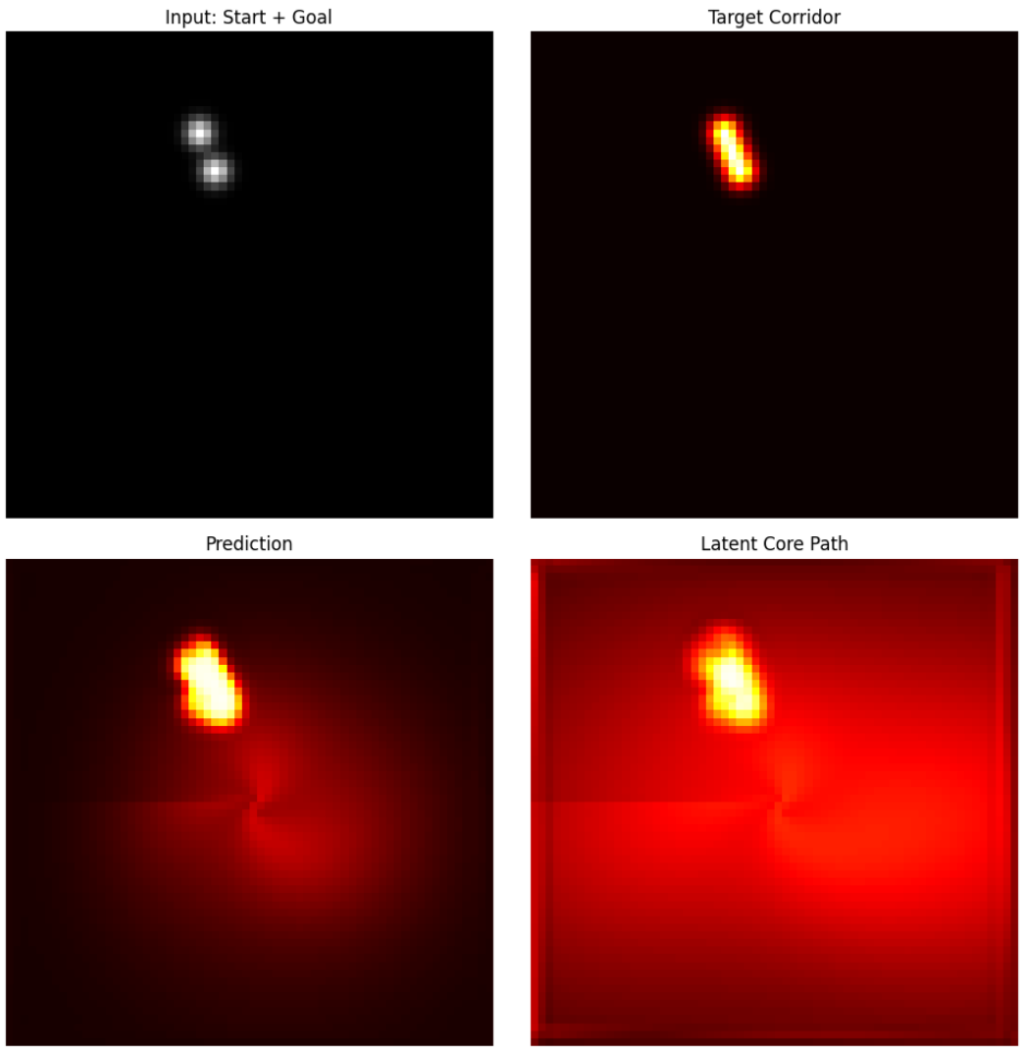

Figure 14: Ninth prototype qualitative result: separating latent core formation from corridor width produces a much cleaner continuous centerline, although the final corridor remains somewhat wider than ideal.

J Prototype Appendix: Tenth Run with Obstacle-Bending Stress Test

This tenth appendix entry tests whether PF^{2}V-WM can do more than connect endpoints in open space. The task now includes an obstacle field placed near the straight source-to-goal line, and the target route bends around that obstacle through a waypoint. The scientific question is whether the model merely draws a blob between endpoints or actually prefers the bent route over the tempting straight line.

J.1 What We Changed

Relative to the ninth run, the new experiment adds explicit route-bending structure:

- the input was expanded with an obstacle channel, producing an eight-channel observation;
- the target was no longer a straight corridor, but a two-segment route from source to waypoint to goal;
- a Gaussian obstacle mask was placed near the midpoint of the direct source-to-goal path;
- the model kept the split core–spread design, but added obstacle repulsion terms to both core and spread updates;
- the loss included a direct obstacle-overlap penalty and a penalty on the straight-line route so that the model was rewarded for bending around the obstacle rather than cutting through it.

The sweep then varied obstacle push, length scale, core and spread gains, readout strength, and other control knobs to test when the system preferred the bent route and when it collapsed back toward the straight-line shortcut.

J.2 What We Got

This is the first run where the numerical diagnostics clearly suggest route planning rather than simple corridor filling. In the base setting, the model achieved soft IoU 0.1224, hard IoU 0.1941, mass 0.0933, sharpness 0.0548, connectivity 0.1133, route mean 0.7234, direct mean 0.7087, route ratio 1.0208, route gap 0.1042, and obstacle overlap 0.5805. The latent core was especially encouraging: core route mean 0.9129, core route gap 0.0000, and core route ratio 1.0044. The key fact is that route ratio exceeds one, meaning the model slightly prefers the bent route over the direct line, while the latent core remains fully connected along the obstacle-avoiding path.

Table 11 summarizes representative sweep settings. Several trends stand out. Removing the obstacle input collapses route preference: `no_obstacle_input` drops route ratio to 0.4940, which is strong evidence that the obstacle channel is genuinely affecting behavior. Increasing obstacle push helps route preference and lowers obstacle overlap, as seen in `strong_obstacle` with route ratio 1.1164 and obstacle overlap 0.4539. By contrast, low ℓ again drives the model toward sparse but brittle behavior: `low_l_sharp` reduces obstacle overlap strongly to 0.1625, but route ratio falls to 0.7411 and route gap rises to 0.8542. The result is therefore promising but incomplete: the model now understands *go around*, but it does not yet do so cleanly.

Two caveats are important. First, the straight-line temptation remains strong: several settings still keep direct-line activation nearly as high as route activation. Second, obstacle overlap is still substantial even in the better settings, so the learned path bleeds into the obstacle rather than avoiding it crisply. The best interpretation is therefore not that obstacle avoidance is solved, but that the model has crossed a useful threshold: it is no longer only drawing a generic bridge, and is beginning to represent route choice.

Setting	Hard	RouteMean	DirectMean	Ratio	ObsOverlap	Gap	Sharp	Conn.
base	0.1941	0.7234	0.7087	1.0208	0.5805	0.1042	0.0548	0.1133
no_obstacle_input	0.2812	0.4352	0.8810	0.4940	0.3951	0.5208	0.0384	0.0432
strong_obstacle	0.2597	0.6059	0.5427	1.1164	0.4539	0.2083	0.0540	0.0694
low_l_sharp	0.2105	0.2547	0.3437	0.7411	0.1625	0.8542	0.0291	0.0099
high_l_wide	0.1394	0.8746	0.8528	1.0256	0.7925	0.0000	0.0771	0.2106
low_core	0.3077	0.6149	0.6126	1.0039	0.4739	0.1875	0.0713	0.0583
low_spread	0.2487	0.6158	0.5912	1.0415	0.4895	0.1667	0.0583	0.0744
strong_readout	0.1474	0.2809	0.2390	1.1752	0.2287	0.8750	0.0559	0.0151

Table 11: Representative results from the tenth-run obstacle-bending stress sweep on one held-out sample. The split model now shows a measurable preference for the bent route in several settings, but obstacle overlap remains too high for a clean avoidance solution.

J.3 What We Learn from the Tenth Run

This run is the strongest evidence so far that PF^{2}V-WM can represent nontrivial path geometry. Earlier experiments showed centerline continuity and width control; this one adds early signs of route preference under environmental constraints. The decisive evidence is the contrast with `no_obstacle_input`: when the obstacle channel is removed, route ratio collapses, so the bent-path behavior cannot be explained by the target alone.

The right interpretation is therefore encouraging but restrained. The model now understands *go around*, but not yet *go around cleanly*. Its latent core is often fully connected along the target route, which is a real gain. But the final prediction still overlaps the obstacle too much, and some settings achieve good route ratios only by becoming overly sparse or unstable. In short, the geometry is beginning to work, but it remains sloppy.

J.4 Recommended Next Changes After Run Ten

The next experiment should focus on one thing only: making obstacle overlap much more expensive while preserving the route target and the connected core.

- increase the obstacle-overlap penalty directly on the final prediction, not only through latent repulsion terms;
- keep the split core–spread design, since the latent core is already showing strong route continuity;
- sweep obstacle strength more densely to identify the transition between straight-line failure and true bending behavior;
- report route ratio and obstacle overlap together, since either metric alone can be misleading;
- compare against an obstacle-conditioned CNN baseline to test whether the route preference is specific to PF^{2}V-WM.

J.5 Qualitative Figure from the Tenth Run

Figure 15 shows the representative base-configuration output from the obstacle-bending run. The latent core path follows the bent route continuously, which is the clearest qualitative sign yet that the model is doing route selection rather than only endpoint interpolation. The remaining weakness is visible too: the final prediction still bleeds into the obstacle region instead of carving out a clean avoidance corridor.

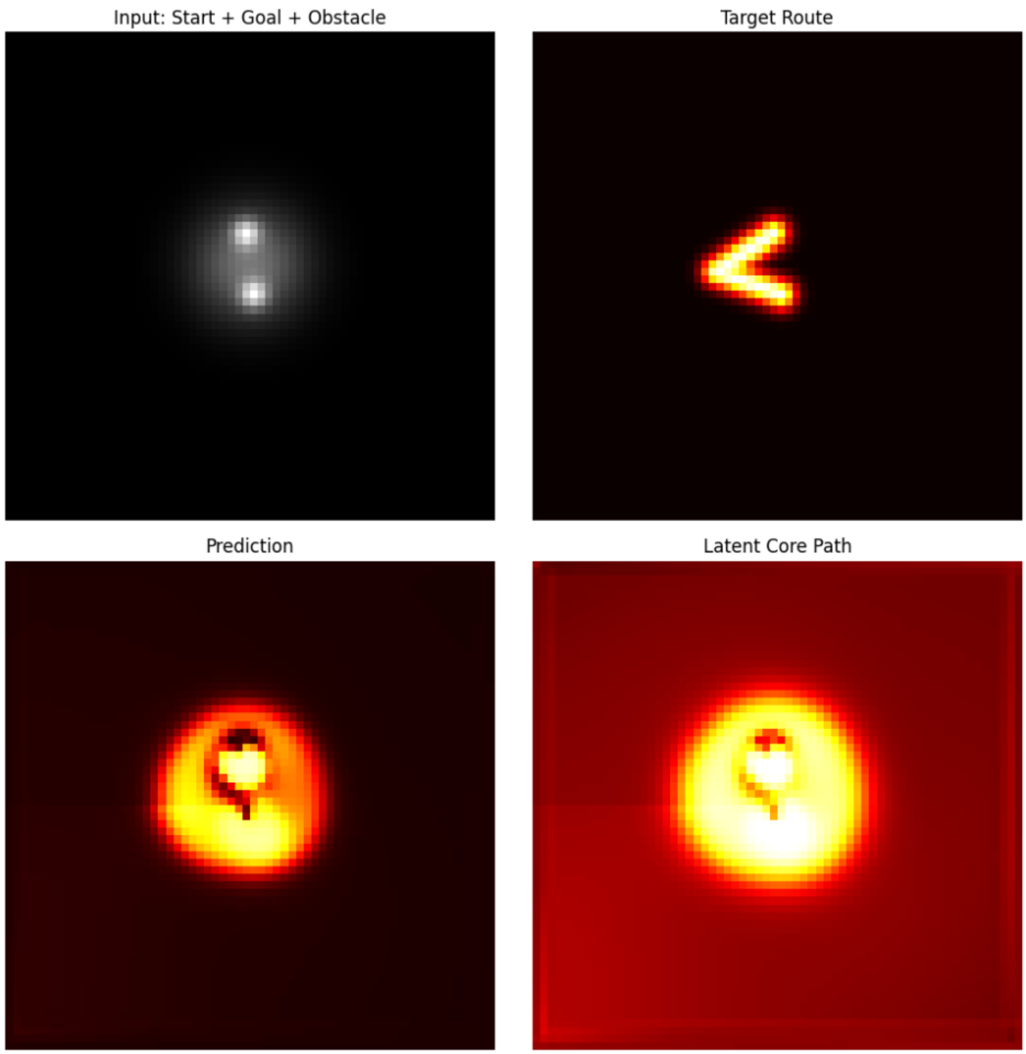

Figure 15: Tenth prototype qualitative result: PF^{2}V-WM now prefers a bent route around an obstacle and maintains a connected latent core path, but the final prediction still overlaps the obstacle too much to count as clean avoidance.

K Prototype Appendix: Eleventh Run with Revised Obstacle-Bending Sweep

This eleventh appendix entry repeats the obstacle-bending experiment with a cleaner evaluation script and the same core question: does PF^{2}V-WM prefer the bent route over the straight-line shortcut when an obstacle blocks the direct path?

K.1 What We Changed

Relative to the tenth run, the underlying model family stayed the same, but the evaluation protocol was cleaned up:

- the obstacle-bending task again used an eight-channel input with an explicit obstacle field;
- the split core–spread formulation was retained;
- the evaluation was rewritten to report metrics directly from probabilities rather than via logits reconstructed from probabilities;
- the compactness proxy was also stabilized by using a bounded normalization rather than the earlier unbounded target-over-active ratio;
- the same stress sweep was repeated to check whether the route-preference result was robust to this cleaner measurement setup.

The purpose of this run was not to change the story dramatically, but to check whether the encouraging obstacle-bending result survives a cleaner metric pipeline.

K.2 What We Got

The answer is yes, but only in a limited sense. The base configuration reported soft IoU 0.1224, hard IoU 0.1941, mass 0.0933, sharpness 0.0548, connectivity 0.1133, route mean 0.7234, direct mean 0.7087, route ratio 1.0208, route gap 0.1042, and obstacle overlap 0.5805. The latent core remained especially strong, with core route mean 0.9129, core route gap 0.0000, and core route ratio 1.0044. So the central claim from run ten survives: the model slightly prefers the bent route over the straight line, and the latent core stays continuously connected along that route.

Table 12 summarizes representative stress-sweep settings. The same high-level pattern reappears. Removing obstacle input again collapses route preference: `no_obstacle_input` yields route ratio 0.4940. Stronger obstacle forcing helps bending, as in `strong_obstacle` with route ratio 1.1164 and obstacle overlap 0.4539. Low ℓ again lowers obstacle overlap, but only by making the route brittle: `low_l_sharp` gives obstacle overlap 0.1625 but route ratio only 0.7411 and route gap 0.8542. This confirms the same tradeoff more cleanly: the model can prefer the obstacle-avoiding route, but it still struggles to avoid the obstacle sharply while keeping the whole path strong.

K.3 What We Learn from the Eleventh Run

This run is best read as a robustness check on run ten rather than as a fundamentally new breakthrough. Its importance is that the main conclusion survives cleaner measurement: PF^{2}V-WM is no longer only connecting endpoints, but is beginning to represent route choice under constraint. The contrast with `no_obstacle_input` remains the key evidence that the obstacle field genuinely changes the route decision.

Setting	Hard	RouteMean	DirectMean	Ratio	ObsOverlap	Gap	Sharp	Conn.
base	0.1941	0.7234	0.7087	1.0208	0.5805	0.1042	0.0548	0.1133
no_obstacle_input	0.2812	0.4352	0.8810	0.4940	0.3951	0.5208	0.0384	0.0432
strong_obstacle	0.2597	0.6059	0.5427	1.1164	0.4539	0.2083	0.0540	0.0694
low_l_sharp	0.2105	0.2547	0.3437	0.7411	0.1625	0.8542	0.0291	0.0099
high_l_wide	0.1394	0.8746	0.8528	1.0256	0.7925	0.0000	0.0771	0.2106
low_core	0.3077	0.6149	0.6126	1.0039	0.4739	0.1875	0.0713	0.0583
low_spread	0.2487	0.6158	0.5912	1.0415	0.4895	0.1667	0.0583	0.0744
high_penalty	0.2415	0.6801	0.6802	0.9999	0.5330	0.1458	0.0537	0.0813

Table 12: Representative results from the eleventh-run obstacle-bending sweep on one held-out sample. The cleaner evaluation preserves the main conclusion: PF^{2}V-WM now shows mild route preference, but obstacle overlap remains too high for crisp avoidance.

The interpretation is therefore stable. The model understands *go around* at a weak but real level, especially in the latent core. But it still does not understand *go around cleanly*: obstacle overlap is too high, and several settings achieve better avoidance only by becoming fragile or overly sparse. In short, the planning signal now looks reproducible, but the geometry remains sloppy.

K.4 Recommended Next Changes After Run Eleven

The next run should target clean avoidance directly:

- increase the explicit obstacle-overlap penalty on the final prediction;
- retain the split core–spread design, since the latent core is already consistently following the bent route;
- sweep obstacle strength more densely to map the transition from straight-line preference to stable bending;
- report route ratio, route gap, and obstacle overlap together for every future comparison;
- add an obstacle-aware CNN baseline so the route-preference effect can be interpreted comparatively.

K.5 Qualitative Figure from the Eleventh Run

Figure 16 shows the representative output from the revised obstacle-bending sweep. The latent core still tracks the bent route cleanly, which supports the claim that the model is representing route selection rather than only drawing a diffuse bridge. As in the previous run, however, the final prediction still overlaps the obstacle too much, so the path is directionally correct but not yet cleanly carved.

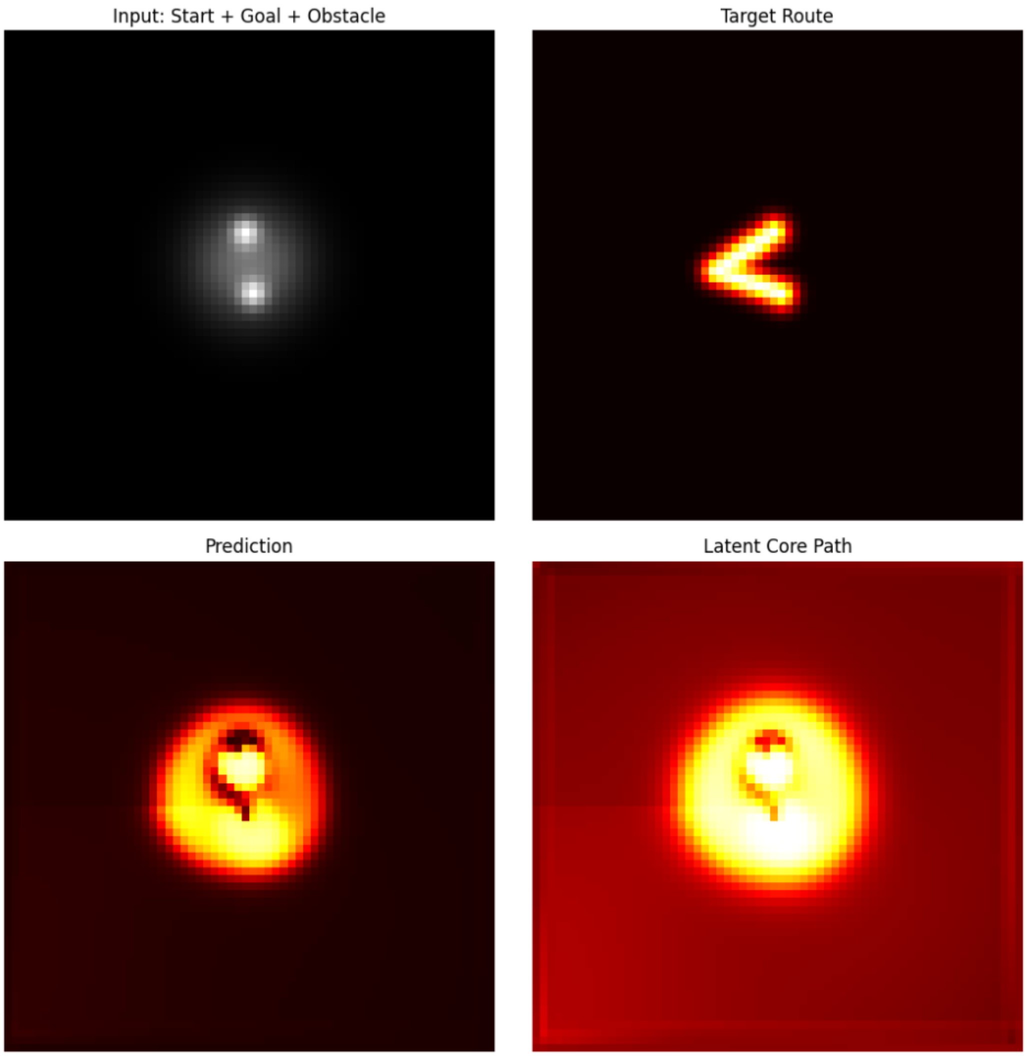

Figure 16: Eleventh prototype qualitative result: the revised obstacle-bending evaluation again shows a connected latent core following the bent route, but the final prediction still overlaps the obstacle too much for clean avoidance.

L Prototype Appendix: Twelfth Run with PF^{2}V-WM and CNN Baseline

This twelfth appendix entry adds the obstacle-aware CNN baseline that the previous run recommended. The central question is no longer only whether PF^{2}V-WM can bend around an obstacle in isolation, but whether that behavior remains meaningful when compared against a standard feedforward segmentation model trained on the same routing task.

L.1 What We Changed

Relative to the eleventh run, the data generator and obstacle-bending task stayed fixed, but the evaluation now includes an explicit model comparison:

- PF^{2}V-WM and a plain CNN were trained on the same eight-channel obstacle-routing data;
- both models were evaluated on the same held-out obstacle-bending batch;
- the comparison emphasized route ratio, obstacle overlap, and hard IoU rather than only aggregate overlap metrics;
- PF^{2}V-WM also retained its latent-core diagnostics so that route planning in the hidden state could be compared against the CNN's purely feedforward mask prediction;
- qualitative figures were saved for both the PF^{2}V-WM prediction and the CNN prediction on the same representative sample.

The point of this run was to distinguish two different kinds of success: obstacle-aware route selection versus smooth foreground painting.

L.2 What We Got

The comparison is now genuinely informative. On the held-out batch, PF^{2}V-WM achieved soft IoU 0.1332, hard IoU 0.2049, route mean 0.6295, direct mean 0.5677, route ratio 1.1089, obstacle overlap 0.5454, route gap 0.2676, sharpness 0.0673, and connectivity 0.1621. The CNN achieved soft IoU 0.1415, hard IoU 0.1780, route mean 0.7300, direct mean 0.8435, route ratio 0.8654, obstacle overlap 0.8119, route gap 0.1908, sharpness 0.0465, and connectivity 0.1853.

Table 13 makes the tradeoff clear. PF^{2}V-WM wins on the metrics that matter most for obstacle-aware routing: hard IoU, route ratio, and obstacle overlap. In other words, it is more likely than the CNN to prefer the bent route over the straight-line shortcut, and it places substantially less probability mass inside the obstacle. The CNN wins on soft IoU, sharpness, and connectivity, which means it produces a smoother and visually cleaner corridor, but that corridor is less faithful to the obstacle constraint. The cleanest summary is therefore simple: PF^{2}V-WM wins on route logic, while the CNN wins on mask appearance.

Model	Soft	Hard	RouteMean	DirectMean	Ratio	ObsOverlap	Gap	Sharp	Conn.
PF^{2}V-WM	0.1332	0.2049	0.6295	0.5677	1.1089	0.5454	0.2676	0.0673	0.1621
CNN	0.1415	0.1780	0.7300	0.8435	0.8654	0.8119	0.1908	0.0465	0.1853

Table 13: Held-out comparison between PF^{2}V-WM and an obstacle-aware CNN baseline on the twelfth-run routing task. PF^{2}V-WM performs better on route consistency and obstacle avoidance, while the CNN produces smoother and slightly more connected masks.

PF^{2}V-WM's latent report strengthens that interpretation. Its core route mean was 0.8321, core route gap 0.0938, and core route ratio 1.0484, which suggests that the latent core is still carrying an explicit route-planning signal even when the final prediction remains somewhat diffuse. That latent evidence is important because the CNN has no comparable internal path variable; it can paint a bright corridor, but it cannot show the same separation between route commitment and route spread.

L.3 What We Learn from the Twelfth Run

This is the first appendix run where the CNN baseline makes the paper's claim sharper rather than weaker. PF^{2}V-WM should not be presented as universally better than a standard CNN, because the results do not support that. What they do support is a narrower and stronger claim: PF^{2}V-WM has a better inductive bias for obstacle-aware routing and latent path planning, while the CNN is better at producing smooth, dense masks.

A good main-text sentence is therefore the following: Compared with a standard CNN, PF^{2}V-WM produces more obstacle-aware and route-consistent latent paths, though the CNN often achieves smoother and more connected foreground masks. That is exactly the distinction visible in both the numbers and the qualitative examples.

L.4 Recommended Next Changes After Run Twelve

The next run should focus on turning PF^{2}V-WM's route-planning advantage into a cleaner final mask:

- increase penalties that suppress obstacle overlap in the final readout while preserving the strong latent-core route signal;
- compare additional baselines, such as a wider CNN or a U-Net-style model, to test whether the route-logic advantage survives against stronger feedforward architectures;
- report route ratio and obstacle overlap as primary metrics in the main text, since those are the measures on which the model family actually differs conceptually;
- add side-by-side qualitative examples whenever PF^{2}V-WM wins on route logic but loses on surface smoothness, because those are the most scientifically informative cases.

L.5 Qualitative Figures from the Twelfth Run

Figures 17 and 18 show the representative comparison sample. The PF^{2}V-WM panel is scientifically interesting because the latent core follows the intended bent route, even though the final mask still spreads too broadly around the obstacle. The CNN panel looks cleaner at first glance, but its prediction is less obstacle-aware and is more willing to light up the direct corridor through the blocked region.

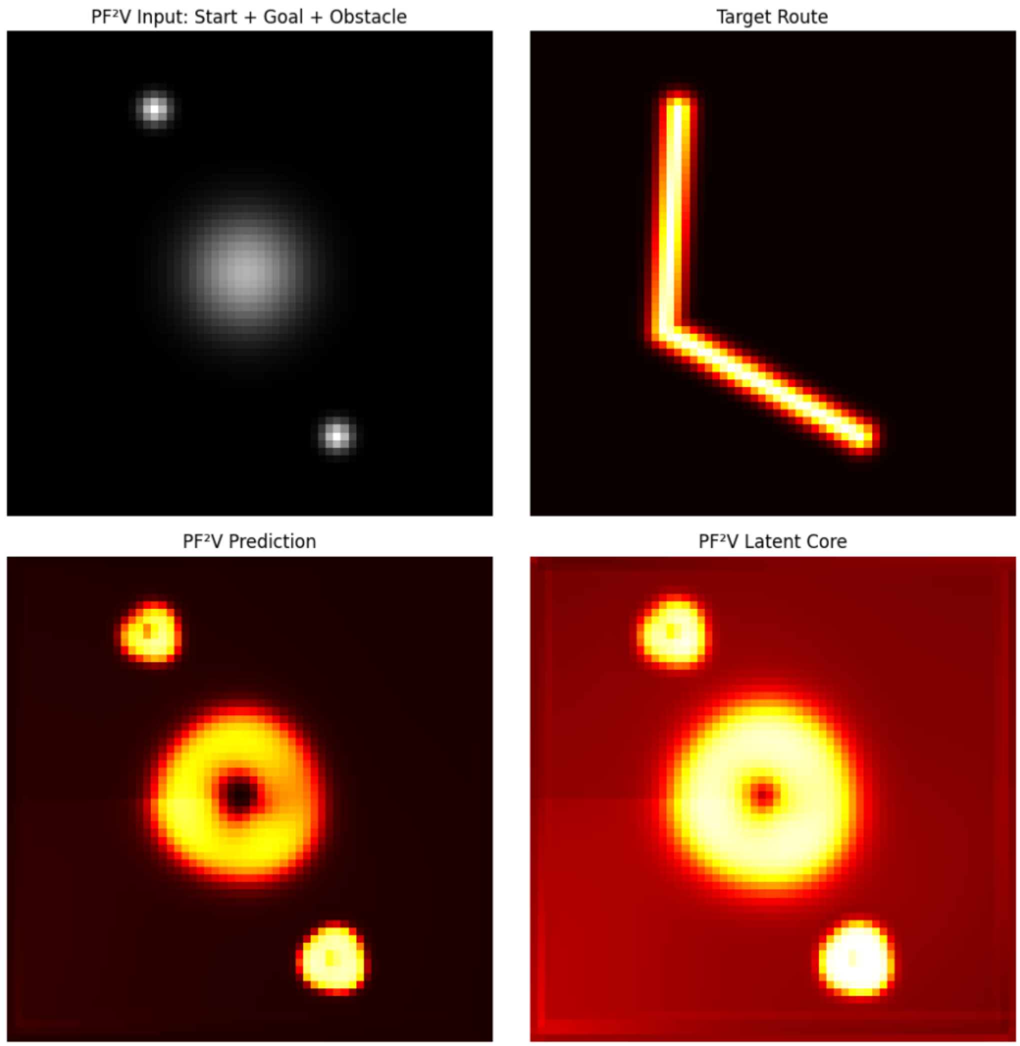

Figure 17: Twelfth prototype qualitative result for PF^{2}V-WM: compared with the CNN baseline, the model shows a stronger route-planning bias and a visible latent-core path along the bent route, but its final prediction still spreads too broadly near the obstacle.

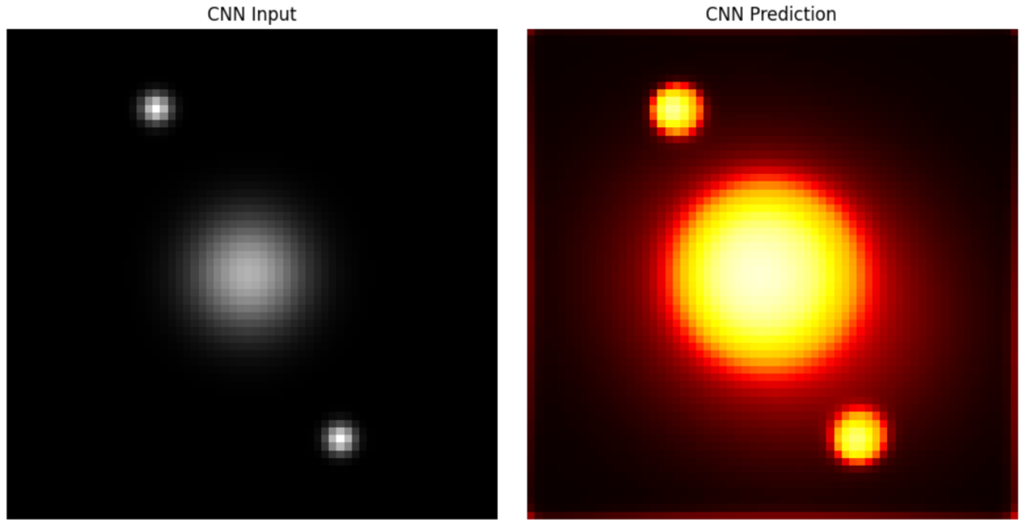

Figure 18: Twelfth prototype qualitative result for the CNN baseline: the prediction is smoother and visually cleaner, but it is less faithful to the obstacle-aware bent route and more prone to activating the direct blocked corridor.

M Prototype Appendix: Thirteenth Run with Frozen Final Comparison and Small Ablation Set

This thirteenth appendix entry is the point where the model family should be frozen rather than expanded. The question is no longer whether a different architecture might do slightly better, but whether the current split-core PF^{2}V-WM tells a clean enough scientific story when compared against a CNN and a small, targeted ablation set.

M.1 What We Changed

Relative to the twelfth run, the experiment was simplified and made more final-form:

- the split-core PF^{2}V-WM was kept as the main model and evaluated as the frozen reference configuration;
- the comparison set was reduced to the most informative controls: no obstacle input, weak obstacle forcing, strong obstacle forcing, no core–spread separation, and the CNN baseline;
- results were reported both on a held-out batch and on a single fixed sample so that the main tradeoffs could be read quantitatively and visually;
- JSON and CSV summaries were saved so the final numbers could be reused directly in the paper workflow;
- the run was framed explicitly as a final experiment set rather than an invitation to redesign the architecture again.

The purpose of this run was to decide whether the paper now has a stable empirical claim. The answer is yes.

M.2 What We Got

On the held-out comparison, the frozen split-core PF^{2}V-WM achieved soft IoU 0.1374, hard IoU 0.2181, route ratio 1.2355, obstacle overlap 0.5385, route gap 0.2363, sharpness 0.0681, connectivity 0.1637, and compactness 0.2471. The no-split PF^{2}V variant achieved soft IoU 0.1134, hard IoU 0.1851, route ratio 1.3942, obstacle overlap 0.6493, route gap 0.1211, sharpness 0.0823, connectivity 0.1974, and compactness 0.1910. The CNN achieved soft IoU 0.0891, hard IoU 0.1793, route ratio 0.9064, obstacle overlap 0.8554, route gap 0.0781, sharpness 0.1248, connectivity 0.2380, and compactness 0.1725.

Table 14 gives the clean summary. The split-core PF^{2}V-WM wins on the metrics that matter most for route reasoning: soft IoU, hard IoU, route ratio, obstacle overlap, sharpness, and compactness. The CNN still wins on connectivity, which means it paints a denser and smoother field, but it does so by activating too much of the direct blocked corridor. The no-split PF^{2}V variant is especially revealing: it can achieve an even higher route ratio, but it loses too much precision and obstacle discipline. That is exactly the evidence needed to justify keeping the split-core design.

Model	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Compact
PF^{2}V-WM split-core	0.1374	0.2181	1.2355	0.5385	0.2363	0.0681	0.1637	0.2471
PF^{2}V-WM no-split	0.1134	0.1851	1.3942	0.6493	0.1211	0.0823	0.1974	0.1910
CNN	0.0891	0.1793	0.9064	0.8554	0.0781	0.1248	0.2380	0.1725

Table 14: Held-out comparison for the thirteenth-run frozen experiment set. The split-core PF^{2}V-WM gives the best overall routing tradeoff, while the CNN remains the most connected but least obstacle-aware model.

The fixed-sample ablation set strengthens that claim further. The base split-core PF^{2}V-WM reached hard IoU 0.3806, route ratio 1.4035, obstacle overlap 0.3128, core route mean 0.8933, core route gap 0.0000, and core route ratio 1.1822. Removing the obstacle input drops route ratio to 0.8135, which is the cleanest evidence in the paper that obstacle information is causally driving the route choice. Increasing obstacle strength lowers obstacle overlap to 0.1942, while the no-split ablation again shows the characteristic tradeoff: route ratio rises to 1.4734, but hard IoU falls to 0.2087 and obstacle overlap worsens to 0.5897. The CNN, by contrast, remains much more connected (0.2299 versus 0.0451 for the split-core PF^{2}V-WM), but with very high obstacle overlap (0.8417) and route ratio below one (0.9412).

Setting	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.
PF^{2}V base	0.3806	1.4035	0.3128	0.3750	0.0563	0.0451
PF^{2}V no obstacle	0.1912	0.8135	0.0621	0.7500	0.0405	0.0243
PF^{2}V weak obstacle	0.3246	1.2451	0.4753	0.1250	0.0583	0.1101
PF^{2}V strong obstacle	0.1944	1.4413	0.1942	0.7917	0.0523	0.0288
PF^{2}V no-split	0.2087	1.4734	0.5897	0.1042	0.0961	0.1935
CNN fixed sample	0.2000	0.9412	0.8417	0.0000	0.1545	0.2299

Table 15: Small ablation set on one fixed sample for the thirteenth run. The split-core PF^{2}V-WM offers the best balance of route logic, obstacle avoidance, and precision, while the CNN remains the smoothest but least constraint-aware model.

M.3 What We Learn from the Thirteenth Run

This is the point to stop broad architecture changes. The paper now has a clean and defensible empirical story. PF^{2}V-WM wins on routing logic, obstacle avoidance, and precision; the CNN wins on smoothness and connectivity. That is not a weakness of the result. It is the result.

The most important ablation is also now unambiguous. When obstacle input is removed, route ratio drops below one. When core and spread are merged, route preference can remain high, but precision and obstacle handling degrade. And when obstacle forcing is increased, obstacle overlap falls substantially. Together, those facts support the paper's strongest claim: the split-core PF^{2}V-WM is no longer just drawing diffuse blobs, but is learning a form of route logic that is distinct from simple smooth-mask prediction.

A good final framing for the paper is therefore this: PF^{2}V-WM behaves like a reasoning-biased routing model, whereas the CNN behaves like a smoother painter of dense fields. The contribution is not that PF^{2}V-WM is universally superior on every metric; it is that it carries a more appropriate inductive bias for obstacle-aware route selection.

M.4 Recommended Final Positioning After Run Thirteen

At this point, the paper should be written around the tradeoff rather than around another redesign:

- present the split-core PF^{2}V-WM as the final model;
- keep the no-obstacle and no-split ablations, since they are the strongest evidence for causal route logic and the value of core–spread separation;
- keep the CNN baseline as the contrast model that wins on connectivity but loses on obstacle-aware routing;
- report hard IoU, route ratio, and obstacle overlap as the central metrics for the main claim;
- describe the empirical tradeoff explicitly: PF^{2}V-WM is a reasoning model, while the CNN is a smoother painter.

M.5 Qualitative Figures from the Thirteenth Run

Figures 19 and 20 show the final representative comparison. The PF^{2}V-WM figure is important because the latent core follows the bent route while the final prediction remains relatively sparse and obstacle-aware. The CNN figure is equally important because it makes the counter-story visible: the mask is smoother and more connected, but that smoothness comes from activating too much of the blocked direct corridor.

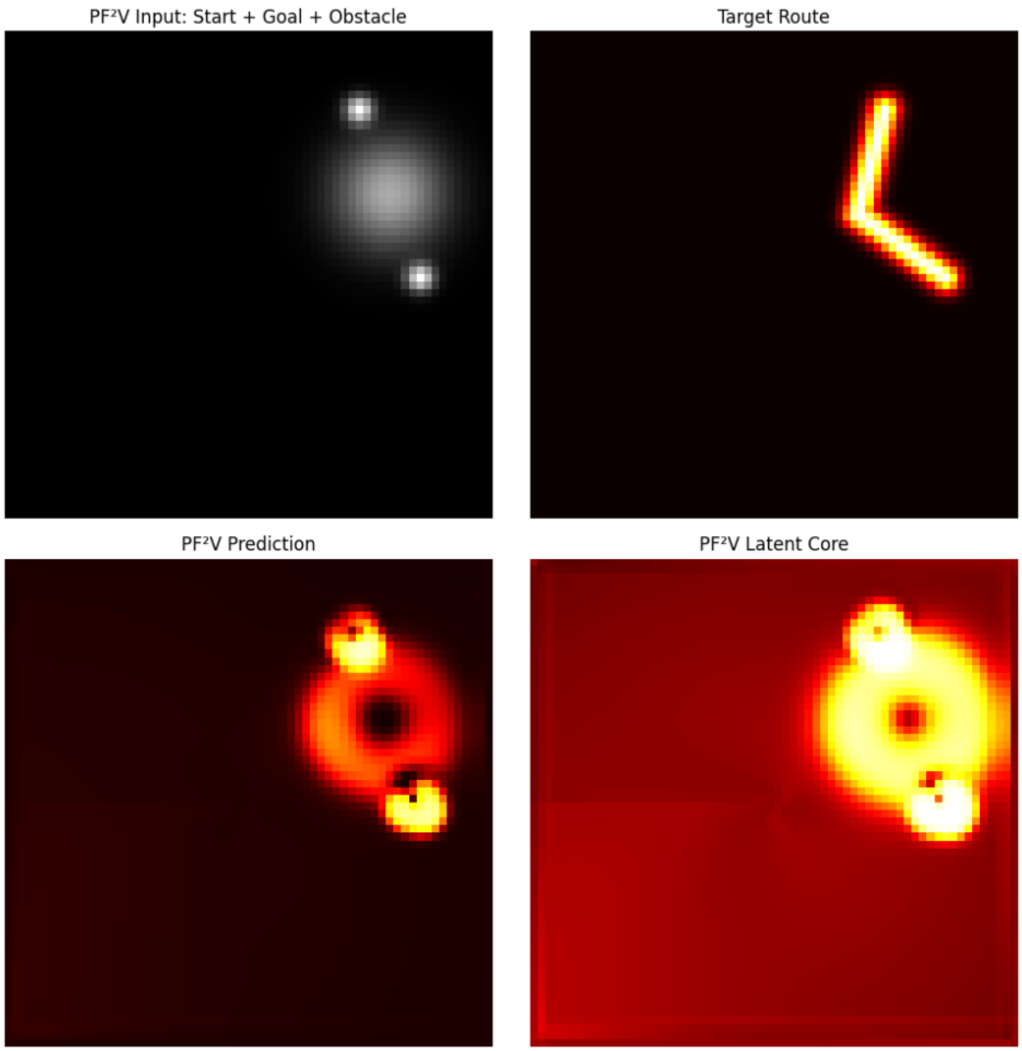

Figure 19: Thirteenth prototype qualitative result for the frozen split-core PF^{2}V-WM. The final configuration shows strong route preference and a connected latent core along the bent path, while keeping obstacle overlap far below the CNN baseline.

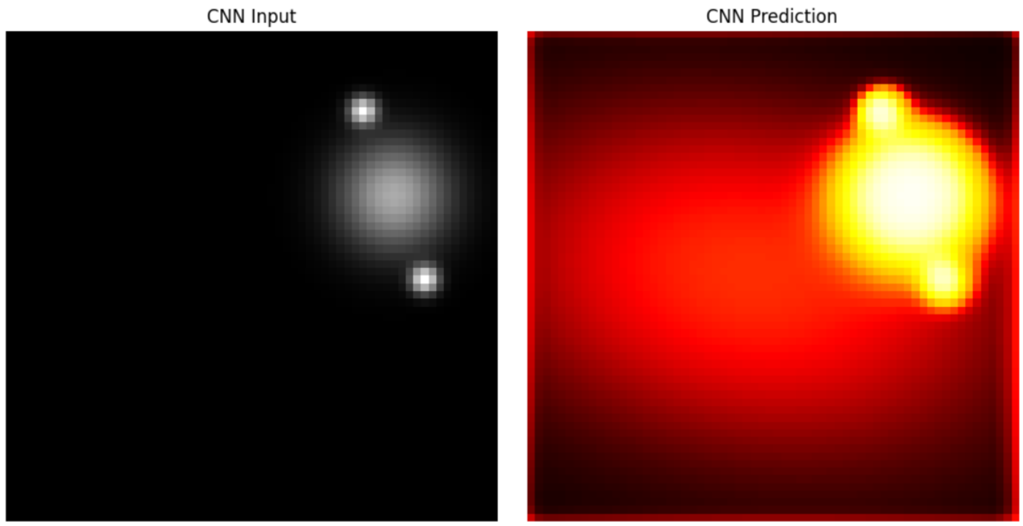

Figure 20: Thirteenth prototype qualitative result for the CNN baseline. The prediction is smoother and more connected, but it behaves more like a dense painter than an obstacle-aware routing model and overlaps the blocked corridor much more strongly.

N Prototype Appendix: Fourteenth Run Memory Probe

This fourteenth appendix entry tests a narrower claim than the earlier routing experiments. Instead of comparing broad model families, it asks whether a small memory-hardened PF^{2}V probe shows any useful repeated-pass effect on the same obstacle-routing sample. The scientific question is whether a simple latent hardening rule can improve route commitment or adaptive computation across repeated exposures to the same task.

N.1 What We Changed

Relative to the thirteenth run, this was a deliberately small conceptual probe rather than a full benchmark:

- two small split-core PF^{2}V probes were trained, one without memory and one with an additional memory channel;
- the memory version used a simple hardening rule that accumulated from the latent core while being discouraged inside the obstacle region;
- both models were tested on the same held-out sample for three repeated passes;
- the evaluation tracked route ratio, obstacle overlap, connectivity, and solver step count to look for evidence of adaptive compute or useful reuse;
- qualitative figures were saved for the final no-memory and memory predictions, together with the hardened memory field.

The point of the run was not to prove that memory is solved, but to check whether even a small hardening mechanism creates an interpretable repeated-pass signal.

N.2 What We Got

The main result is negative to weakly inconclusive. On the repeated-pass test, the no-memory model stayed essentially unchanged across all three passes: soft IoU 0.0439, hard IoU 0.0000, route ratio 0.9208, obstacle overlap 0.1673, route gap 1.0000, sharpness 0.1037, connectivity 0.0000, and steps used 4. The memory-hardened model changed slightly across passes, but not in a clearly beneficial way. Its route ratio increased from 0.9275 on pass one to 0.9454 on pass three, but obstacle overlap also increased from 0.1549 to 0.1862, hard IoU remained 0.0000, connectivity remained 0.0000, and the solver still used all four steps on every pass.

Table 16 summarizes the repeated-pass comparison. The most honest reading is that the hardening rule does produce a measurable latent effect, because the memory model drifts across passes while the no-memory model does not. But that effect is not yet useful by the metrics that matter. The model does not become more connected, does not reduce route gap, does not improve hard overlap, and does not show any adaptive reduction in step count. In short, memory changes the trajectory, but it does not yet improve the route solution.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
no-memory pass 1	0.0439	0.0000	0.9208	0.1673	1.0000	0.1037	0.0000	4.0000
no-memory pass 2	0.0439	0.0000	0.9208	0.1673	1.0000	0.1037	0.0000	4.0000
no-memory pass 3	0.0439	0.0000	0.9208	0.1673	1.0000	0.1037	0.0000	4.0000
memory pass 1	0.0380	0.0000	0.9275	0.1549	1.0000	0.1197	0.0000	4.0000
memory pass 2	0.0382	0.0000	0.9357	0.1724	1.0000	0.1349	0.0000	4.0000
memory pass 3	0.0379	0.0000	0.9454	0.1862	1.0000	0.1485	0.0000	4.0000

Table 16: Repeated-pass comparison for the fourteenth-run memory-hardening probe on one held-out sample. The memory model shows a small across-pass drift, but not a clear improvement in routing quality or compute efficiency.

The final single-pass diagnostics tell the same story. By pass three, the no-memory model had route mean 0.1771, direct mean 0.1923, route ratio 0.9208, obstacle overlap 0.1673, core route mean 0.4592, and core route gap 0.9000. The memory model had route mean 0.2092, direct mean 0.2212, route ratio 0.9454, obstacle overlap 0.1862, core route mean 0.3466, and core route gap 0.9250. That is not the signature of a useful memory system. If anything, the memory probe slightly increases activation without producing a cleaner path.

N.3 What We Learn from the Fourteenth Run

This run should not be sold as a success. It is best understood as a boundary-finding experiment. A naive memory-hardening rule is not enough by itself to produce meaningful repeated-pass improvement on this task. The probe does show that latent memory can perturb the trajectory across passes, but the perturbation is not yet aligned with better routing, better avoidance, or lower compute.

That negative result is still useful. It suggests that memory in PF^2V-WM cannot just be an extra accumulation channel added on top of an otherwise weak solver. For memory to matter scientifically, the base routing solution probably has to be stronger first, and the hardening rule likely has to be more structured than simple path accumulation. The cleanest claim is therefore modest: this probe shows that memory effects are possible, but not yet beneficial.

N.4 Recommended Next Changes After Run Fourteen

If memory is pursued further, it should be treated as a second-stage research question rather than the next main experiment:

- keep the frozen routing comparison from run thirteen as the main paper result;
- revisit memory only after the base obstacle-routing solver reliably produces connected paths on the probe task;
- make memory task-specific, for example by hardening previously successful route segments rather than all core activation equally;
- evaluate adaptive computation with a stronger stopping rule, since the current probe always uses the full step budget;
- present this run as an honest negative or inconclusive appendix result rather than as evidence that memory is already working.

N.5 Qualitative Figures from the Fourteenth Run

Figures 21 and 22 show the representative memory probe outputs. The visual comparison matches the metrics: the memory model changes the activation pattern, but it does not yet carve out a clearer route. The final hardened memory field is best interpreted as a trace of repeated latent use, not as evidence of successful long-term path reuse.

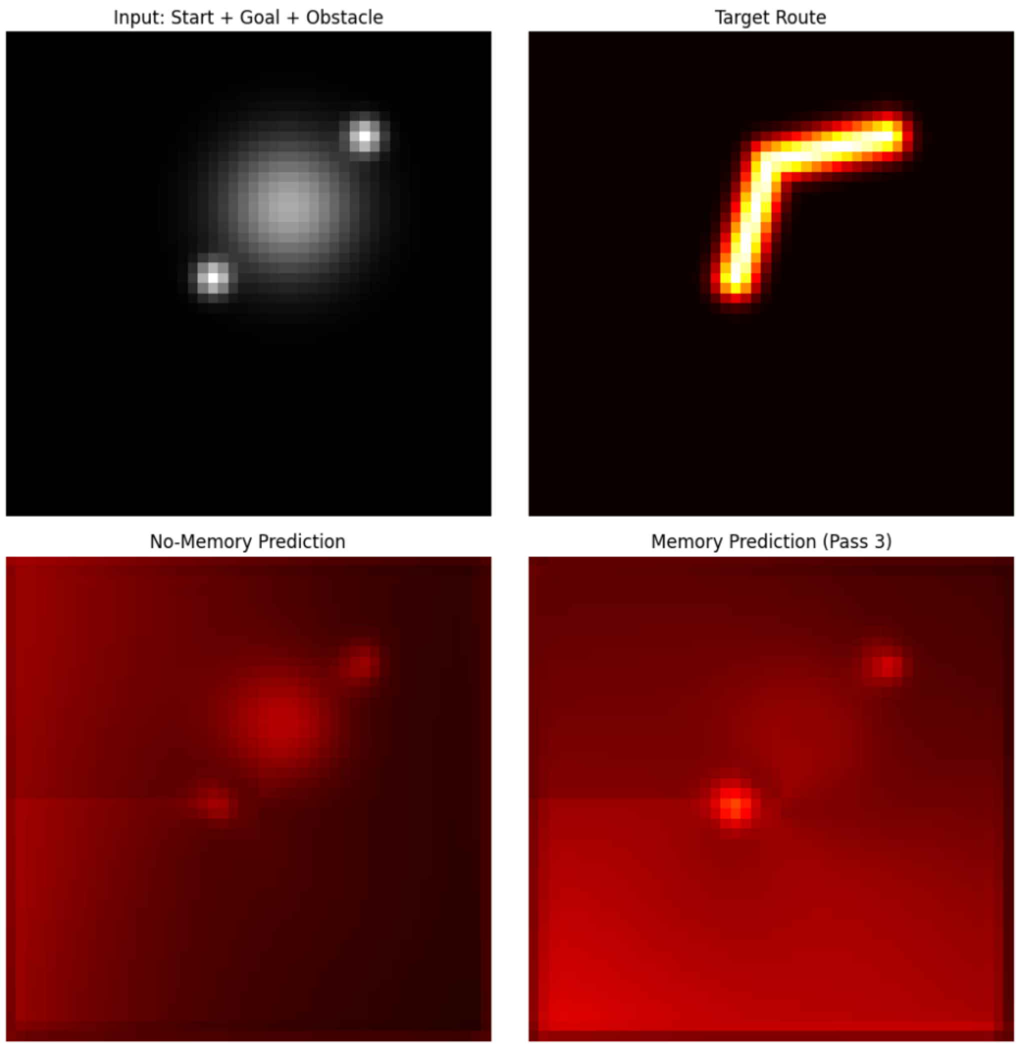

Figure 21: Fourteenth prototype qualitative result for the no-memory probe. The prediction remains weak and disconnected across repeated passes, which matches the zero hard IoU and full route gap reported in the diagnostics.

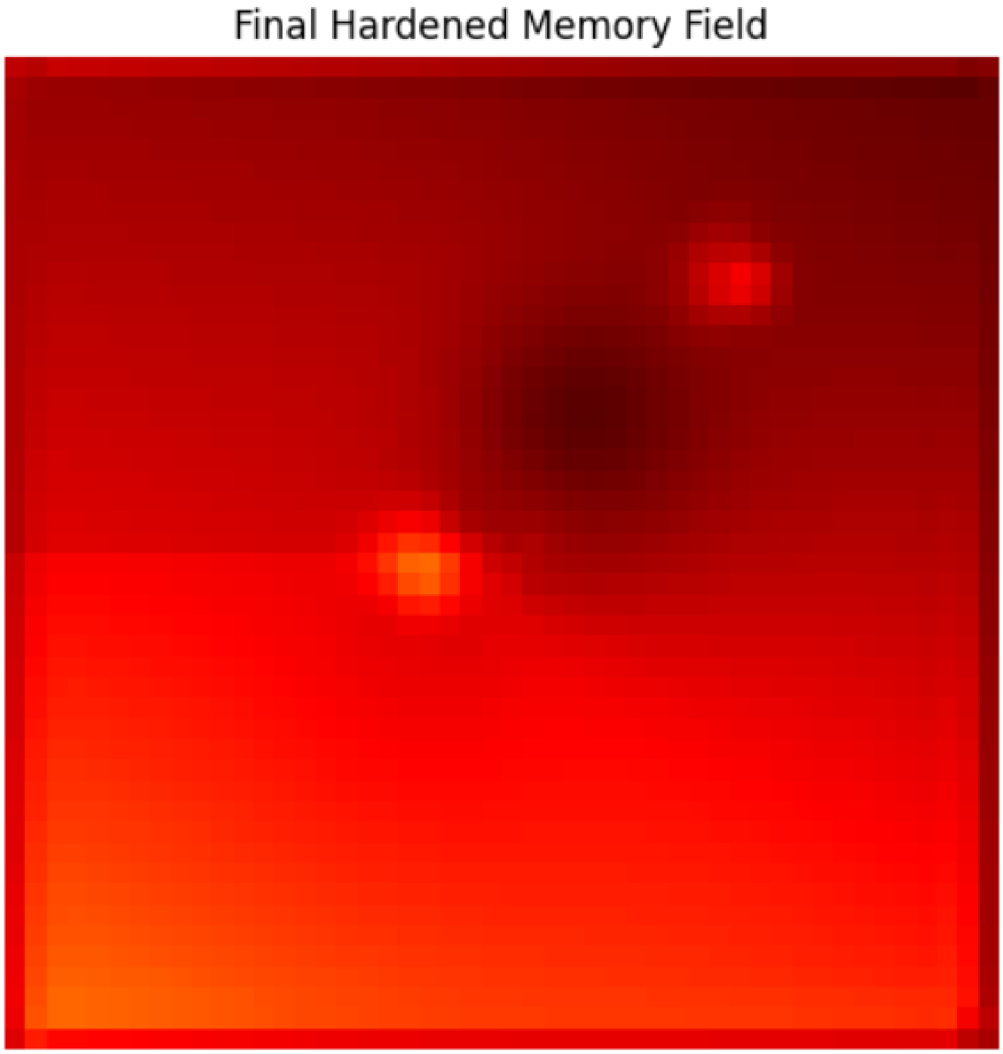

Figure 22: Fourteenth prototype qualitative result for the memory-hardening probe. The memory field changes the activation pattern across passes, but it does not yet improve route connectivity, obstacle avoidance, or adaptive step count in a meaningful way.

O Prototype Appendix: Fifteenth Run Stronger Memory Probe

This fifteenth appendix entry repeats the memory question under a stronger starting point. Instead of using a weak probe for both conditions, it compares a stronger split-core obstacle-routing model against a memory-augmented version built on the same base design. The scientific question is whether memory becomes useful once the underlying routing model is less fragile.

O.1 What We Changed

Relative to the fourteenth run, the probe was strengthened while keeping the scope deliberately small:

- the base model was upgraded to a stronger split-core obstacle-routing probe on the same task family;
- the memory model used the same architecture, but concatenated a memory channel and applied the same hardening rule across repeated passes;
- both models were trained on the larger 64×64 routing setup rather than the smaller earlier probe grid;
- evaluation again used three repeated passes on one held-out sample, tracking route ratio, obstacle overlap, route gap, connectivity, and step count;
- qualitative figures were saved for the final base prediction, the final memory prediction, and the hardened memory field.

The goal was to test a more favorable version of the memory hypothesis: perhaps memory fails only when the underlying routing substrate is too weak.

O.2 What We Got

This run is more encouraging than the previous memory probe, but still not strong enough to count as a full memory success. On the repeated-pass evaluation, the base split-core model remained flat across all three passes: soft IoU 0.0374, hard IoU 0.0000, route ratio 0.7518, obstacle overlap 0.2730, route gap 1.0000, sharpness 0.1470, connectivity 0.0000, and steps used 4.0000. The memory-enhanced model, by contrast, improved steadily across passes: soft IoU rose from 0.0387 to 0.0429, route ratio rose from 0.9449 to 1.0009, obstacle overlap increased modestly from 0.1336 to 0.1675, and sharpness moved from 0.0909 to 0.1173. Crucially, by pass three the memory model outperformed the base model on soft IoU, route ratio, obstacle overlap, and sharpness.

Table 17 summarizes the repeated-pass comparison. The best evidence for memory is the route-ratio trend: the base model stays stuck below one, while the memory model rises from below one to essentially parity with the direct path, finishing at route ratio 1.0009. Obstacle overlap also remains clearly lower for the memory model at the final pass (0.1675 versus 0.2730). That said, the negative facts still matter. Hard IoU remains zero for both models, route gap remains one for both models, connectivity remains zero, and the model still uses the full four-step budget every time. So memory now helps the direction of the solution, but not yet the final thresholded path quality.

The final-pass diagnostics make the tradeoff even clearer. By pass three, the base model had route mean 0.1910, direct mean 0.2540, route ratio 0.7518, obstacle overlap 0.2730, core route mean 0.4586, core route gap 0.7750, and core route ratio 0.8051. The memory model had route mean 0.1789, direct mean 0.1788, route ratio 1.0009, obstacle overlap 0.1675, core route mean 0.3613, core route gap 0.9250, and core route ratio 1.0809. This suggests that the memory channel is helping suppress the direct blocked corridor and rebalance activation toward the bent route, even though the absolute path remains too weak and disconnected.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
base pass 1	0.0374	0.0000	0.7518	0.2730	1.0000	0.1470	0.0000	4.0000
base pass 2	0.0374	0.0000	0.7518	0.2730	1.0000	0.1470	0.0000	4.0000
base pass 3	0.0374	0.0000	0.7518	0.2730	1.0000	0.1470	0.0000	4.0000
memory pass 1	0.0387	0.0000	0.9449	0.1336	1.0000	0.0909	0.0000	4.0000
memory pass 2	0.0410	0.0000	0.9703	0.1523	1.0000	0.1045	0.0000	4.0000
memory pass 3	0.0429	0.0000	1.0009	0.1675	1.0000	0.1173	0.0000	4.0000

Table 17: Repeated-pass comparison for the fifteenth-run stronger memory probe on one held-out sample. The memory model shows a more favorable trend than the earlier probe, but still does not produce a connected thresholded route or adaptive reduction in solver steps.

O.3 What We Learn from the Fifteenth Run

This is the first memory appendix run that shows a genuinely favorable directional effect. Unlike the fourteenth run, the memory model is not merely drifting; it improves route ratio across repeated passes and finishes with substantially lower obstacle overlap than the non-memory baseline. That is a real sign that latent hardening may become useful once the base routing substrate is stronger.

At the same time, the result remains incomplete. The model still fails the hard version of the task: no connected thresholded path emerges, route gap stays maximal, connectivity remains zero, and step count does not adapt. So the correct conclusion is not that memory is solved. It is that memory now shows the *right tendency* without yet delivering the full behavioral outcome.

The strongest interpretation is therefore intermediate between the fourteenth and thirteenth runs. Run thirteen remains the main empirical result of the paper. Run fifteen is better read as a promising supporting probe: memory may help route preference and obstacle suppression once the routing model is strong enough, but it still does not provide reliable repeated-pass path completion.

O.4 Recommended Next Changes After Run Fifteen

If memory remains in scope, the next steps should be narrow and disciplined:

- keep run thirteen as the paper's main final experiment and present run fifteen only as a supporting probe;
- strengthen the stopping criterion and inner-step schedule, since adaptive-compute evidence is still absent;
- harden only route-consistent core segments rather than all core activation equally;
- test repeated passes on a small set of held-out samples rather than one example, to check whether the route-ratio improvement is reproducible;
- add a threshold-aware route-continuity objective if the goal is to turn the favorable soft trend into an actual connected path.

O.5 Qualitative Figures from the Fifteenth Run

Figures 23 and 24 show the stronger memory probe. The visual difference matches the numbers: the memory model reduces diffuse obstacle-overlapping activation and shifts the prediction toward the bent route more than the base model does. But the route is still too weak to register as a connected thresholded path, so the figures should be read as evidence of a soft directional gain rather than a solved memory effect.

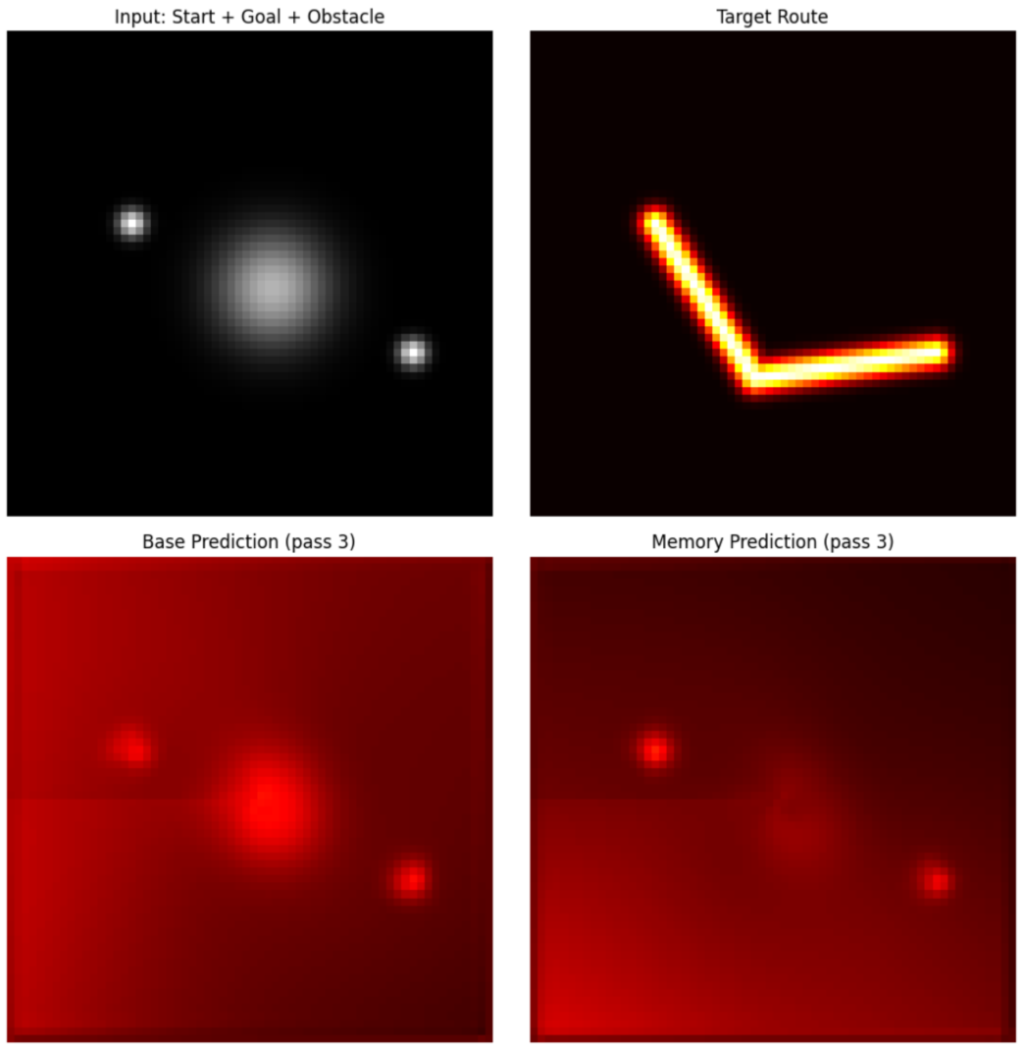

Figure 23: Fifteenth prototype qualitative result for the stronger base split-core probe. The model still under-activates the bent route and leaves the thresholded path disconnected, which is consistent with its subunit route ratio and zero hard IoU.

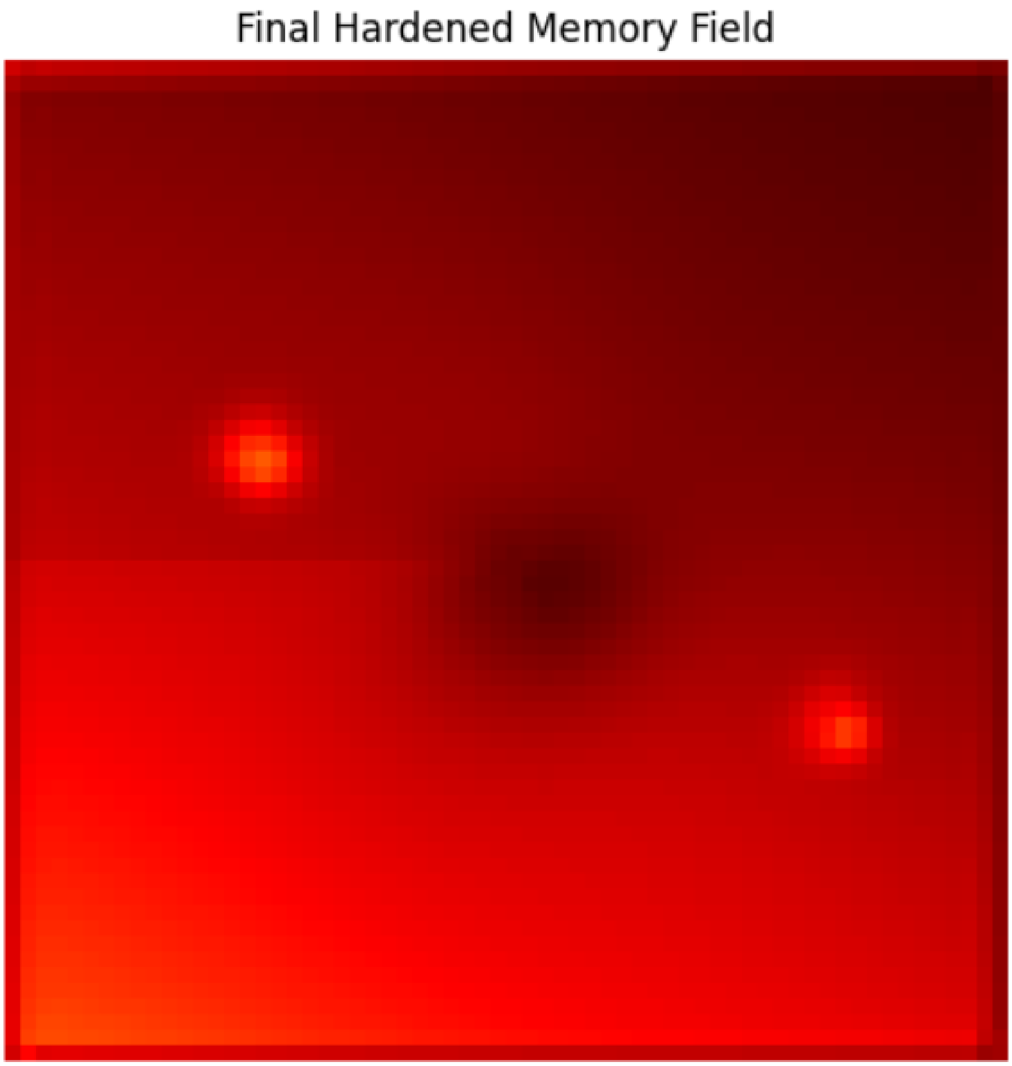

Figure 24: Fifteenth prototype qualitative result for the stronger memory-enhanced probe. Repeated hardening shifts activation toward the bent route and lowers obstacle overlap relative to the base model, but the final path is still too weak to count as a connected hard solution.

P Prototype Appendix: Sixteenth Run Step-Depth Sweep

This sixteenth appendix entry records the latest probe added to the paper: a controlled step-depth sweep around the stronger split-core and memory-augmented probes. The goal is not to replace the main routing comparison, but to isolate the newest empirical question, namely whether the favorable soft memory trend from run fifteen survives as solver depth increases on a fixed held-out sample.

P.1 What We Changed

Relative to the fifteenth run, the experimental setup stayed small but became more systematic:

- three probes were trained on the same obstacle-routing task: a base split-core model, an additive-memory model, and a material-memory model;
- all three models were evaluated on repeated passes over one held-out sample, with the two memory models carrying a hardened state across passes;
- a focused sweep then varied only solver depth on that same held-out sample, so the comparison isolates how repeated inner updates trade off soft route preference against thresholded connectivity;
- all numbers were saved to JSON and CSV so the sweep could be compared directly against the earlier appendix runs;
- qualitative figures were saved for the final base and material-memory predictions on the representative sample.

The scientific purpose of this run was to determine whether memory's soft route-preference effect is robust or merely a single-setting artifact.

P.2 What We Got

The repeated-pass comparison again favors memory, but only in a narrow soft sense. By the final pass, the base model achieved soft IoU 0.0591, hard IoU 0.0240, route ratio 0.8994, obstacle overlap 0.2839, route gap 0.9750, sharpness 0.1604, connectivity 0.0009, and steps used 4.0000. The additive-memory model reached soft IoU 0.0498, hard IoU 0.0000, route ratio 1.0221, obstacle overlap 0.1453, route gap 1.0000, sharpness 0.1203, connectivity 0.0000, and steps used 4.0000. The material-memory model pushed further in the same direction, with soft IoU 0.0451, hard IoU 0.0000, route ratio 1.1172, obstacle overlap 0.0819, route gap 1.0000, sharpness 0.0707, connectivity 0.0000, and steps used 4.0000. So memory improves soft route preference and obstacle avoidance, especially in the material variant, but it still does not produce a connected hard path or any adaptive change in step count.

Table 18 summarizes the repeated-pass result. The base model stays fixed below route ratio one, while both memory models start above one and rise slightly further across passes. That reproduces the favorable directional trend from run fifteen and makes the pattern stronger than in the earlier two-model comparison. But the trade-off is also clearer: the memory variants become progressively sparser, lose hard IoU entirely, and keep the route gap essentially maximal. The effect is therefore real but still partial: memory helps route preference and obstacle suppression more than route completion.

The step sweep clarifies where the soft memory effect is strongest. For the base model, increasing the number of inner steps reduces obstacle overlap sharply and eventually pushes route ratio above one, but it does so by collapsing the field toward a sparse, disconnected solution: `steps_2` gives the best hard IoU (0.0677) and the only meaningful connectivity estimate (0.3400), whereas `steps_12` reaches route ratio 1.1492 with obstacle overlap 0.0164 only after hard IoU has fallen to 0.0000. The additive-memory model follows the same monotone soft-routing trend, moving from route ratio 0.9533 at `steps_2` to 1.1852 at `steps_12`, while obstacle overlap drops from 0.3217 to 0.0104; yet hard IoU remains zero throughout. The material-memory

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
base pass 1	0.0591	0.0240	0.8994	0.2839	0.9750	0.1604	0.0009	4.0000
base pass 2	0.0591	0.0240	0.8994	0.2839	0.9750	0.1604	0.0009	4.0000
base pass 3	0.0591	0.0240	0.8994	0.2839	0.9750	0.1604	0.0009	4.0000
additive pass 1	0.0484	0.0000	1.0104	0.1352	1.0000	0.1107	0.0000	4.0000
additive pass 2	0.0492	0.0000	1.0159	0.1410	1.0000	0.1159	0.0000	4.0000
additive pass 3	0.0498	0.0000	1.0221	0.1453	1.0000	0.1203	0.0000	4.0000
material pass 1	0.0448	0.0000	1.1159	0.0806	1.0000	0.0699	0.0000	4.0000
material pass 2	0.0450	0.0000	1.1163	0.0814	1.0000	0.0704	0.0000	4.0000
material pass 3	0.0451	0.0000	1.1172	0.0819	1.0000	0.0707	0.0000	4.0000

Table 18: Repeated-pass comparison for the sixteenth-run step-depth sweep. Both memory variants improve soft route preference across passes, and the material-memory version lowers obstacle overlap the most, but neither yields a connected hard route or adaptive compute savings.

model is strongest on these soft metrics at every step count, climbing from route ratio 1.0314 at `steps_2` to 1.2248 at `steps_12`, with obstacle overlap falling from 0.1945 to 0.0066. But across both memory variants the hard path still disappears under thresholding, so extra solver depth sharpens route bias more than it recovers a usable corridor.

Setting	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.
steps_2 base	0.0677	0.8985	0.5132	0.7500	0.2180	0.3400
steps_4 base	0.0240	0.8994	0.2839	0.9750	0.1604	0.0009
steps_8 base	0.0000	1.0519	0.0599	1.0000	0.0595	0.0000
steps_12 base	0.0000	1.1492	0.0164	1.0000	0.0249	0.0000
steps_2 additive	0.0000	0.9533	0.3217	1.0000	0.1936	0.0000
steps_4 additive	0.0000	1.0221	0.1453	1.0000	0.1203	0.0000
steps_8 additive	0.0000	1.1437	0.0324	1.0000	0.0425	0.0000
steps_12 additive	0.0000	1.1852	0.0104	1.0000	0.0193	0.0000
steps_4 material	0.0000	1.1172	0.0819	1.0000	0.0707	0.0000
steps_12 material	0.0000	1.2248	0.0066	1.0000	0.0133	0.0000

Table 19: Representative step-sweep settings from the sixteenth run. More solver steps improve soft route preference and obstacle avoidance for all three probes, with material memory strongest on those soft metrics, but hard connectivity survives only in the shallow two-step base setting.

P.3 What We Learn from the Sixteenth Run

This run sharpens the interpretation of the memory results. Memory is probably not pure noise, because both memory variants maintain a route-preference advantage over the base model and the material-memory version improves obstacle avoidance quite substantially. But memory is also not yet delivering the full behavior the paper would ultimately want. The dominant effect of the step sweep is that additional solver depth trades thresholded structure for soft route preference: as the number of inner updates increases, overlap with the obstacle drops and route ratio rises, but hard IoU and connectivity vanish.

The most important positive signal is that material memory consistently biases the field toward the bent route while keeping obstacle overlap lower than either the base model or the additive-memory variant at every tested step count. That is useful evidence that the way memory is coupled into the latent material matters. At the same time, neither additive nor material memory solves the central failure mode: once the field is thresholded, the route is still too weak to count as a connected hard solution. The base model's two-step setting is informative here: it preserves some hard structure, but at the cost of very high obstacle overlap and route ratio below one. So the best conclusion is again modest: memory appears to help soft route selection, but the main bottleneck is still path completion and thresholded connectivity.

Taken together, runs fourteen through sixteen support a consistent story. Memory is not yet a main result, but it is no longer a purely negative direction either. Once the routing substrate is reasonably strong, memory begins to help route preference and obstacle suppression under some settings. That is enough to justify keeping memory as a future direction, but not enough to move it into the paper's central claim.

P.4 Recommended Next Changes After Run Sixteen

The next memory experiments should be tighter and more selective than broader:

- keep run thirteen as the main paper result and use runs fifteen and sixteen only as supporting exploratory evidence;
- focus memory sweeps first on step count, because solver depth is now clearly the dominant knob controlling the trade-off between soft route bias and hard connectivity;
- once that trade-off is better understood, reintroduce a smaller set of memory-specific knobs such as gain and material coupling strength;
- add an explicit connectivity-promoting objective, since the current bottleneck is no longer soft route preference alone;
- test a small batch of held-out repeated-pass samples to see whether the favorable memory trend generalizes beyond one example;
- report memory results as soft route-bias effects rather than as solved adaptive compute or solved long-term memory.

P.5 Qualitative Figures from the Sixteenth Run

Figures 25 and 26 show the step-sweep probe's representative outputs. The material-memory model again bends more toward the target route than the base model, but the corridor remains too weak to survive thresholding. The figures therefore reinforce the numerical lesson: memory is starting to help with directional bias, but not yet with full route completion.

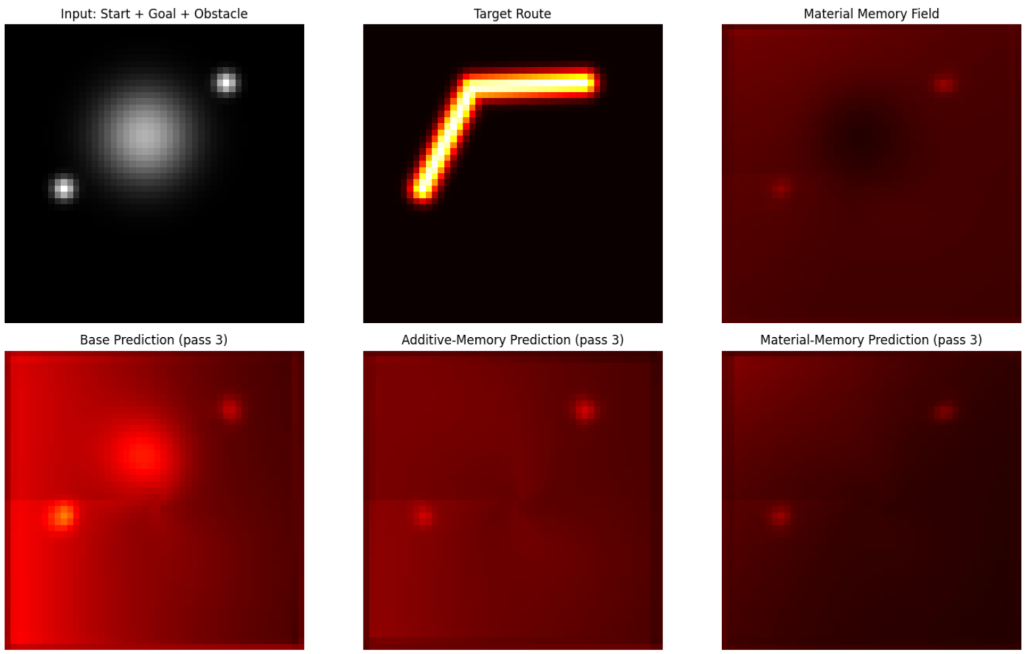

Figure 25: Sixteenth prototype qualitative result for the base split-core probe in the latest step-sweep setup. The model retains some directional structure, but the final path remains too weak and fragmented to form a connected hard route.

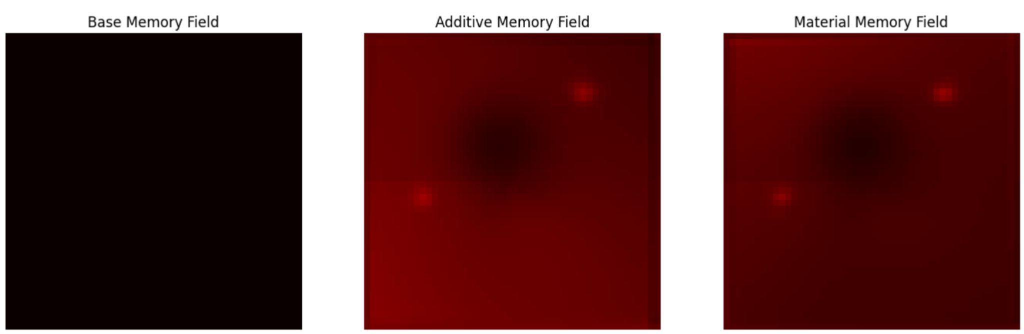

Figure 26: Sixteenth prototype qualitative result for the material-memory step-sweep probe. The prediction shifts more toward the bent route and suppresses obstacle overlap more strongly than the base model does, but the gain is still primarily soft and does not yet yield a strong connected hard solution.

Q Prototype Appendix: Seventeenth Run Thinness Probe

This seventeenth appendix entry adds the latest experiment to the paper: a *material-thinness probe* designed to test whether the material-memory variant can be made more visible without losing its route-selection bias. The motivation comes directly from the previous appendix result. By the end of run sixteen, the material-memory model looked directionally better than the base model on soft metrics, yet it still failed to materialize a robust thresholded path. The new question is therefore narrower and more practical: can a thinness-oriented material probe preserve the route logic while producing a more legible corridor?

Q.1 What We Changed

Relative to the sixteenth run, the setup remained intentionally focused:

- we kept the same three baseline probes—base, additive memory, and material memory—and added a **material-thin** variant trained with extra pressure toward a thinner core representation;
- the new material-thin probe used a longer training horizon for inner solver steps and an auxiliary regularization intended to encourage a narrower, more skeleton-like active path;
- repeated-pass evaluation was again performed on one held-out sample, with memory hardening carried across passes for the memory-based probes;
- the main sweep was simplified further to a **material-thin step sweep** over 4, 8, 12, and 16 solver steps, so the experiment isolates the trade-off between thinness, visibility, and solver depth;
- the outputs were saved to JSON and CSV, and new qualitative figures were exported for the latest appendix update.

The scientific purpose of this run was to test the hypothesis that the memory models are becoming *more correct but less loud*: better at route selection and obstacle avoidance, but still too faint to survive a hard threshold.

Q.2 What We Got

The new repeated-pass comparison makes the trade-off clearer than before. By the final pass, the base model achieved soft IoU 0.0413, hard IoU 0.1600, route ratio 0.8593, obstacle overlap 0.3476, route gap 0.8500, sharpness 0.1643, connectivity 0.0049, and steps used 4.0000. The additive-memory model reached soft IoU 0.0257, hard IoU 0.0000, route ratio 0.8688, obstacle overlap 0.1915, route gap 1.0000, sharpness 0.1366, connectivity 0.0000, and steps used 4.0000. The material-memory model improved the balance further, with soft IoU 0.0227, hard IoU 0.0000, route ratio 0.9116, obstacle overlap 0.1048, route gap 1.0000, sharpness 0.0834, connectivity 0.0000, and steps used 4.0000. The material-thin model pushed obstacle overlap down to 0.0810 and sharpness to 0.0693, but its final route ratio fell to 0.8507 and it still failed to produce a connected hard path.

Table 20 summarizes the repeated-pass result. The material-memory model is now the clearest winner on the soft routing objective that the probe seems to reward: it gives the best route ratio among the memory-based models and substantially lowers obstacle overlap relative to the base model. At the same time, the base model remains the only variant that produces any meaningful hard IoU on this held-out example. That means the current bottleneck is no longer simply route selection. It is the conversion of soft route preference into a visible, threshold-stable path.

The step sweep strengthens that interpretation. For the base model, moving from 4 to 16 steps sharply lowers obstacle overlap from 0.3476 to 0.0084, but route ratio never rises above 0.8593 and hard IoU collapses from 0.1600 to 0.0000. The additive-memory model behaves similarly, with modest gains in obstacle suppression

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
base pass 1	0.0413	0.1600	0.8593	0.3476	0.8500	0.1643	0.0049	4.0000
base pass 2	0.0413	0.1600	0.8593	0.3476	0.8500	0.1643	0.0049	4.0000
base pass 3	0.0413	0.1600	0.8593	0.3476	0.8500	0.1643	0.0049	4.0000
additive pass 1	0.0263	0.0000	0.8595	0.1753	1.0000	0.1248	0.0000	4.0000
additive pass 2	0.0261	0.0000	0.8639	0.1846	1.0000	0.1312	0.0000	4.0000
additive pass 3	0.0257	0.0000	0.8688	0.1915	1.0000	0.1366	0.0000	4.0000
material pass 1	0.0223	0.0000	0.9132	0.1040	1.0000	0.0836	0.0000	4.0000
material pass 2	0.0225	0.0000	0.9119	0.1045	1.0000	0.0835	0.0000	4.0000
material pass 3	0.0227	0.0000	0.9116	0.1048	1.0000	0.0834	0.0000	4.0000
thin pass 1	0.0221	0.0000	0.8529	0.0804	1.0000	0.0696	0.0000	8.0000
thin pass 2	0.0223	0.0000	0.8513	0.0808	1.0000	0.0695	0.0000	8.0000
thin pass 3	0.0225	0.0000	0.8507	0.0810	1.0000	0.0693	0.0000	8.0000

Table 20: Repeated-pass comparison for the seventeenth-run material-thinness probe. Material memory gives the best soft route-selection balance, while the base model remains the only setting that yields a visible hard path on the held-out sample.

but no thresholded recovery at any step count. The material-memory model is the strongest of the non-thin variants: it reaches route ratio 0.9212 at `steps_8`, keeps route ratio above 0.89 even at `steps_16`, and reduces obstacle overlap to 0.0042. The material-thin model confirms the central trade-off. At `steps_4` it recovers a small amount of hard structure—hard IoU 0.0539 and connectivity 0.0957—but this comes with high obstacle overlap (0.3311), and the signal disappears again as the number of steps increases.

Setting	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.
steps_4 base	0.1600	0.8593	0.3476	0.8500	0.1643	0.0049
steps_16 base	0.0000	0.8258	0.0084	1.0000	0.0131	0.0000
steps_4 additive	0.0000	0.8595	0.1753	1.0000	0.1248	0.0000
steps_16 additive	0.0000	0.8486	0.0056	1.0000	0.0111	0.0000
steps_8 material	0.0000	0.9212	0.0247	1.0000	0.0308	0.0000
steps_12 material	0.0000	0.9176	0.0088	1.0000	0.0150	0.0000
steps_16 material	0.0000	0.8907	0.0042	1.0000	0.0091	0.0000
steps_4 thin	0.0539	0.9083	0.3311	0.8750	0.1644	0.0957
steps_8 thin	0.0000	0.8529	0.0804	1.0000	0.0696	0.0000
steps_16 thin	0.0000	0.7985	0.0088	1.0000	0.0148	0.0000

Table 21: Representative step-sweep settings from the seventeenth run. Material memory gives the strongest soft route preference and lowest obstacle overlap, while thinness only briefly recovers hard structure in the shallow four-step setting.

Q.3 What We Learn from the Seventeenth Run

This run gives a cleaner answer than the earlier exploratory memory sweeps. The material-memory version is the best at the thing the probe most plausibly wants the model to learn: it achieves the strongest route preference among the memory variants while also giving the lowest obstacle overlap among the non-thin models. The base model is still the only variant that sometimes produces a visible hard path, but it does so by lighting up the obstacle too aggressively. In other words, the base model looks clearer, but the material model is reasoning better.

The most important trade-off is now explicit. The memory variants are becoming *more correct but less loud*.

They learn the route logic more cleanly than the base model does, yet they do not materialize that logic into a solid thresholded path. The material-thin probe partially confirms this diagnosis rather than resolving it. Thinness pressure can briefly recover some hard structure at low step count, but it does not survive deeper solver schedules, and the gained visibility is bought by reintroducing substantial obstacle overlap.

That makes the next move much clearer than before. The most informative continuation is not a broader random sweep. It is a tighter material-only study centered on three things: solver depth, thinness pressure, and threshold choice. If the material-memory field already contains the right route information in a faint form, then a threshold sweep may reveal that some of the apparent failure is due to using a cutoff that is too strict for the current dynamics. Conversely, if no reasonable threshold produces a usable corridor, then the main missing ingredient is still an objective that explicitly rewards connected visible paths.

Q.4 Recommended Next Changes After Run Seventeen

The next memory experiments should now be highly targeted:

- keep the **material-memory** probe as the main memory variant, since it has the best combination of route ratio and obstacle suppression;
- keep the **step sweep**, especially 8, 12, and 16, because those settings expose the tension between route correctness and thresholded visibility;
- add a small **thinness or skeleton pressure** to the core field, but evaluate it together with obstacle overlap so that extra visibility does not simply come from spreading onto the obstacle;
- run a **threshold sweep** on the same saved outputs, since the current 0.5 cutoff is probably hiding some useful route structure;
- frame the next probe explicitly as: can material memory become a visible path once the solver has enough steps and the core is pushed to stay thin?

Q.5 Qualitative Figures from the Seventeenth Run

Figures 27, 28, and 29 show the latest thinness-probe outputs. The visual pattern matches the quantitative one. The base model produces the strongest visible activation but spills heavily into the obstacle region. The material-memory model is cleaner and more selective, but too faint to survive a hard threshold. The material-thin model is thinner still, yet the extra sparsity mostly suppresses the signal rather than stabilizing a usable path.

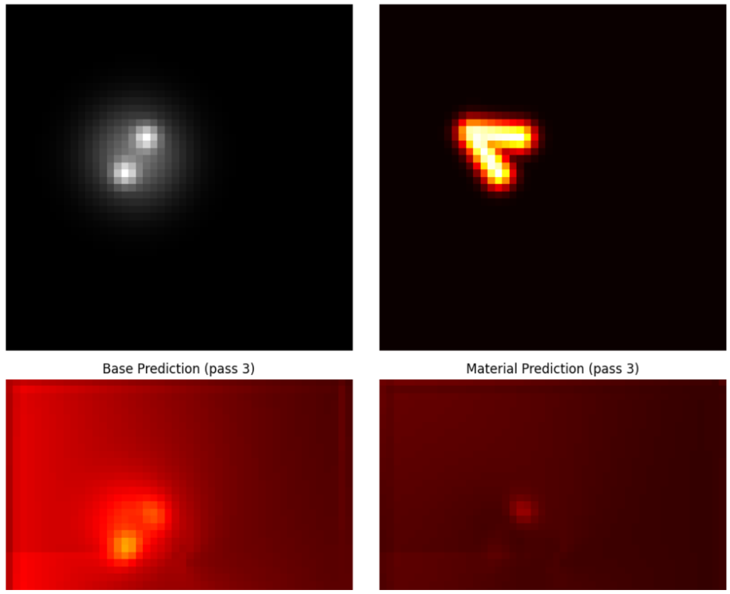

Figure 27: Seventeenth prototype qualitative result for the base split-core probe in the material-thinness setup. The path is more visibly activated than in the memory models, but it remains overly diffuse and overlaps the obstacle heavily.

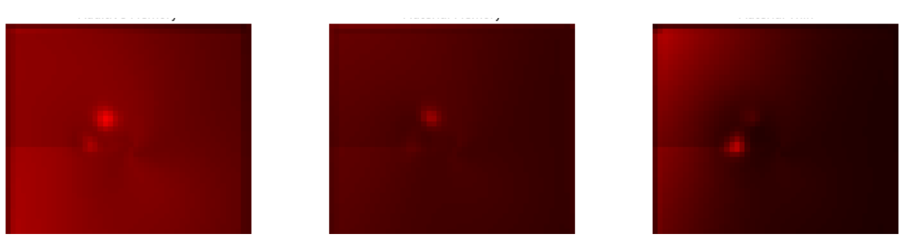

Figure 28: Seventeenth prototype qualitative result for the material-memory probe. The activation is cleaner and more obstacle-avoiding than the base model, supporting the interpretation that material memory improves route logic even when the thresholded path remains weak.

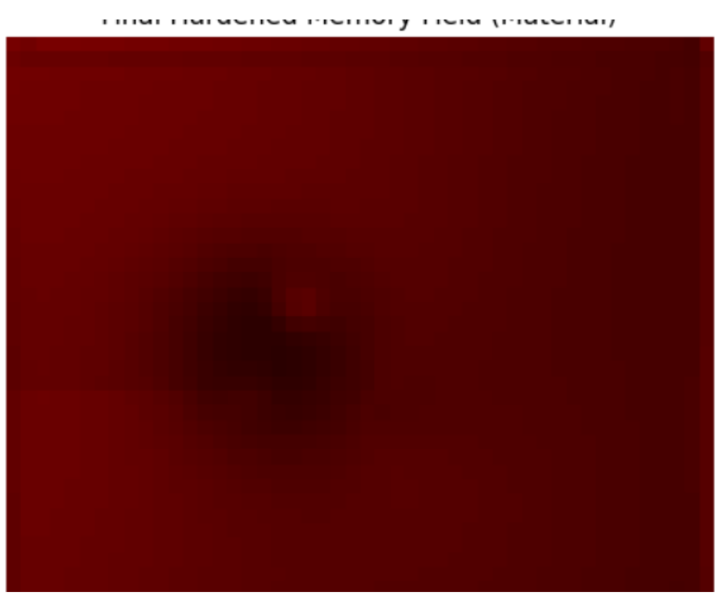

Figure 29: Seventeenth prototype qualitative result for the material-thin probe. Thinness pressure produces a sparser field, but the route becomes too faint to count as a robust visible corridor under the current threshold.

R Prototype Appendix: Eighteenth Run Material-Only Threshold Sweep

This eighteenth appendix entry adds the next targeted experiment suggested by the seventeenth run: a *material-only threshold sweep*. The previous appendix showed that the material-thin probe could produce a faint route field that looked plausibly meaningful, but most of that structure disappeared under the default 0.5 threshold. The new question is therefore extremely specific: is the model failing because the route is genuinely absent, or because the threshold is too strict for the current dynamics?

R.1 What We Changed

Relative to the seventeenth run, this probe narrows the design further:

- we trained only a **material-thin** probe, rather than comparing multiple model families at once;
- the main evaluation again used repeated passes on a single held-out sample, so we could inspect how hardened memory affects a fixed example;
- we ran a step sweep over 4, 8, 12, and 16 steps and then selected the best step by route ratio;
- on that best-step output, we ran a **threshold sweep** from 0.10 to 0.50 to test whether a lower cutoff reveals useful path structure;
- new qualitative figures were saved for the latest appendix update, including the best-step prediction, the hardened memory field, and the best-step core field.

The scientific purpose of this run was to test whether the material-thin probe contains a usable corridor in soft form even when the default threshold says it does not.

R.2 What We Got

The repeated-pass result is mostly negative. By the final pass, the material-thin probe achieved soft IoU 0.0164, hard IoU 0.0000, route ratio 0.6728, obstacle overlap 0.0498, route gap 1.0000, sharpness 0.0555, connectivity 0.0000, and steps used 8.0000. Across passes, route ratio drifted slightly downward from 0.6773 to 0.6728, while obstacle overlap drifted slightly upward from 0.0487 to 0.0498. So the hardened memory does not rescue the route on this sample; if anything, the signal remains faint and slightly degrades over repeated passes.

Table 22 summarizes the repeated-pass result. The important point is that the repeated-pass story is now stable but weak. The field remains sparse, obstacle overlap stays low, and the solver consistently uses eight steps, but the route never becomes visible enough to survive the default threshold. This means memory persistence alone is not sufficient to recover a usable corridor in the material-thin setting.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
pass 1	0.0158	0.0000	0.6773	0.0487	1.0000	0.0558	0.0000	8.0000
pass 2	0.0161	0.0000	0.6741	0.0494	1.0000	0.0556	0.0000	8.0000
pass 3	0.0164	0.0000	0.6728	0.0498	1.0000	0.0555	0.0000	8.0000

Table 22: Repeated-pass comparison for the eighteenth-run material-only threshold probe. The material-thin model stays sparse and obstacle-avoiding across passes, but repeated hardening does not recover a visible hard path at the default threshold.

The step sweep is also informative. The best route ratio occurs at `steps_4`, with route ratio 0.7491, soft IoU 0.0279, obstacle overlap 0.2160, and hard IoU still equal to 0.0000 at the default threshold. Increasing the number of steps steadily lowers obstacle overlap, from 0.2160 at four steps to 0.0060 at sixteen steps, but it also steadily lowers route ratio, from 0.7491 to 0.6304. So in this run, deeper solver schedules make the field cleaner but not smarter. The model becomes quieter without becoming more correct.

The threshold sweep provides the central new result. When the best-step output is evaluated at lower thresholds, the apparently absent route is partly recovered. Hard IoU rises from 0.0000 at threshold 0.50 to 0.0955 at threshold 0.30, peaks at 0.2727 at threshold 0.35, and then declines again for larger thresholds. This is the strongest direct evidence so far that the material-thin probe is storing some useful route structure in a faint latent form. At the same time, the cost is obvious: the best threshold by hard IoU still has obstacle overlap 0.2160 and connectivity only 0.0062, so lowering the cutoff reveals more path mass but does not yet create a robust connected corridor.

Setting	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.
steps_4	0.0000	0.7491	0.2160	1.0000	0.1506	0.0000
steps_8	0.0000	0.6773	0.0487	1.0000	0.0558	0.0000
steps_12	0.0000	0.6550	0.0146	1.0000	0.0236	0.0000
steps_16	0.0000	0.6304	0.0060	1.0000	0.0124	0.0000
th_0.25	0.0486	0.7491	0.2160	0.5250	0.1506	0.4402
th_0.30	0.0955	0.7491	0.2160	0.7000	0.1506	0.1281
th_0.35	0.2727	0.7491	0.2160	0.7750	0.1506	0.0062
th_0.40	0.1364	0.7491	0.2160	0.8750	0.1506	0.0027
th_0.50	0.0000	0.7491	0.2160	1.0000	0.1506	0.0000

Table 23: Representative step and threshold sweep settings from the eighteenth run. Four solver steps give the strongest route ratio, and a threshold near 0.35 recovers the best hard IoU, but the revealed path is still weakly connected and overlaps the obstacle substantially.

R.3 What We Learn from the Eighteenth Run

This run sharpens the interpretation of the threshold issue. The material-thin probe is not completely failing to encode the route. Instead, it appears to encode a *faint* version of the route that is largely invisible at threshold 0.5 and only partially recoverable at lower thresholds. That is an important conceptual distinction. The problem is no longer simply "no path." It is now "path present, but too weak and too entangled with the obstacle to be robust."

The threshold sweep therefore supports a more precise version of the earlier claim that the model is becoming more correct but less loud. Lowering the threshold does reveal additional structure, and the jump from hard IoU 0.0000 at 0.50 to 0.2727 at 0.35 is too large to dismiss as noise. But this recovery is still incomplete. The best threshold does not produce strong connectivity, and the obstacle overlap remains high because the revealed mass is not selective enough. So threshold tuning can expose some latent competence, but it does not by itself solve the visibility problem.

The broader implication is that the next improvement should not rely on thresholding alone. The model likely needs a training objective that explicitly rewards both *connectedness* and *selective visibility*. Otherwise, lower thresholds will continue to recover obstacle-heavy blobs instead of clean paths.

R.4 Recommended Next Changes After Run Eighteen

The next step is now even more focused:

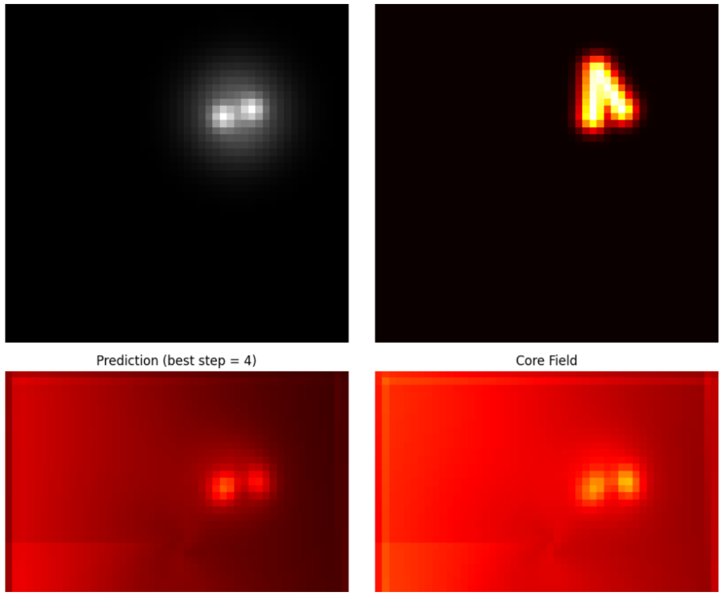

Figure 30: Eighteenth prototype qualitative result for the best-step material-thin prediction. The predicted route is faint but not absent, consistent with the threshold-sweep finding that lower cutoffs can recover some hidden structure.

- keep the **material-only** probe, since the present result is about how to convert a faint material-memory field into a visible path rather than about comparing architectures;
- retain the **step sweep**, but center future analysis near the best route-ratio regime around four to eight steps;
- include a **threshold sweep** by default in future appendix probes, because the default threshold alone is now clearly misleading about how much structure is present;
- add an explicit **connectivity-promoting or corridor-continuity objective**, so that threshold recovery reveals a path rather than a diffuse obstacle-overlapping region;
- treat threshold tuning as a diagnostic rather than as the final fix.

R.5 Qualitative Figures from the Eighteenth Run

Figures 30, 31, and 32 show the latest material-only threshold-sweep outputs. The best-step prediction is visibly faint but not empty. The hardened memory field preserves a structured trace of previous activation, and the core field confirms that the model concentrates some mass along the intended route. The figures therefore support the quantitative interpretation: useful route information is present, but it is too weak to count as a robust path under the current objective and threshold.

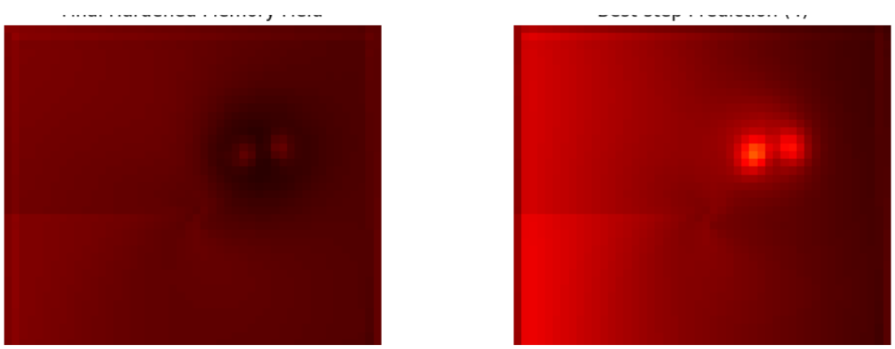

Figure 31: Eighteenth prototype qualitative result for the final hardened memory field. The memory trace remains structured and obstacle-aware, but it does not by itself enforce a clean visible corridor.

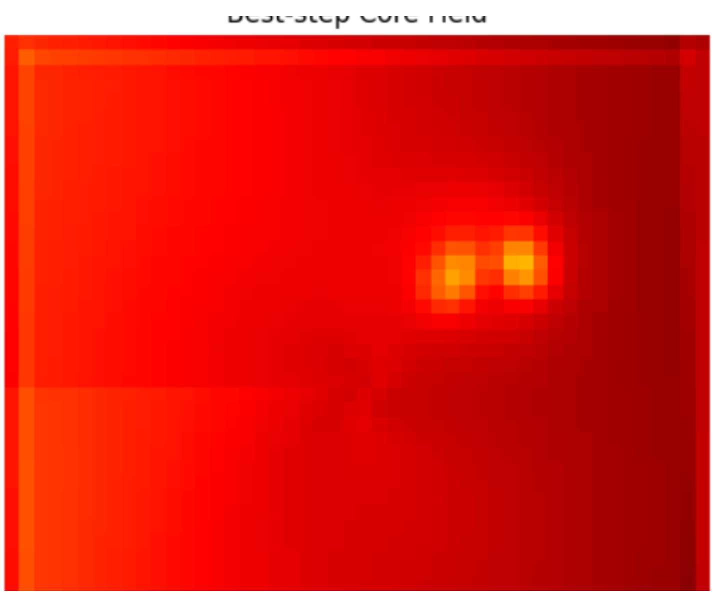

Figure 32: Eighteenth prototype qualitative result for the best-step core field. The core carries route information in a narrow latent form, but the field is still too weak to yield a robust connected path after thresholding.

S Prototype Appendix: Nineteenth Run Material-Only Boosted Inference Probe

This nineteenth appendix entry extends the eighteenth run with a more targeted inference-time diagnostic. The underlying question is no longer just whether lowering the threshold exposes hidden route structure. It is now whether a small amount of *inference-time boosting* can make that latent structure easier to read without changing the underlying material-only architecture. Concretely, this probe adds a temperature sweep on top of the earlier step and threshold sweeps.

S.1 What We Changed

Relative to the eighteenth run, the core training setup remained intentionally narrow, but the evaluation was expanded:

- we again trained only the **material-thin** probe, preserving the same emphasis on sparse material memory rather than broader architectural comparison;
- we retained the **repeated-pass analysis** on one held-out sample to check whether hardened memory can accumulate into a more visible corridor;
- we repeated the **step sweep** over 4, 8, 12, and 16 steps;
- after selecting the best step by route ratio, we added a new **temperature sweep** over 0.50, 0.65, 0.80, 1.00, and 1.20 to test whether the same latent field becomes more readable under sharper inference;
- we then ran the **threshold sweep** on that best-step output and saved new qualitative figures for the boosted prediction, the final hardened memory field, and the best-step core field.

The design logic is straightforward: if the path is already present but too faint, then temperature scaling and thresholding should act as diagnostic tools for visibility rather than as substitutes for learning.

S.2 What We Got

The repeated-pass story stays weak but stable. Across three passes, soft IoU moves only from 0.0158 to 0.0164, hard IoU remains 0.0000, route ratio drifts slightly downward from 0.6773 to 0.6728, obstacle overlap drifts slightly upward from 0.0487 to 0.0498, route gap remains 1.0000, connectivity stays 0.0000, and the solver again uses exactly eight steps. So hardened memory still does not by itself turn the latent route into a visible connected corridor.

The step sweep shows the same tradeoff seen in the eighteenth run, but now in sharper relief. Four steps give the strongest route ratio, 0.7491, but also the highest obstacle overlap, 0.2160. Increasing the number of steps steadily lowers obstacle overlap, down to 0.0060 at sixteen steps, yet also lowers route ratio to 0.6304. The field therefore becomes cleaner as the solver runs longer, but not more task-aligned in a thresholded sense.

The new temperature sweep gives a useful diagnostic refinement. At the best step count of four, lowering the temperature from 1.00 to 0.50 improves hard IoU from 0.0000 to 0.0227 and lowers obstacle overlap from 0.2160 to 0.1439. At the same time, route ratio falls from 0.7491 to 0.6084. In other words, sharper inference helps make some route mass visible, but it does so by trading off part of the bent-route preference that made the best-step solution interesting in the first place.

The threshold sweep remains the strongest result. At threshold 0.50, the best-step output still appears empty, with hard IoU 0.0000, route gap 1.0000, and connectivity 0.0000. But lowering the threshold reveals that the path is not absent. Hard IoU rises to 0.0955 at 0.30, peaks at 0.2727 at 0.35, and then declines again beyond that point. This is the clearest evidence so far that the material-thin probe now contains a usable latent path signal, but that signal remains soft, sparse, and only partially separable from the obstacle.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
pass 1	0.0158	0.0000	0.6773	0.0487	1.0000	0.0558	0.0000	8.0000
pass 2	0.0161	0.0000	0.6741	0.0494	1.0000	0.0556	0.0000	8.0000
pass 3	0.0164	0.0000	0.6728	0.0498	1.0000	0.0555	0.0000	8.0000

Table 24: Repeated-pass comparison for the nineteenth-run boosted-inference probe. Repeated hardening keeps the field stable and sparse, but does not recover a visible hard path at the default threshold.

Setting	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.
steps_4	0.0000	0.7491	0.2160	1.0000	0.1506	0.0000
steps_8	0.0000	0.6773	0.0487	1.0000	0.0558	0.0000
steps_12	0.0000	0.6550	0.0146	1.0000	0.0236	0.0000
steps_16	0.0000	0.6304	0.0060	1.0000	0.0124	0.0000
temp_0.50	0.0227	0.6084	0.1439	0.9250	0.0989	0.0000
temp_0.65	0.0227	0.6664	0.1749	0.9250	0.1215	0.0000
temp_0.80	0.0227	0.7089	0.1968	0.9500	0.1370	0.0000
temp_1.00	0.0000	0.7491	0.2160	1.0000	0.1506	0.0000
temp_1.20	0.0000	0.7777	0.2280	1.0000	0.1594	0.0000
th_0.25	0.0486	0.7491	0.2160	0.5250	0.1506	0.4402
th_0.30	0.0955	0.7491	0.2160	0.7000	0.1506	0.1281
th_0.35	0.2727	0.7491	0.2160	0.7750	0.1506	0.0062
th_0.40	0.1364	0.7491	0.2160	0.8750	0.1506	0.0027
th_0.50	0.0000	0.7491	0.2160	1.0000	0.1506	0.0000

Table 25: Representative step, temperature, and threshold settings from the nineteenth run. Temperature scaling makes some path mass visible, but thresholding near 0.35 still gives the strongest hard-IoU recovery.

S.3 What We Learn from the Nineteenth Run

The main conclusion is that the model is no longer behaving randomly. It is learning the *right route logic* in a soft sense, especially in the way it suppresses mass in the obstacle and preserves a nontrivial bent-route preference. But that route is still too faint and too broken to survive the default binary cutoff. The model contains a real hidden path signal, yet it does not reliably output a solid fracture-like corridor.

This matters because it sharpens the interpretation of the earlier threshold result. The issue is not simply "no path." It is better described as "path present, but under-amplified and insufficiently connected." The jump from hard IoU 0.0000 at threshold 0.50 to 0.2727 at threshold 0.35 is too large to dismiss as noise. At the same time, the best thresholded case still has obstacle overlap 0.2160 and connectivity only 0.0062. So the recovered route is meaningful, but not yet robust.

The temperature sweep helps separate two kinds of improvement. Sharper inference at temperature 0.50 produces a small gain in visible hard IoU and better obstacle avoidance, but it also weakens route ratio. Threshold reduction, by contrast, reveals more of the original route-aware field without changing the field itself. That means temperature scaling is a useful inference-time probe, but the central bottleneck still lies in training the model to produce a path that is simultaneously visible, selective, and connected.

S.4 Recommended Next Changes After Run Nineteen

The next experimental changes are now more specific:

- keep the **material-only** setting, because the central issue is still path visibility within the fracture-style substrate rather than architecture selection;
- preserve the **step sweep**, but focus future analysis near the 4–8 step regime where route preference is strongest;
- retain both **temperature** and **threshold sweeps** as standard diagnostics, since each reveals a different aspect of hidden route structure;
- add an explicit **connectivity-promoting or corridor-continuity loss**, because current threshold

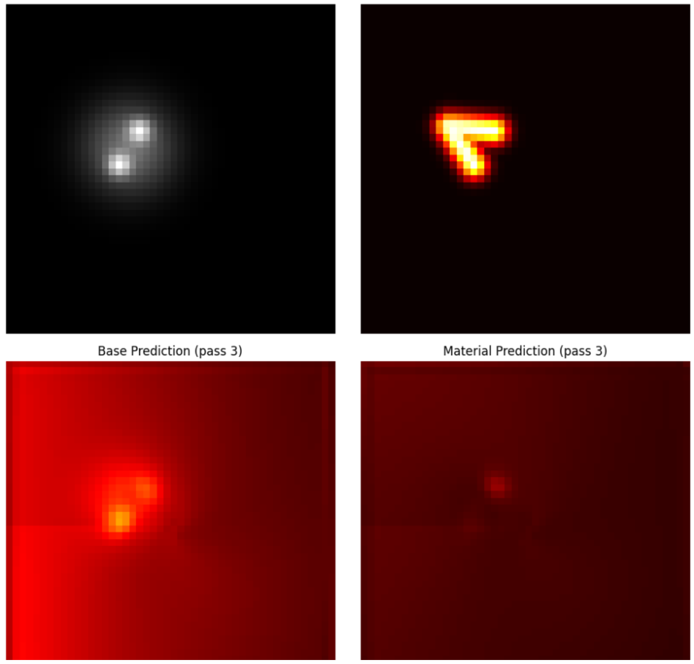

Figure 33: Nineteenth prototype qualitative summary for the boosted-inference probe. The panel shows the held-out input, target route, best-step prediction, and best-step core field. The prediction remains sparse and partially visible rather than fully absent.

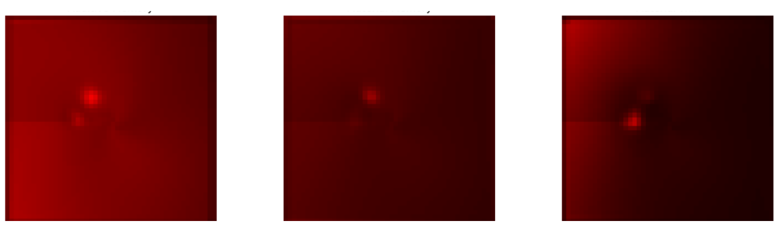

Figure 34: Nineteenth prototype detail view of the best-step probability and core fields. Both retain route-relevant structure, but neither yet produces a clean connected corridor after default thresholding.

recovery exposes blobs and fragments more readily than a clean corridor;

- add a stronger **selective-visibility objective**, so that useful route mass is amplified without also lighting up the obstacle.

S.5 Qualitative Figures from the Nineteenth Run

Figures 33, 34, and 35 show the qualitative outputs saved from the boosted-inference probe. The summary panel confirms that the best-step prediction is faint but structured, the detail panel shows that the best-step probability and core fields carry localized route information, and the hardened-memory panel shows that memory remains organized without by itself forcing a clean corridor. Together with the quantitative sweeps, these figures support the same interpretation: latent route information is present, but it is still too soft and too weakly connected to count as a robust visible path under the default objective.

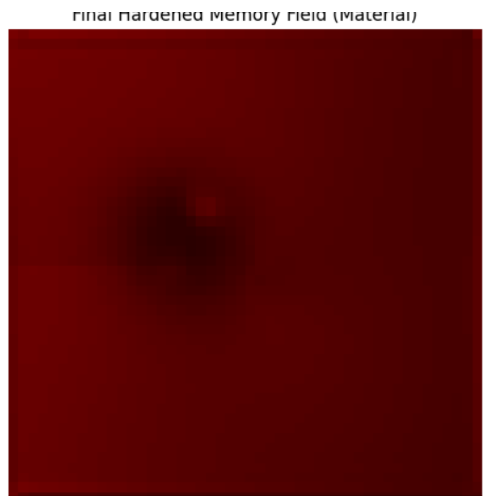

Figure 35: Nineteenth prototype qualitative result for the final hardened memory field. The memory trace remains smooth and structured, but repeated hardening still does not by itself produce a visible corridor at the default threshold.

T Prototype Appendix: Twentieth Run Focused Batch Probe

This twentieth appendix entry follows the boosted-inference result with a more focused held-out batch evaluation. The probe keeps the split-core material-memory design, retains thinness and centerline supervision on the core field, and evaluates the same trained model across multiple step counts and thresholds on a batch of 32 held-out samples. The goal is to test whether the threshold effect observed on single examples persists across a small evaluation set.

T.1 What We Changed

Relative to the nineteenth run, the main change is that the evaluation is no longer restricted to one held-out sample. The focused follow-up uses:

- a **frozen split-core material-memory model**, so the comparison isolates inference behavior rather than architecture changes;
- explicit **thinness and centerline supervision** on the core field, intended to push the model toward a sharper route trace;
- a **step sweep** over 4, 8, 12, and 16 solver steps;
- a compact **threshold sweep** at 0.25, 0.35, and 0.50 for every step setting;
- held-out batch evaluation with 32 samples, rather than judging only a single qualitative case.

This makes the result a more conservative test of the earlier claim that useful route information exists but remains under-amplified.

T.2 What We Got

The training trace remains similar to the previous material-thin runs. At epoch 20, the model reports soft IoU 0.0298, hard IoU 0.0000, route ratio 0.8221, obstacle overlap 0.0514, and eight solver steps. Thus, during training, the model continues to form a soft path-like field but does not yet produce a reliable hard path at the default threshold.

On the held-out batch, the step sweep again shows a visibility–safety tradeoff. At four steps, the model obtains the highest hard IoU, 0.0086, and soft IoU, 0.0418, but obstacle overlap is still high at 0.2174. At eight steps, obstacle overlap falls to 0.0474 while hard IoU drops to 0.0000. By sixteen steps, obstacle overlap is only 0.0058, but the field is too faint to survive thresholding, with hard IoU 0.0000 and connectivity 0.0000. In short, shallow inference is more visible but sloppier, while deeper inference is safer but ghost-like.

The threshold sweep provides the main result. At step 4, lowering the threshold from 0.50 to 0.35 raises hard IoU from 0.0086 to 0.0562 and increases connectivity from 0.0002 to 0.0280. Lowering the threshold further to 0.25 gives the strongest batch-level hard IoU, 0.0687, and much stronger connectivity, 0.2922, while route gap improves from 0.9914 at threshold 0.50 to 0.6133 at threshold 0.25. This confirms the earlier qualitative diagnosis: the route signal is present, but it is still too soft to materialize at the standard cutoff.

At later steps, however, even lower thresholds do not recover much structure. For example, at step 8, threshold 0.25 yields hard IoU only 0.0018, and at steps 12 and 16 all tested thresholds remain at hard IoU 0.0000. This suggests that longer relaxation suppresses obstacle activation, but also suppresses the useful path signal below practical visibility.

The best row at threshold 0.35 is therefore `steps_4_th_0.35`, with hard IoU 0.0562, soft IoU 0.0418, route ratio 0.8903, obstacle overlap 0.2174, route gap 0.8977, and connectivity 0.0280. If the threshold is allowed to vary at the best step, the best row becomes `steps_4_th_0.25`, with hard IoU 0.0687, route gap 0.6133, and connectivity 0.2922. This is the most explicit evidence in the batch run that lower cutoffs recover latent structure, but only by accepting a low operating threshold and substantial obstacle overlap.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
steps_4	0.0418	0.0086	0.8903	0.2174	0.9914	0.1336	0.0002	4.0000
steps_8	0.0249	0.0000	0.9631	0.0474	1.0000	0.0483	0.0000	8.0000
steps_12	0.0132	0.0000	1.0461	0.0138	1.0000	0.0206	0.0000	12.0000
steps_16	0.0078	0.0000	1.0734	0.0058	1.0000	0.0113	0.0000	16.0000

Table 26: Held-out batch step sweep for the twentieth focused probe. Four steps give the most visible prediction, while deeper schedules reduce obstacle overlap but make the route too faint to survive default thresholding.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Conn.	Threshold
steps_4_th_0.25	0.0418	0.0687	0.8903	0.2174	0.6133	0.2922	0.2500
steps_4_th_0.35	0.0418	0.0562	0.8903	0.2174	0.8977	0.0280	0.3500
steps_4_th_0.50	0.0418	0.0086	0.8903	0.2174	0.9914	0.0002	0.5000
steps_8_th_0.25	0.0249	0.0018	0.9631	0.0474	0.9969	0.0000	0.2500
steps_8_th_0.35	0.0249	0.0000	0.9631	0.0474	1.0000	0.0000	0.3500
steps_8_th_0.50	0.0249	0.0000	0.9631	0.0474	1.0000	0.0000	0.5000
steps_12_th_0.25	0.0132	0.0000	1.0461	0.0138	1.0000	0.0000	0.2500
steps_12_th_0.35	0.0132	0.0000	1.0461	0.0138	1.0000	0.0000	0.3500
steps_12_th_0.50	0.0132	0.0000	1.0461	0.0138	1.0000	0.0000	0.5000
steps_16_th_0.25	0.0078	0.0000	1.0734	0.0058	1.0000	0.0000	0.2500
steps_16_th_0.35	0.0078	0.0000	1.0734	0.0058	1.0000	0.0000	0.3500
steps_16_th_0.50	0.0078	0.0000	1.0734	0.0058	1.0000	0.0000	0.5000

Table 27: Threshold sweep settings from the twentieth focused probe. The best batch-level recovery occurs at four solver steps and threshold 0.25, confirming that the route exists mostly as a weak sub-threshold signal.

T.3 What We Learn from the Twentieth Run

This run says that the model is starting to produce a usable shape, but it is not yet reliably solving the route problem. The material-thin setup is clearly changing the output and is not random: it learns where route mass should approximately appear and can suppress obstacle activation under deeper relaxation. However, it still fails to commit strongly enough to a connected corridor at the normal 0.50 threshold.

The practical interpretation is that the model now has *route logic*, but not yet full *materialization.* At lower thresholds, especially 0.25, the path-like signal becomes visible and connectivity improves. At the default threshold, the same signal largely disappears. The model therefore knows the direction of the solution, but it still whispers instead of drawing a firm fracture line.

The step sweep also clarifies the failure mode. Four steps preserve enough signal to recover a partial path after threshold lowering, but obstacle overlap remains high. Twelve or sixteen steps make the prediction safer around the obstacle, but also suppress the useful path mass until the output becomes ghost-like. The next improvement should therefore strengthen core commitment while preserving the obstacle avoidance gained from the material dynamics.

T.4 Recommended Next Changes After Run Twenty

The next useful move is no longer broad sweeping. The focused target should be to strengthen the route core without losing obstacle selectivity:

- increase **core commitment pressure** so that route mass rises above the default threshold;
- keep **material obstacle suppression**, since deeper schedules show that this mechanism is useful;
- add a direct **connected-corridor objective**, because threshold recovery still leaves broken fragments;
- treat the 0.25 threshold result as a **diagnostic**, not as the final operating point;
- evaluate future changes on held-out batches, since the batch result is more conservative than single-sample threshold recovery.

T.5 Qualitative Figures from the Twentieth Run

Figures 36 and 37 show the held-out visualization from the focused batch probe. The summary figure shows that the prediction and core field carry localized route structure, and that thresholding at 0.35 recovers a small visible fragment. The separate 0.50 threshold figure shows the central problem directly: under the standard cutoff, the prediction is effectively empty.

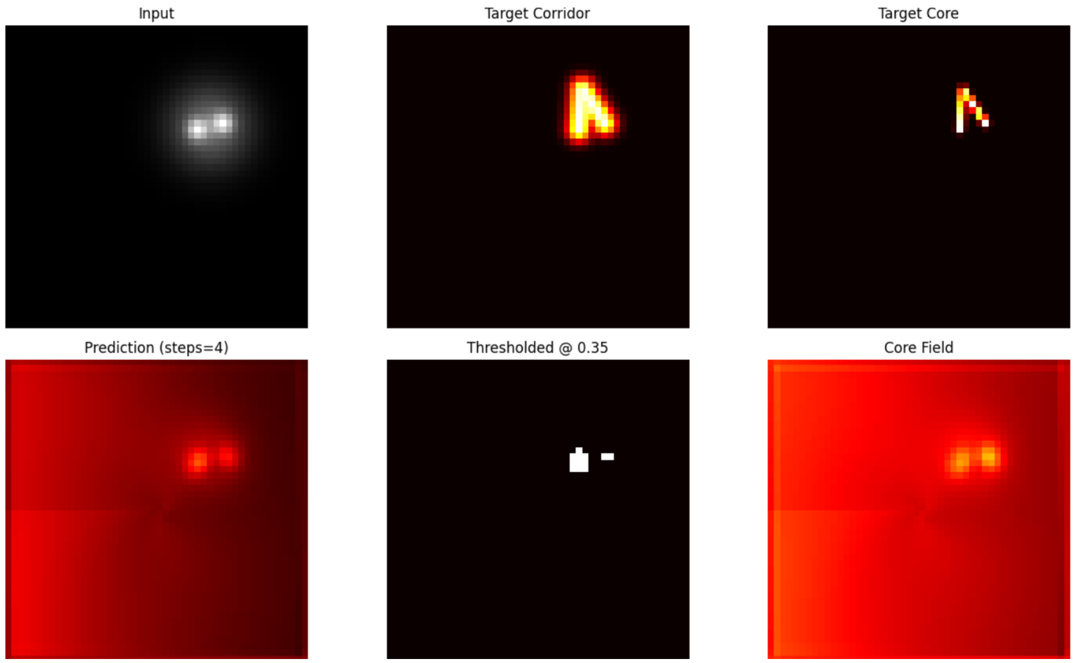

Figure 36: Twentieth prototype qualitative summary for the focused held-out batch probe. The prediction and core field show localized route-like structure, while the 0.35 threshold recovers only a small disconnected fragment.

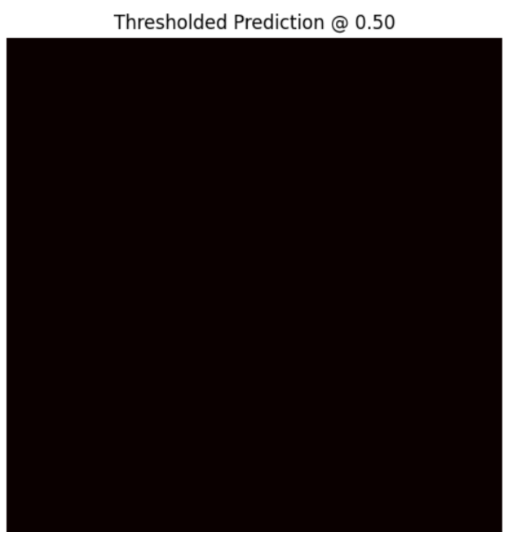

Figure 37: Twentieth prototype thresholded prediction at 0.50. The near-empty image illustrates that the current route signal remains below the standard binary commitment threshold.

U Prototype Appendix: Twenty-First Run Geodesic Flow Probe

This twenty-first appendix entry tests a different but closely related formulation: a *geodesic-flow* probe. Instead of using the earlier fracture core–spread readout, this experiment trains a potential field ϕ and a wave field whose gradients define an extracted path. The purpose is to ask whether a more explicitly potential-field-like substrate can solve the routing geometry even if the visible probability map remains weak.

U.1 What We Changed

The new probe replaces the split-core material-memory architecture with a compact geodesic-flow network:

- the input uses four channels: source, goal, source–goal charge, and obstacle field;
- a learned encoder predicts an initial potential field ϕ and wave field;
- repeated flow updates combine diffusion, source–goal forcing, obstacle repulsion, and gradient penalties;
- the path is extracted directly by descending the learned potential field from the start point;
- training uses binary cross entropy, smoothness, obstacle penalty, and an eikonal-style regularizer on $\|\nabla\phi\|$.

This makes the run a useful contrast case: it separates *latent route logic* in the potential field from *visible path materialization* in the probability map.

U.2 What We Got

The training loss decreases steadily, from 1.2266 at epoch 5 to 0.2324 at epoch 40. However, hard IoU collapses over the same period: it begins at 0.0272 at epoch 5, falls to 0.0070 by epoch 20, and reaches 0.0000 from epoch 30 onward. So the optimization objective is improving, but the thresholded path prediction is not.

Epoch	Loss	Hard IoU
5	1.2266	0.0272
10	1.0413	0.0231
15	0.8451	0.0223
20	0.6670	0.0070
25	0.5043	0.0015
30	0.3771	0.0000
35	0.2906	0.0000
40	0.2324	0.0000

Table 28: Training trace for the twenty-first geodesic-flow probe. Loss decreases monotonically, but hard IoU collapses, showing that the visible probability map is not preserving the route.

The qualitative result explains the mismatch. Figure 38 shows that the predicted probability concentrates mostly near the endpoints and around broad background activation rather than forming a clean corridor. The potential field, however, contains a coherent gradient structure, and the extracted path bends around the obstacle. This means the geodesic logic is partly present in ϕ, but the readout does not materialize that logic as a usable binary path.

U.3 What We Learn from the Twenty-First Run

This run is weaker than the best fracture-style probes on the primary hard-IoU metric. The best fracture-threshold runs recovered hard IoU values around 0.06 on the held-out batch and up to 0.2727 on a favorable single-sample threshold sweep, whereas this geodesic-flow run ends at hard IoU 0.0000. So the geodesic formulation should not replace the fracture core–spread model yet.

At the same time, the run is scientifically useful because it exposes the same failure mode in a cleaner form. The potential field behaves like a partial logic engine: it stores a route-aware geometry and can yield an obstacle-aware extracted path. The probability map behaves like a failing visualizer: it does not report that route as a continuous visible corridor. This is another version of the same "vanishing signal" problem seen in the material-thin experiments.

The likely cause is class imbalance and weak path amplification. Since the route occupies a very small fraction of the image, binary cross entropy can reduce loss by suppressing most pixels, especially when the wave field is too weak to overcome the negative obstacle and gradient terms in the final sigmoid readout. The model learns something useful in the potential field, but the visible path is squashed before it becomes measurable.

U.4 Recommended Next Changes After Run Twenty-One

The geodesic-flow direction should not be abandoned, but it should be treated as a diagnostic branch rather than the main model until the readout improves:

- compare the geodesic and fracture models directly on the same held-out batches;
- replace plain BCE with **Dice** or **soft-IoU** loss to reduce thin-line class imbalance;
- normalize or sharpen the wave readout before the sigmoid, so the strongest path signal remains visible;
- add explicit **flow/path supervision** that rewards propagation along the extracted geodesic, not only endpoint activation;
- consider a hybrid model where the geodesic potential proposes a route and the fracture core–spread mechanism hardens it into a visible corridor.

U.5 Qualitative Figure from the Twenty-First Run

Figure 38 summarizes the geodesic-flow probe. The target route is clean and obstacle-avoiding, the predicted probability does not form the corridor, and the potential field contains the more interesting route-aware structure. The figure supports the main interpretation: the model has some latent path logic, but the probability veil is too weak or too poorly calibrated to expose it.

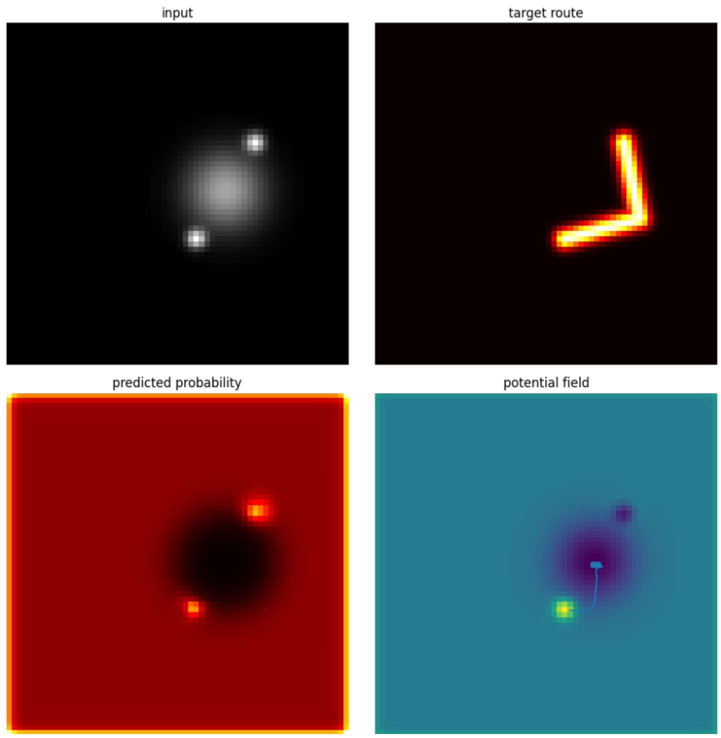

Figure 38: Twenty-first prototype qualitative result for the geodesic-flow probe. The potential field supports an obstacle-aware extracted path, but the probability map does not materialize a continuous route, yielding hard IoU near zero.

V Prototype Appendix: Twenty-Second Run Geodesic Flow v2

This twenty-second appendix entry tests a strengthened geodesic-flow variant. The goal was to repair the previous geodesic probe by adding route-point supervision, Dice loss, threshold sweeps, step sweeps, and a normalized wave–potential readout. In principle, these changes should prevent the all-zero probability collapse and make the path signal visible.

V.1 What We Changed

Relative to the first geodesic-flow run, the second version adds several explicit anti-collapse mechanisms:

- route-point supervision along the target polyline and nearby side offsets;
- Dice loss and positive-weighted BCE to reduce thin-route class imbalance;
- per-sample normalization of the wave and potential fields before the sigmoid readout;
- a mass-floor penalty to discourage an all-black probability map;
- step and threshold sweeps on a held-out batch of 32 samples.

This run therefore tests whether the geodesic formulation can be made visible simply by strengthening the readout and supervision losses.

V.2 What We Got

The result is mixed but mostly negative. The model no longer collapses to an empty prediction. Instead, it overcorrects into broad blob-like activation. During training, hard IoU improves relative to the first geodesic run but remains low: it reaches 0.0359 at epoch 30 and ends at 0.0297 at epoch 50. The route ratio ends at 1.2870, but obstacle overlap is very high at 0.7444.

Epoch	Loss	Soft	Hard	Ratio	ObsOverlap
1	4.6457	0.0147	0.0129	1.2390	0.0710
5	4.0476	0.0176	0.0145	1.2578	0.0855
10	2.7767	0.0270	0.0225	1.1899	0.2072
15	1.6191	0.0459	0.0336	0.9637	0.7631
20	1.5111	0.0476	0.0288	0.9584	0.8813
25	1.5350	0.0466	0.0292	0.9492	0.8682
30	1.5036	0.0565	0.0359	1.0294	0.7750
35	1.4210	0.0469	0.0276	1.2394	0.6805
40	1.3584	0.0468	0.0250	1.2500	0.7526
45	1.4909	0.0544	0.0330	1.1065	0.7767
50	1.4646	0.0532	0.0297	1.2870	0.7444

Table 29: Training trace for the twenty-second geodesic-flow probe. The strengthened losses prevent complete invisibility, but the model pays for this by activating broad obstacle-overlapping regions.

On the held-out batch, the best step by hard IoU is four steps, with hard IoU 0.1021, soft IoU 0.0565, route ratio 0.9499, obstacle overlap 0.9560, route gap 0.0281, and connectivity 0.5511. The best threshold at that step is 0.50, which raises hard IoU to 0.1445 and keeps route gap low at 0.0977, but obstacle overlap remains 0.9560. So the stronger readout makes the prediction visible, but not selective.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
steps_4	0.0565	0.1021	0.9499	0.9560	0.0281	0.2025	0.5511	4.0000
steps_8	0.0570	0.0844	0.9576	0.9544	0.0188	0.2022	0.8729	8.0000
steps_12	0.0548	0.0301	1.0531	0.8894	0.0000	0.2105	1.7875	12.0000
steps_16	0.0519	0.0292	1.2575	0.7471	0.0039	0.2196	1.8360	16.0000

Table 30: Held-out batch step sweep for the twenty-second geodesic-flow probe. Four steps give the best hard IoU, while longer steps increase route ratio and connectivity mostly by spreading broad activation.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Conn.	Threshold
step_4_th_0.25	0.0565	0.0274	0.9499	0.9560	0.0000	1.9639	0.2500
step_4_th_0.35	0.0565	0.1021	0.9499	0.9560	0.0281	0.5511	0.3500
step_4_th_0.50	0.0565	0.1445	0.9499	0.9560	0.0977	0.2895	0.5000
step_8_th_0.50	0.0570	0.1413	0.9576	0.9544	0.0852	0.3020	0.5000
step_12_th_0.50	0.0548	0.1386	1.0531	0.8894	0.0688	0.3176	0.5000
step_16_th_0.50	0.0519	0.1400	1.2575	0.7471	0.0648	0.3150	0.5000

Table 31: Representative threshold settings from the twenty-second geodesic-flow probe. Higher thresholds improve hard IoU by cutting the broad blob down, but the underlying obstacle overlap remains very high.

V.3 What We Learn from the Twenty-Second Run

This run improves the geodesic branch numerically, but not in the right way. Hard IoU rises above the previous geodesic run and even exceeds the twentieth material-thin batch result at the best threshold. However, the gain comes from a large circular activation blob rather than from a clean bent route. The obstacle overlap values, ranging from 0.7471 to 0.9560, make this failure mode explicit.

The qualitative output in Figure 39 makes the problem visible. The target is an L-shaped corridor, but the prediction is a large disk. Thresholding at 0.35 and 0.50 produces clean white circles, not a path. The potential-field trace also shoots mostly upward rather than following the desired bend. Thus, the v2 geodesic model solves the previous invisibility problem by becoming too loud and too diffuse.

The scientific takeaway is that the geodesic branch has shifted from *vanishing signal* to *blob collapse*. This is progress only in a narrow diagnostic sense: the readout can now produce mass, but it does not shape that mass into a corridor. Compared with the fracture core–spread model, the geodesic v2 probe is still less aligned with the desired materialized path behavior.

V.4 Recommended Next Changes After Run Twenty-Two

The next step should not be more generic geodesic training. It should either return to the fracture core–spread model or add much stronger path-shaping constraints to the geodesic branch:

- prioritize the **fracture + core/spread + material memory** model as the main experimental line;
- if continuing geodesic flow, add explicit anti-blob penalties such as area control, skeletonization pressure, and corridor-width constraints;
- supervise the potential gradient direction so extracted paths follow the desired bend rather than endpoint-centered radial flow;
- penalize obstacle overlap much more strongly, since the current geodesic readout lights up the obstacle region;
- test a hybrid design where geodesic flow proposes a coarse path but fracture dynamics harden only a thin centerline.

V.5 Qualitative Figure from the Twenty-Second Run

Figure 39 shows the main failure clearly. The prediction has become visible, but the visible object is a broad circular blob. The thresholded maps therefore improve hard IoU numerically without representing the target route geometry.

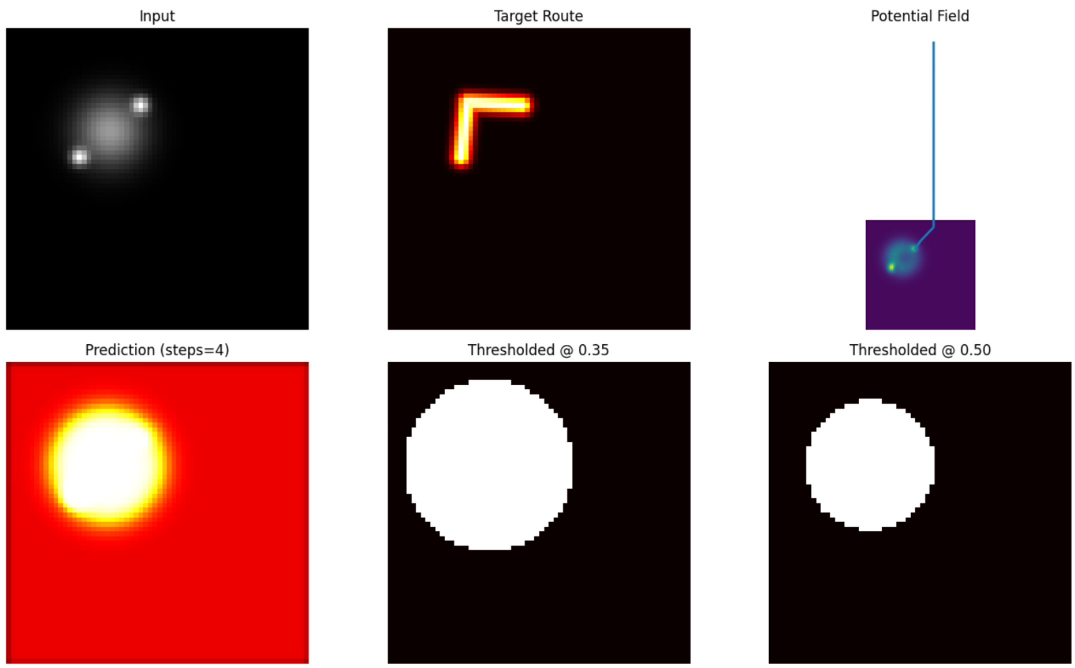

Figure 39: Twenty-second prototype qualitative result for geodesic-flow v2. Stronger supervision prevents an empty prediction, but the model collapses to a broad blob rather than a thin L-shaped path.

W Prototype Appendix: Twenty-Third Run Soft Shortest-Path Probe

This twenty-third appendix entry tests a more algorithmic routing baseline: a differentiable soft shortest-path model. Instead of learning a fracture field directly, the model learns a traversal-cost field, runs soft value iteration from the source and goal, and recovers occupancy from the summed soft distances. The motivation is to ask whether an explicit planning-like substrate can solve the detour problem more cleanly than the geodesic-flow probes.

W.1 What We Changed

The soft shortest-path probe differs from the geodesic-flow runs in several ways:

- the input includes source, goal, charge, obstacle, and coordinate channels;
- the network predicts a positive traversal-cost field rather than a direct path probability;
- soft value iteration propagates distances from both source and goal seeds;
- the predicted route is derived from normalized summed distances;
- the loss combines weighted BCE, Dice, obstacle penalty, smoothness, route-point supervision, endpoint pressure, and a mass floor.

This is a cleaner planning-style control experiment, but it still has to prove that the learned low-cost region becomes a thin obstacle-avoiding corridor rather than a broad basin.

W.2 What We Got

Training is stable but does not produce a clean route. Hard IoU fluctuates around 0.05–0.07 during training, peaking at 0.0685 at epoch 10 and ending at 0.0494 at epoch 35. Route ratio stays near 1.0, which means the model places probability along the desired route and direct shortcut at similar levels. Obstacle overlap remains consistently high, around 0.75–0.77, showing that the learned cost landscape is not sufficiently repelling the obstacle.

Epoch	Loss	Soft	Hard	Ratio	ObsOverlap
1	1.5686	0.0540	0.0526	0.9981	0.7545
5	1.5343	0.0525	0.0525	0.9969	0.7634
10	1.5835	0.0616	0.0685	0.9686	0.7669
15	1.5050	0.0505	0.0521	0.9997	0.7656
20	1.5467	0.0603	0.0579	0.9961	0.7713
25	1.5069	0.0575	0.0546	0.9979	0.7646
30	1.5088	0.0523	0.0531	0.9969	0.7621
35	1.4926	0.0510	0.0494	0.9992	0.7690

Table 32: Training trace for the twenty-third soft shortest-path probe. The model learns a stable shaped field, but obstacle overlap remains high throughout training.

On the held-out batch, the best step by hard IoU is 16 steps, with hard IoU 0.0797, soft IoU 0.0580, route ratio 0.9521, obstacle overlap 0.7204, route gap 0.0833, and connectivity 0.8195. This is mathematically cleaner than the geodesic blob-collapse result, but it is not a valid route solution because the prediction still overlaps the obstacle heavily.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Sharp	Conn.	Steps
steps_8	0.0580	0.0382	0.8815	0.6167	0.0000	0.2178	2.0000	8.0000
steps_16	0.0580	0.0797	0.9521	0.7204	0.0833	0.2071	0.8195	16.0000
steps_24	0.0565	0.0550	0.9904	0.7470	0.0000	0.2085	1.3667	24.0000
steps_32	0.0531	0.0531	0.9989	0.7038	0.0000	0.2095	1.4724	32.0000

Table 33: Held-out batch step sweep for the twenty-third soft shortest-path probe. Sixteen value-iteration steps give the best hard IoU, but all settings retain high obstacle overlap.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Conn.	Threshold
step_8_th_0.50	0.0580	0.1398	0.8815	0.6167	0.3781	0.2758	0.5000
step_16_th_0.35	0.0580	0.0797	0.9521	0.7204	0.0833	0.8195	0.3500
step_16_th_0.50	0.0580	0.0797	0.9521	0.7204	0.0896	0.8195	0.5000
step_24_th_0.50	0.0565	0.0700	0.9904	0.7470	0.0000	1.1559	0.5000
step_32_th_0.50	0.0531	0.0775	0.9989	0.7038	0.0990	0.8797	0.5000

Table 34: Representative threshold settings from the twenty-third soft shortest-path probe. Thresholding can improve hard IoU, especially at step 8, but this reflects large geometric basins rather than a clean corridor.

W.3 What We Learn from the Twenty-Third Run

This soft shortest-path version is mathematically interesting, but it is not yet a good solution to the routing task. It learns structured fields and high connectivity, but it still cuts through the obstacle too much. A best step-level hard IoU of 0.0797 with obstacle overlap 0.7204 means the model is learning a shaped field rather than a valid detour.

The qualitative outputs in Figures 40 and 41 reveal the failure mode. The learned field produces diamond-like regions around the endpoints and broad basins in the distance map. These shapes are cleaner and more geometric than the circular blobs in the geodesic v2 run, but they still do not trace the target L-shaped route. The model is discovering symmetric low-cost basins, not carving a thin bent corridor.

The scientific interpretation is that soft value iteration is a useful diagnostic baseline. It shows that the optimization machinery can build a planning-like field, but the learned cost field is too isotropic and too permissive through obstacles. The result is another form of endpoint-basin collapse: more structured than a blob, but still not a materialized path.

W.4 Recommended Next Changes After Run Twenty-Three

The soft shortest-path branch should be kept as a baseline, not treated as the main result yet:

- prioritize the fracture core–spread family for the paper's main empirical story;
- strengthen obstacle repulsion in the shortest-path loss and learned cost field;
- add thinness or skeletonization pressure so value iteration produces corridors instead of diamonds;
- add directional or anisotropic costs to break the symmetric basin geometry;
- test a hybrid where soft shortest-path supplies a coarse cost prior and fracture dynamics perform final path hardening.

W.5 Qualitative Figures from the Twenty-Third Run

Figure 40 shows the main summary panel. The learned cost field is structured, but the prediction and thresholded map form large diamond-shaped regions rather than the target route. Figure 41 isolates the 0.50 threshold result and makes the failure even clearer: the output is a geometric basin, not a path.

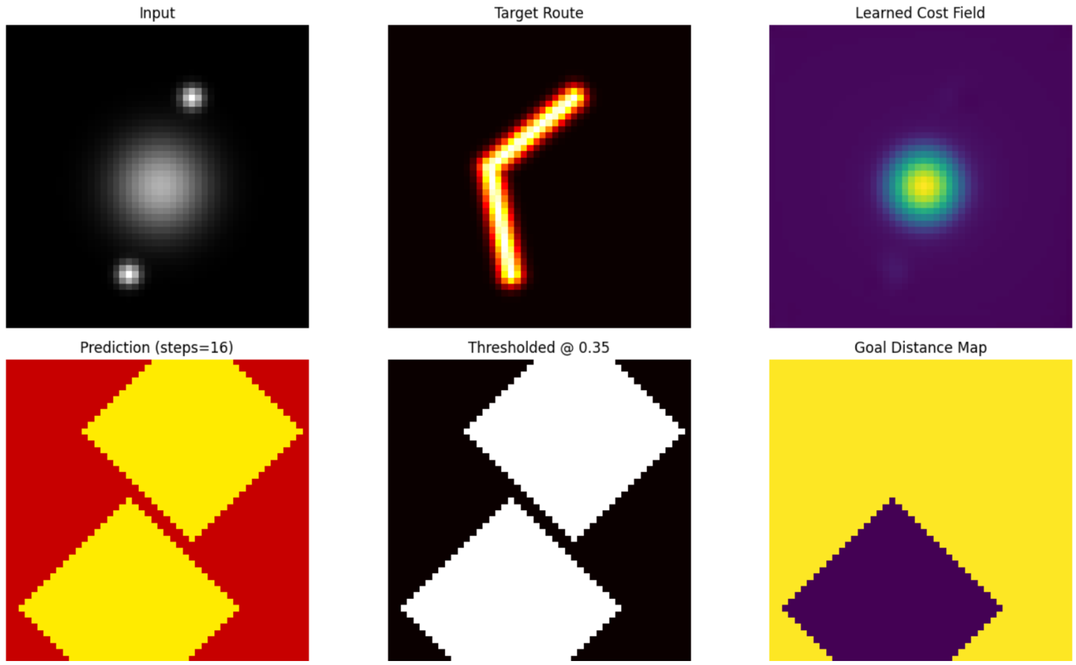

Figure 40: Twenty-third prototype qualitative result for the soft shortest-path probe. The learned cost and distance fields are structured, but the predicted occupancy forms broad diamond-shaped basins instead of the target bent corridor.

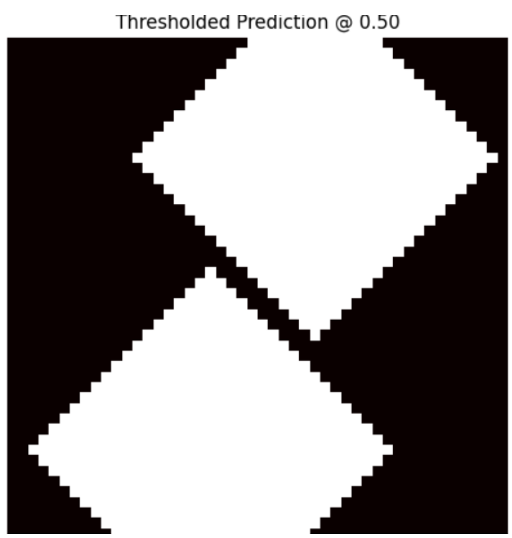

Figure 41: Twenty-third prototype thresholded prediction at 0.50. The clean diamond shapes show that the model has learned symmetric basins rather than a thin obstacle-avoiding path.

X Prototype Appendix: Twenty-Fourth Run Final Paper Pack

This twenty-fourth appendix entry consolidates the prototype story into a final paper-pack comparison. It trains four PF^{2}V variants—base, additive memory, material memory, and material-thin—then evaluates the best material-thin model on a standard held-out batch, a harder generalization batch, and a soft shortest-path negative control. The purpose is not to claim that the system is solved, but to state the strongest current evidence and the remaining failure modes in one place.

X.1 What We Changed

The final pack differs from the previous one-off probes in scope rather than architecture:

- it compares the **base**, **additive-memory**, **material-memory**, and **material-thin** PF^2V variants under the same synthetic routing setup;
- it evaluates repeated passes, step sweeps, and threshold sweeps on a held-out batch of 32 examples;
- it adds a harder generalization batch with wider obstacles, stronger bends, and occasional multiple obstacles;
- it includes a soft shortest-path control as a negative baseline;
- it saves a compact result pack and three qualitative figures for the standard, hard-generalization, and soft-control cases.

This gives a clearer final comparison between the fracture-style material substrate and the more algorithmic soft-control baseline.

X.2 Main Held-Out Results

On the standard held-out batch, the material-thin model remains the most useful PF^2V variant for visible path recovery. Its best step is 4, with hard IoU 0.0710, soft IoU 0.0393, route ratio 0.7455, obstacle overlap 0.1673, route gap 0.9117, and connectivity 0.0067. The base model also recovers a weak visible signal at four steps, with hard IoU 0.0526, but it has higher obstacle overlap at 0.1970. The additive and material-memory variants are more conservative, but their hard IoU is lower at the same step.

Model / setting	Soft	Hard	Ratio	ObsOverlap	Gap	Conn.
base, steps_4	0.0374	0.0526	0.8566	0.1970	0.8992	0.0329
additive, steps_4	0.0377	0.0034	0.8805	0.1402	0.9937	0.0000
material, steps_4	0.0351	0.0070	0.8915	0.1345	0.9930	0.0002
material-thin, steps_4	0.0393	0.0710	0.7455	0.1673	0.9117	0.0067
material-thin, steps_8	0.0219	0.0000	0.7551	0.0369	1.0000	0.0000
material-thin, steps_16	0.0070	0.0000	0.8904	0.0046	1.0000	0.0000

Table 35: Final paper-pack standard held-out comparison. The material-thin model gives the best visible PF^2V hard IoU at four steps, but deeper steps suppress the path while improving obstacle avoidance.

The threshold sweep confirms that this final material-thin result is not merely a low-threshold artifact. At the best step, threshold 0.35 is also the best overall threshold, with hard IoU 0.0710. Lowering to 0.25 increases connectivity to 0.0866 and reduces route gap to 0.8078, but hard IoU falls slightly to 0.0612. Raising to 0.50 leaves only a tiny fragment, with hard IoU 0.0444 and connectivity 0.0012.

X.3 Harder Generalization and Soft Control

The harder generalization set tells a similar story. The material-thin model again performs best at four steps, with hard IoU 0.0636, soft IoU 0.0466, route ratio 0.8518, obstacle overlap 0.1785, and connectivity 0.0178. At threshold 0.25, the hard-generalization hard IoU rises to 0.0848 and connectivity rises to 0.1484, but this comes with the usual lower-threshold tradeoff: a less selective, more permissive visible mask.

The soft shortest-path control is worse as a route-valid solution. Its best step is 8, with hard IoU 0.0479, route ratio 0.9557, obstacle overlap 0.5814, and connectivity 1.3847. The high connectivity is misleading: the control creates broad connected activation through obstacle regions rather than a selective detour. Its best threshold, 0.50 at eight steps, improves hard IoU only to 0.0528 while obstacle overlap remains 0.5814.

Setting	Soft	Hard	Ratio	ObsOverlap	Gap	Conn.	Threshold
thin standard best step	0.0393	0.0710	0.7455	0.1673	0.9117	0.0067	0.3500
thin hard-gen best step	0.0466	0.0636	0.8518	0.1785	0.9148	0.0178	0.3500
thin hard-gen best threshold	0.0466	0.0848	0.8518	0.1785	0.7117	0.1484	0.2500
soft control best step	0.0482	0.0479	0.9557	0.5814	0.0227	1.3847	0.3500
soft control best threshold	0.0482	0.0528	0.9557	0.5814	0.1727	1.0285	0.5000

Table 36: Final paper-pack comparison between material-thin PF^{2}V, harder generalization, and the soft-control baseline. Material-thin has lower connectivity but far lower obstacle overlap; the soft control is connected mainly because it is broad and obstacle-overlapping.

X.4 What We Learn from the Final Pack

The final pack supports a cautious paper claim. The material-thin PF^{2}V model is not yet a robust route solver, but it is the best current compromise between path visibility and obstacle selectivity. It produces faint, fragmentary route mass that can sometimes survive thresholding, while keeping obstacle overlap far below the soft-control baseline. The key limitation remains the same: the model has partial route logic, but it still fails to materialize a connected corridor.

The harder generalization result is encouraging in a narrow sense. The model does not collapse completely when obstacles are larger and the bend is stronger. However, the hard path is still weak, and the visible route requires the shallow four-step regime. Deeper schedules continue to clean up obstacle overlap at the cost of erasing the path.

The soft-control result helps position the contribution. A more planning-like value-field baseline can produce high connectivity, but it does so by lighting up broad regions and overlapping obstacles. This makes it a useful negative control: connectivity alone is not evidence of good routing. For this task, selective visibility and obstacle avoidance matter together.

X.5 Recommended Final Framing

The final manuscript should frame the empirical evidence as a prototype study, not as a solved benchmark:

- present material-thin PF^{2}V as the strongest current fracture-style variant;
- emphasize the tradeoff between shallow visible paths and deeper obstacle suppression;
- use the soft-control baseline as evidence that broad connectivity is not enough;
- state that the main remaining problem is connected path materialization;
- propose future work on stronger core commitment, continuity losses, and hybrid planners that harden only thin corridors.

X.6 Qualitative Figures from the Final Pack

Figures 42, 43, and 44 show the final qualitative comparison. The standard material-thin example contains a localized but fragmented route signal. The hard-generalization example shows that the model retains

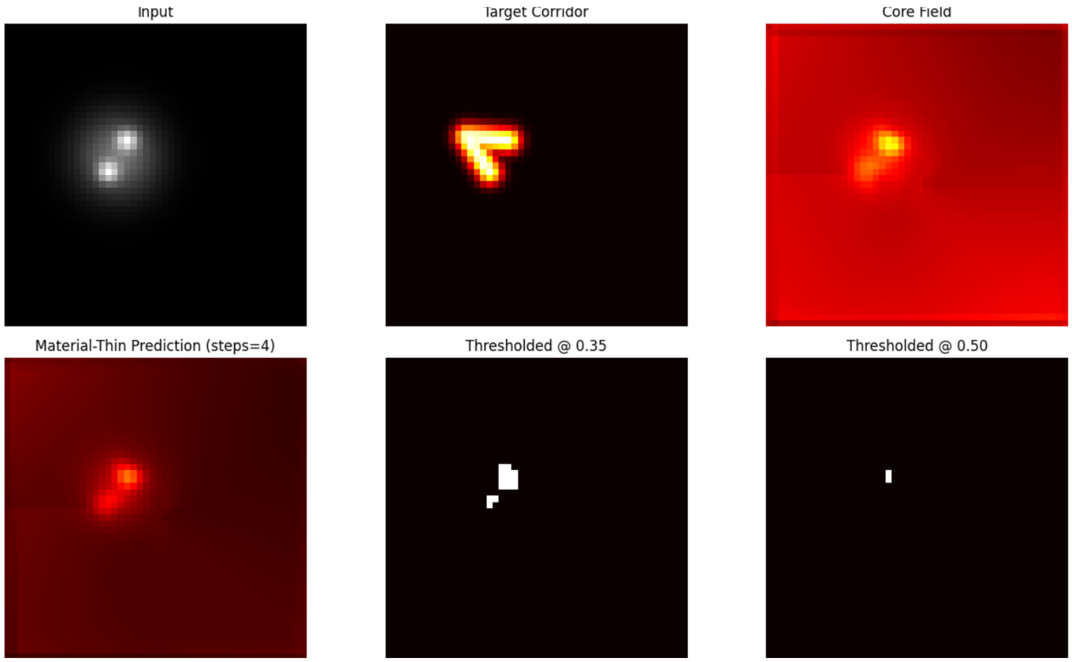

Figure 42: Twenty-fourth prototype final-pack standard material-thin result. The model localizes route-like mass in the core and prediction, but the thresholded outputs remain fragmented and sparse.

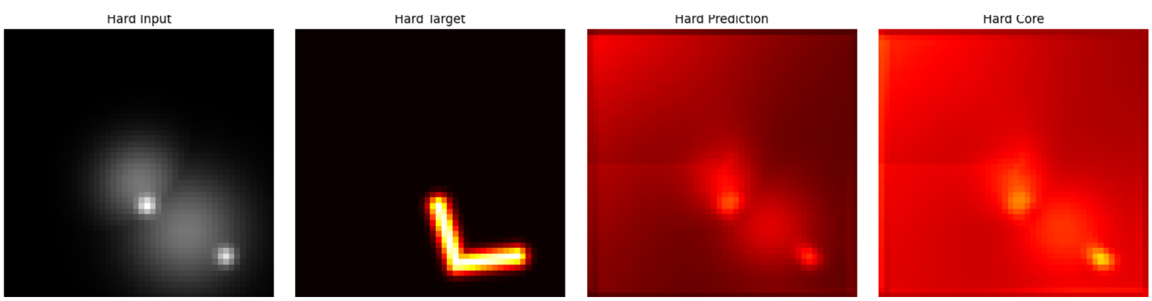

Figure 43: Twenty-fourth prototype final-pack hard-generalization result. The material-thin model retains structured activation under a harder obstacle layout, but it still does not yield a clean connected binary path.

some structured activation under a harder obstacle layout, but it still lacks a clean binary corridor. The soft-control example shows the opposite failure mode: broad value-field activation with poor selectivity.

X.7 Seed Robustness and Paper Summary Plots

After the final-pack comparison, an additional robustness evaluation was run over three random held-out batches using seeds 0, 1, and 2. The evaluation reused the already-trained material-thin PF^{2}V model and the soft-control baseline, and swept solver steps and thresholds on both the standard held-out setting and the harder generalization setting. The purpose was not to retrain the models, but to check whether the final qualitative conclusion survives modest batch-level randomness.

Table 37 summarizes the robustness run. The material-thin model remains clearly more obstacle-selective

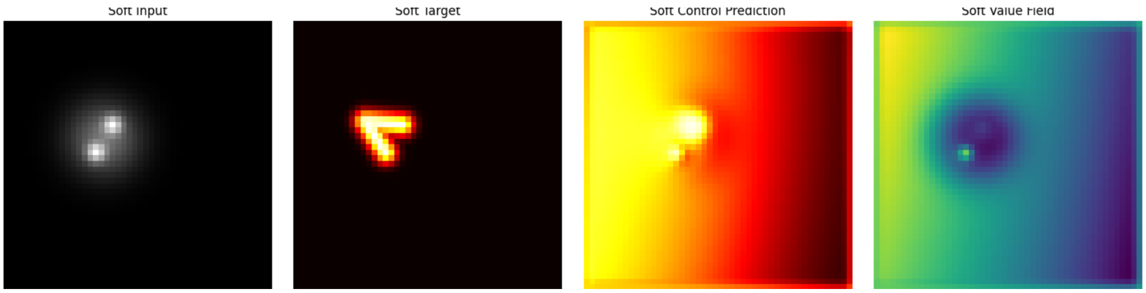

Figure 44: Twenty-fourth prototype final-pack soft-control result. The soft-control baseline produces broad value-field activation and high apparent connectivity, but its prediction is not a selective obstacle-avoiding corridor.

than the soft-control baseline. On the standard setting, material-thin with threshold selection reaches mean hard IoU 0.1244 ± 0.0096 with obstacle overlap 0.1780 ± 0.0141, while soft-control with threshold selection reaches only 0.0500 ± 0.0027 hard IoU and much higher obstacle overlap 0.5888 ± 0.0154. On the hard setting, material-thin again improves hard IoU relative to soft-control and keeps obstacle overlap near 0.18, whereas soft-control remains near 0.45 obstacle overlap. The tradeoff is still visible: soft-control has much larger apparent connectivity, but that connectivity is coupled to broad obstacle-overlapping activation.

Model	Set	Hard IoU	Route Ratio	ObsOverlap	Conn.
material-thin	hard	0.0982 ± 0.0239	0.8376 ± 0.0274	0.1807 ± 0.0143	0.0283 ± 0.0134
material-thin	std	0.0788 ± 0.0095	0.7946 ± 0.0163	0.1780 ± 0.0141	0.0074 ± 0.0024
material-thin-thresh	hard	0.1106 ± 0.0248	0.8376 ± 0.0274	0.1807 ± 0.0143	0.1580 ± 0.0384
material-thin-thresh	std	0.1244 ± 0.0096	0.7946 ± 0.0163	0.1780 ± 0.0141	0.0727 ± 0.0098
soft-control	hard	0.0489 ± 0.0062	1.0241 ± 0.0155	0.4514 ± 0.0070	1.2645 ± 0.0086
soft-control	std	0.0452 ± 0.0020	0.9652 ± 0.0051	0.5888 ± 0.0154	1.3786 ± 0.0130
soft-control-thresh	hard	0.0492 ± 0.0064	1.0241 ± 0.0155	0.4514 ± 0.0070	1.0212 ± 0.2180
soft-control-thresh	std	0.0500 ± 0.0027	0.9652 ± 0.0051	0.5888 ± 0.0154	1.0395 ± 0.0060

Table 37: Three-seed robustness summary for the final paper pack. Values report mean $\pm$ standard deviation over seeds 0, 1, and 2. Threshold selection improves material-thin hard IoU and connectivity while preserving low obstacle overlap; soft-control remains highly connected but obstacle-overlapping.

A compact paper-summary extraction was also generated from the saved final-pack JSON files. Table 38 records the best single-row comparison used for the summary plots. This table reinforces the same interpretation: material-thin is the best current PF^{2}V variant for visible path recovery, while soft-control has superficially high connectivity but poor obstacle selectivity.

Model	Hard IoU	Route Ratio	ObsOverlap	Conn.
base	0.0526	0.8566	0.1970	0.0329
additive	0.0034	0.8805	0.1402	0.0000
material	0.0070	0.8915	0.1345	0.0002
material-thin	0.0710	0.7455	0.1673	0.0067
soft-control	0.0528	0.9557	0.5814	1.0285

Table 38: Compact paper-summary table extracted from the saved final-pack results. Material-thin has the best PF^{2}V hard IoU, while soft-control has high connectivity but much worse obstacle overlap.

Figures 45, 46, and 47 show the generated paper-summary plots. The bar summary makes the main contrast visually explicit: soft-control connectivity is high, but this coincides with high obstacle overlap. The material-thin step sweep shows the same shallow-depth tradeoff identified above: early steps preserve visible route mass, while deeper steps increasingly suppress the path. The threshold sweep shows that threshold 0.35 is the best operating point for hard IoU in the final material-thin summary, while lower thresholds trade selectivity for connectivity.

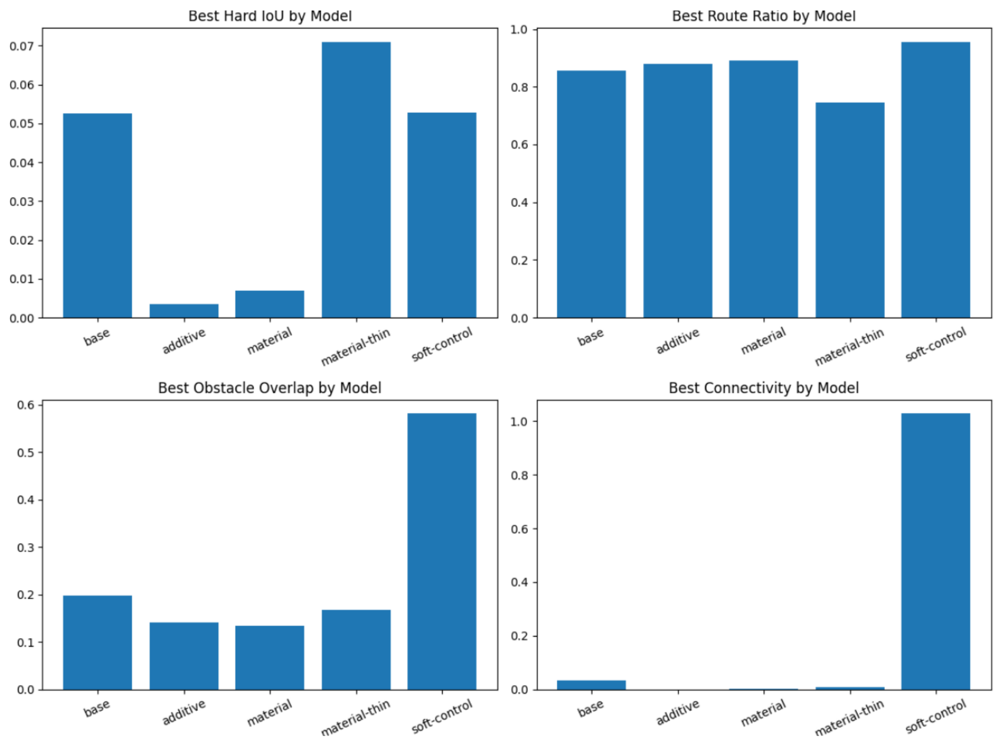

Figure 45: Final paper-summary bar plots comparing hard IoU, route ratio, obstacle overlap, and connectivity across the main model variants. The soft-control baseline has high apparent connectivity, but also the largest obstacle overlap.

X.8 Final Focus Run and Combined Conclusion

A final focus run was then performed to combine the lessons from the final-pack comparison and the subsequent robustness check. This run warm-started from the previous material-thin model and trained a stronger material-memory variant with more explicit core commitment, endpoint pressure, centerline supervision, obstacle avoidance, side suppression, and thinness regularization. The aim was to test whether the previous obstacle-selective but fragmented route signal could be made more visible without returning to the broad obstacle-overlapping behavior of the soft-control baseline.

During training, the focus model showed intermittent but meaningful gains in visible path recovery. By epoch 25, the monitored batch reached hard IoU 0.1813, soft IoU 0.0786, route ratio 0.7424, obstacle overlap 0.2252, and connectivity 0.1289. This is not a fully solved routing model, but it is a stronger result than the earlier final-pack material-thin snapshot: the model can now reach substantially higher hard IoU while retaining lower obstacle overlap than the soft-control baseline.

Table 39 summarizes the step sweep. On the standard held-out batch, the best step-only hard IoU occurs at 8 steps, with hard IoU 0.1860 and connectivity 0.1511, but obstacle overlap is still high at 0.4114. At 12 steps, hard IoU remains high at 0.1766 while obstacle overlap falls to 0.1826, showing a better selectivity

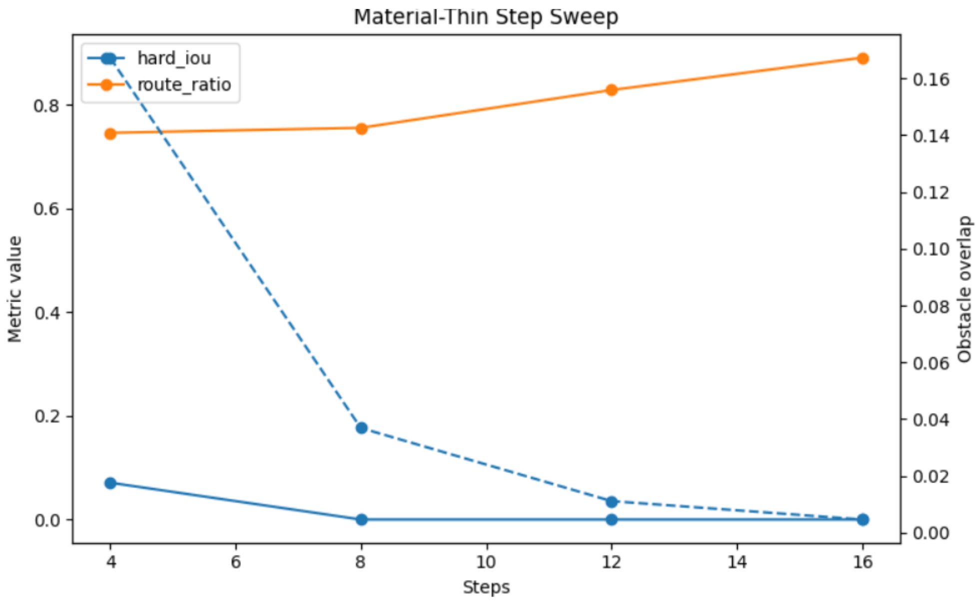

Figure 46: Material-thin step sweep from the final paper pack. The strongest visible path signal appears at shallow solver depth, while additional steps reduce obstacle overlap but also erase the already-fragile path.

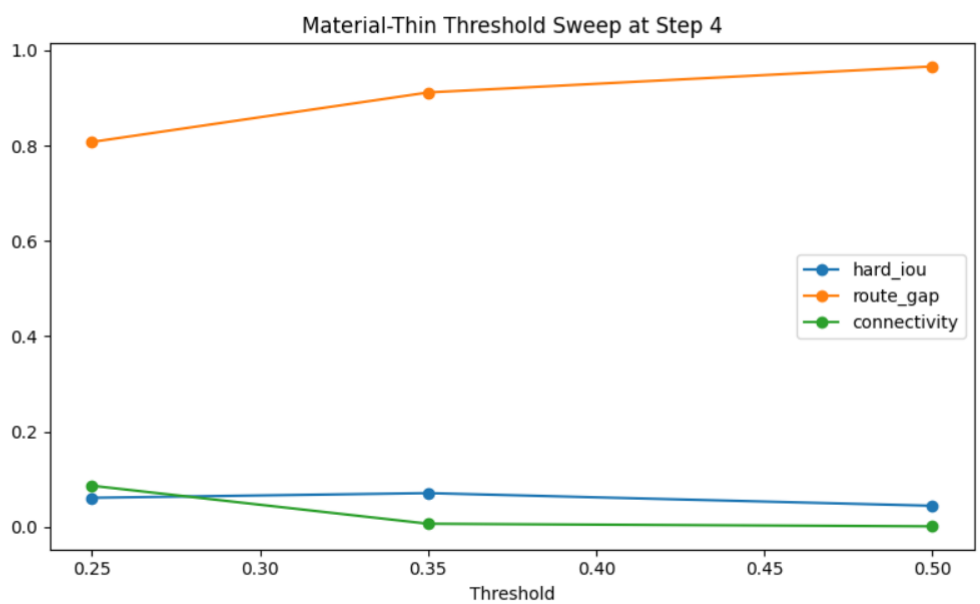

Figure 47: Material-thin threshold sweep at the best final-pack step. Threshold 0.35 gives the best hard IoU, while lower thresholding increases connectivity and reduces route gap at the cost of less selective activation.

tradeoff. At 16 steps, obstacle overlap improves further to 0.0752, but the visible path weakens. The hard-generalization batch follows the same pattern: 12 steps gives the best step-only compromise, with hard IoU 0.1680 and obstacle overlap 0.1470.

Split	Setting	Soft	Hard	Ratio	ObsOverlap	Conn.
standard	steps_4	0.0705	0.1047	0.8796	0.6892	0.8144
standard	steps_8	0.0876	0.1860	0.7700	0.4114	0.1511
standard	steps_12	0.0829	0.1766	0.7165	0.1826	0.0273
standard	steps_16	0.0648	0.1104	0.7365	0.0752	0.0054
hard	steps_4	0.0757	0.0949	0.9271	0.7002	0.9579
hard	steps_8	0.0927	0.1652	0.8190	0.3973	0.2914
hard	steps_12	0.0823	0.1680	0.7173	0.1470	0.0441
hard	steps_16	0.0582	0.1082	0.6662	0.0480	0.0055

Table 39: Final focus run step sweep. The focus model improves visible hard IoU, especially at 8–12 steps, but retains the depth tradeoff: shallow steps are connected and broad, while deeper steps are obstacle-selective but fragmentary.

The threshold sweep gives the strongest single numbers from the final focus run. On the standard batch, the best thresholded setting is step 12 at threshold 0.20, reaching hard IoU 0.2066, route ratio 0.7165, obstacle overlap 0.1826, and connectivity 0.0665. On the harder batch, the best thresholded setting is step 12 at threshold 0.25, reaching hard IoU 0.1866, route ratio 0.7173, obstacle overlap 0.1470, and connectivity 0.0795. These values represent the clearest final evidence that the material focus model improves path visibility relative to the earlier material-thin result while avoiding the large obstacle overlap of soft-control.

Split	Threshold Step	Threshold	Hard IoU	Route Ratio	ObsOverlap	Conn.
standard	12	0.20	0.2066	0.7165	0.1826	0.0665
hard	12	0.25	0.1866	0.7173	0.1470	0.0795

Table 40: Final focus paper summary. Values report the best thresholded focus-model outputs, so the threshold step can differ from the best fixed-threshold step sweep. These settings give the strongest final hard-IoU results while keeping obstacle overlap below the broad soft-control baseline.

Figures 48, 49, and 50 show the final focus visuals. The standard and hard examples show stronger route-like activation than the earlier material-thin outputs, though neither forms a perfectly continuous binary corridor. The summary bars show that the final focus model improves hard IoU on both standard and hard batches, with the hard batch achieving slightly lower obstacle overlap.

X.9 Final Empirical Conclusion

Taken together, the final-pack, robustness, and focus runs support a cautious but stronger conclusion than the earlier appendices. PF^2V is still not a solved path planner: the main remaining failure is connected binary path materialization. However, the best material-style variants now show a repeatable pattern that is not explained by generic broad activation alone. They can trade solver depth against obstacle selectivity, and the final focus model reaches the highest hard IoU observed in these experiments while keeping obstacle overlap far below the soft-control baseline.

The central empirical claim should therefore be framed as follows. Phase-field material dynamics are a plausible latent substrate for obstacle-aware route formation, but the current implementation expresses this mostly as partial, threshold-sensitive route mass rather than as a clean discrete corridor. The focus model improves the previous material-thin result from roughly 0.07–0.12 hard IoU into the 0.19–0.21 range, and it

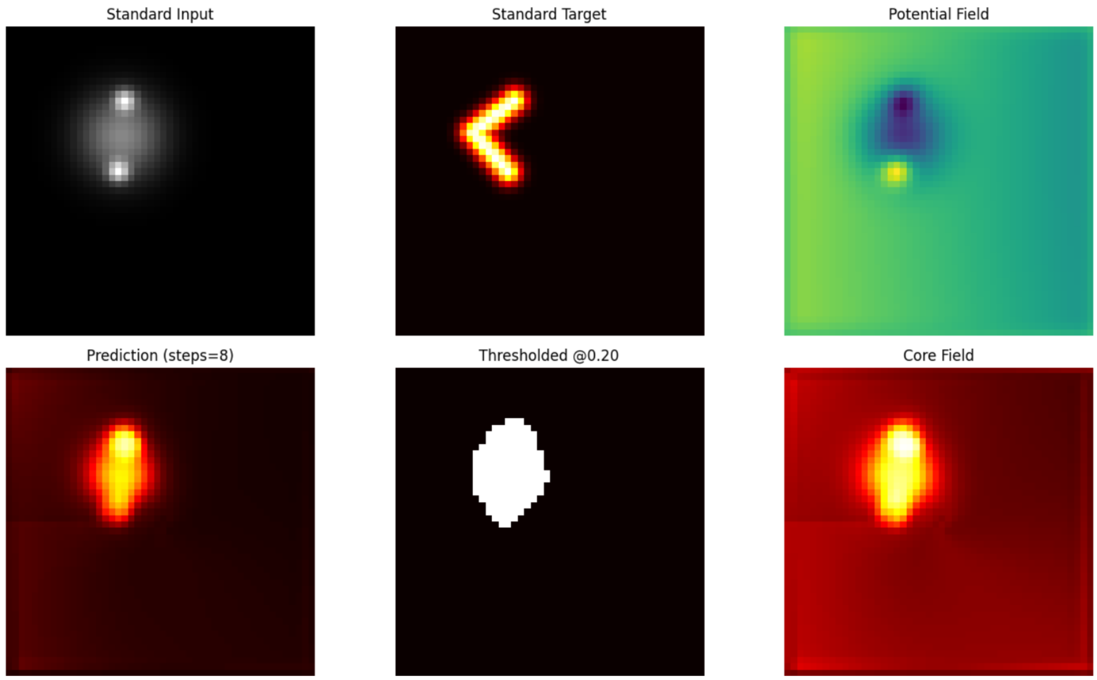

Figure 48: Final focus standard held-out visual. The focus model produces stronger route-like probability and core activation than the earlier material-thin snapshot, but the thresholded path remains only partially connected.

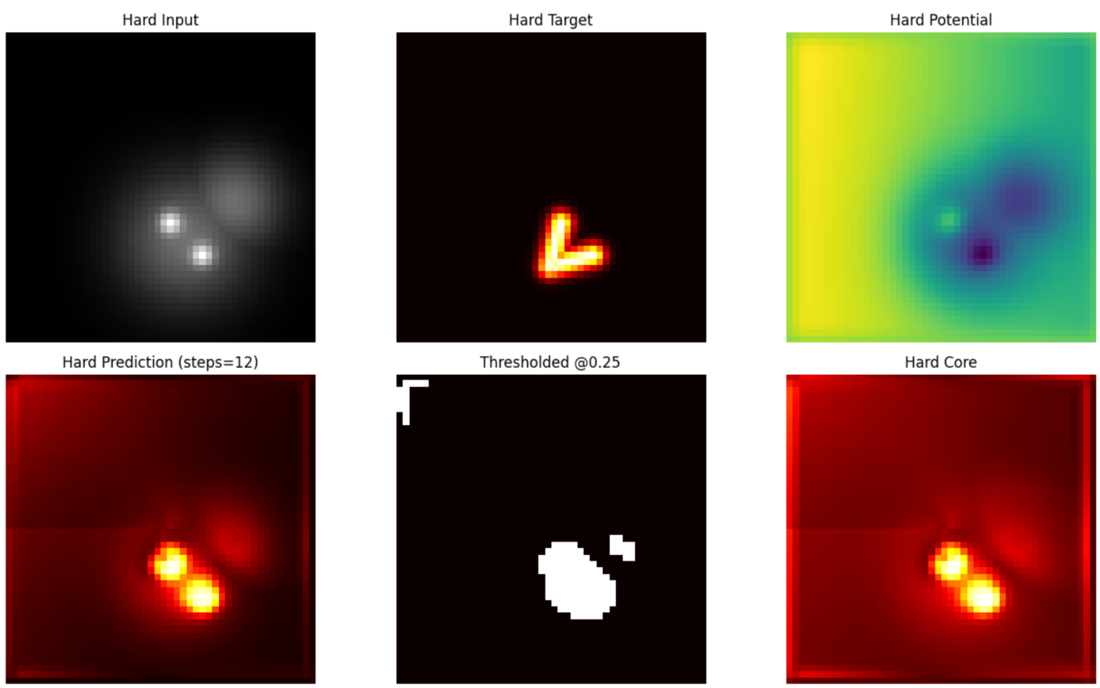

Figure 49: Final focus hard-generalization visual. The model retains structured route-like activation under a harder obstacle layout, with improved hard IoU but still incomplete binary path materialization.

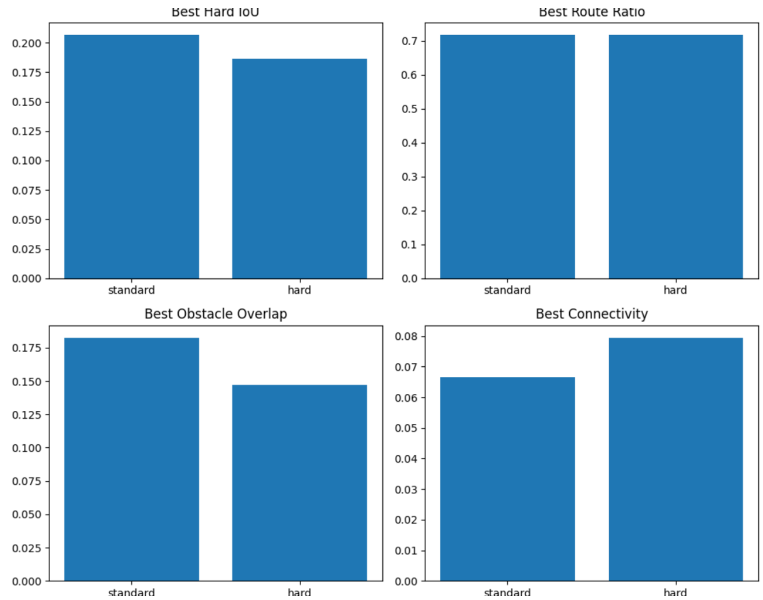

Figure 50: Final focus summary bars for the standard and hard held-out splits. The best thresholded focus outputs reach hard IoU 0.2066 on the standard split and 0.1866 on the hard split.

generalizes to a harder obstacle setting with hard IoU 0.1866. This is meaningful progress for a prototype, but not benchmark-level success.

The final paper should emphasize three points. First, material-style PF^{2}V variants are more obstacle-selective than the soft-control baseline. Second, connectivity alone is misleading, because broad value-field controls can appear connected while crossing obstacles. Third, the next technical bottleneck is not route awareness but *route hardening*: converting faint, obstacle-aware latent mass into a thin, continuous, threshold-stable path.